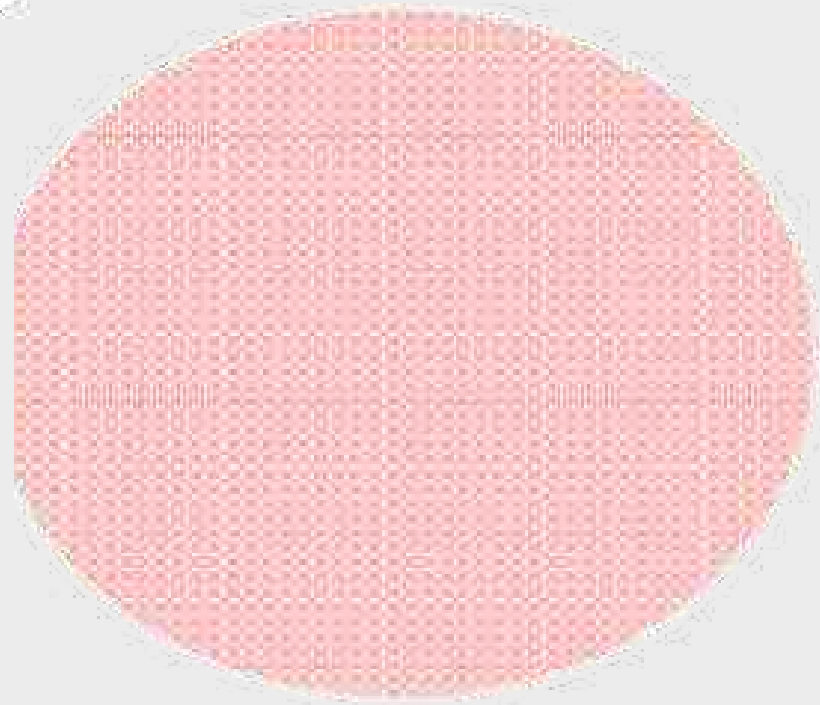




BIM
PORTFOLIO

Ar.Bilal Ahmad



BIM Architect | Architectural Designer | Interior Designer

Aspiring architect with a strong passion for integrating science and art to create impactful spaces. Experienced in conceptual design, construction documentation, and material selection. Skilled in collaborating with multidisciplinary teams, participating in live projects, and conducting site visits. Currently pursuing a diploma in Building Information Modeling (BIM) to strengthen technical expertise and collaborative workflows.

 bilalahmadk37@gmail.com

 +92 319 5791745

 <https://www.linkedin.com/in/ar-bilal-ahmad>

 Riyadh, Saudi Arabia

EDUCATION

Bachelor of Architecture
University of Engineering & Technology (UET),
Peshawar – Abbottabad Campus

- October 2020 - July 2025
- Achieved a CGPA of 3.29 demonstrating strong academic performance in architecture

WORK EXPERIENCE

Senior Architect

MAS-DESIGNS ARCHITECTURE FIRM

 November 2025-July 2026  Lahore

A creative architecture studio focused on innovative design solutions and real-world project execution.

- Working as a Senior Architect, I contribute to commercial and residential projects from concept to execution. I have experience in minimalist design, design development, and planning, along with strong knowledge of material selection. I have also worked as an Interior Designer and Site Supervisor, handling site management and coordination. Additionally, I am involved in project management, ensuring smooth and timely project execution.

BIM Architect

ARCHILANCE LLC

 March 2025-October 2025  Lahore

- Developed and coordinated BIM models using Autodesk Revit for residential and interior projects. Created parametric Revit families, detailed architectural drawings, and RCPs while utilizing BIM 360/ACC for collaboration and model coordination. Contributed to 16+ projects, ensuring accuracy, quality, and timely delivery.

Trainee Architect

MMAD-DESIGN ARCHITECTURE FIRM

 Jan 2023-Feb 2023  Lahore

- Assisted in concept development, architectural drawings, and site visits.
- Gained hands-on experience in design and construction processes.

Junior Architect

DESIGNS (Architects Engineers Designers)

 January 2024 - February 2024  Lahore

- Assisted with architectural drawings, presentations, material boards, and site visits.
- Gained practical exposure to construction and technical coordination.

Junior Architect

GALLERIA DESIGNS ARCHITECTURE FIRM

 January 2025 - February 2025  Lahore

- Prepared working drawings and developed interior/exterior material selections.
- Contributed to residential and commercial design development.

CERTIFICATION

IAP (Institute of Architects Pakistan)

Registered as an Affiliate Member of IAP on 24th April 2025, reflecting my professional commitment to architectural practice, continuous learning, and engagement with Pakistan's leading architectural community.

IAPEX Peshawar Chapter (Building Materials & Product Exhibition & RCESS-II)

Volunteer – IAPEX Peshawar Chapter

Recognized for active, dedicated participation and valuable contribution as a volunteer.

IAPEX Lahore Chapter 11.24 (Building Materials Exhibition)

Participated in the Building Materials Exhibition, gaining exposure to innovative construction materials and industry trends.

Alumni Event – UET Abbottabad

Participated as part of the Media & Graphics Team, contributing to event branding, visual content creation, and overall presentation design.

COURSE

BIM (Building Information Modeling)

 September 2015 - June 2026

Completed a 1-year BIM diploma focused on 3D modeling, construction documentation, BIM coordination, and digital project delivery.

Skills.

Implemented ISO 19650 BIM workflows and managed Revit models from LOD 300–500. Experienced in parametric modeling, quantity take-offs, façade and interior detailing, RCP and lighting layouts, and multidisciplinary Architecture, Structure & MEP coordination. Skilled in Lumion, Enscape, and V-Ray for 3D visualization.

SOFTWARE SKILLS

Drafting & 3D : AutoCAD, Revit, Navisworks,

3D Visualization: Lumion, Enscape, D5 Render, V-Ray

Graphic Tools: Adobe Photoshop, MS PowerPoint

Design & Coordination: Concept Development, Material Selection, Client Coordination, Team Collaboration

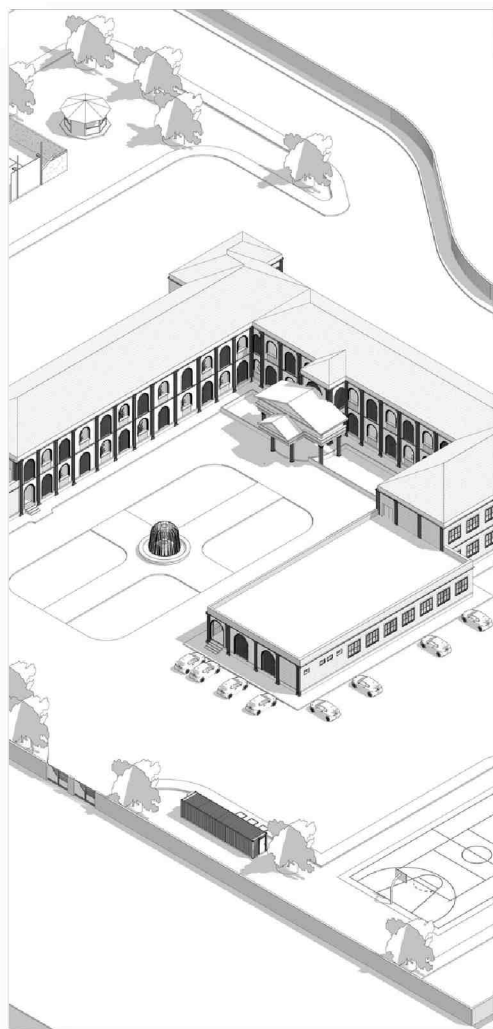
LANGUAGE

Pastho	Native	
English	Advanced	
Pastho	Proficient	

CONTENTS

PROJECTS

01



GOVERNMENT
DEGREE COLLEGE

02



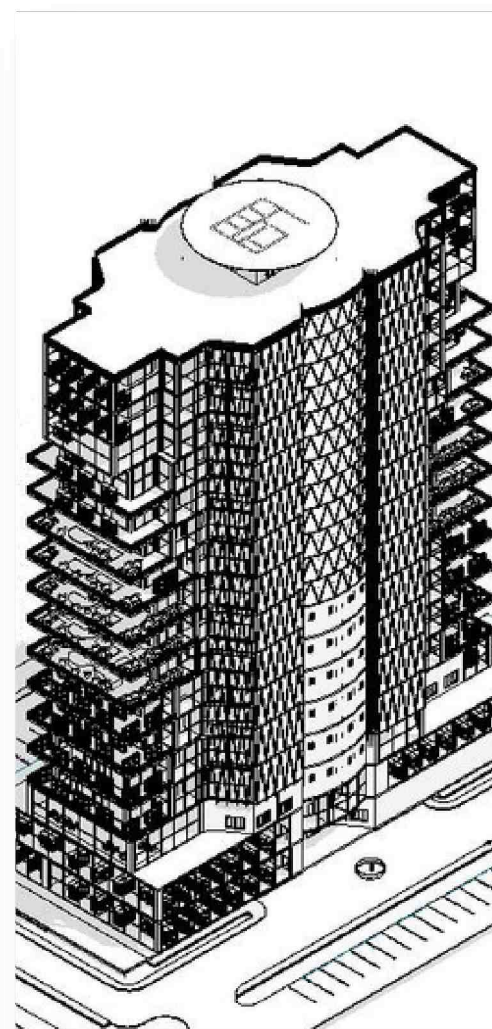
INTERACTIVE
RETREAT

03



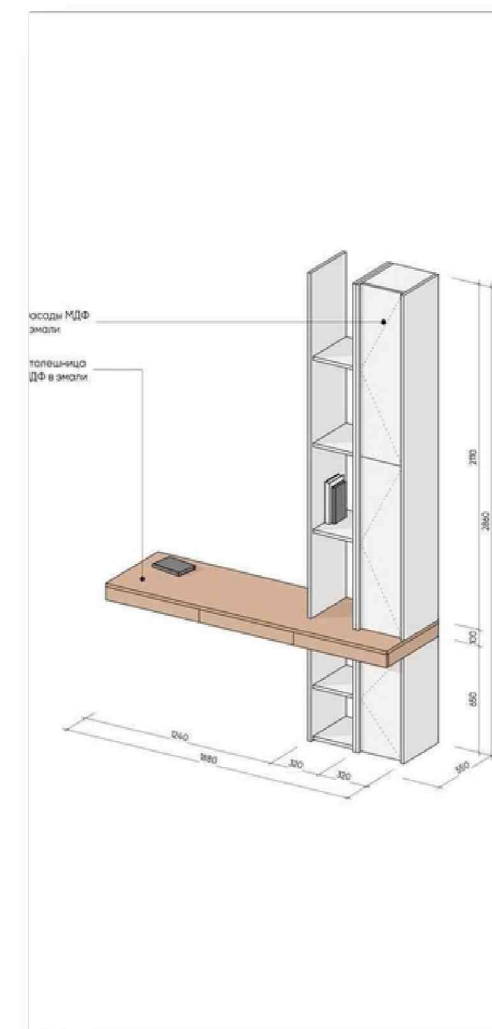
RIISING THE TECH

04



VERTICAL
FARMING

05



DETAILS AND
FAMILIES WORK

YEAR:
2026

LOCATION :
PESHAWAR

TYPE : EDUCATIONAL
BUILDING

01

GOVERNMENT COLLEGE

This project is a courtyard-based Government College Building designed to provide a functional, comfortable, and well-organized educational environment. The design is centered around a spacious internal courtyard that acts as the social and environmental core of the building, promoting natural daylight, ventilation, and visual connectivity. The building accommodates classrooms, laboratories, administrative offices, staff rooms, support spaces, and multipurpose areas, with clearly defined zoning and efficient circulation. Covered corridors and open spaces ensure easy movement and interaction, while the overall planning focuses on accessibility, functionality, student comfort, and a sustainable learning environment.



DESIGN CONCEPT

The design draws inspiration from the idea of "Learning in a Courtyard," where the built form embraces a central open space that becomes the heart of the institution.

The architecture promotes openness, connectivity, and community interaction while maintaining privacy and security.



Create a learning environment that encourages interaction and community.



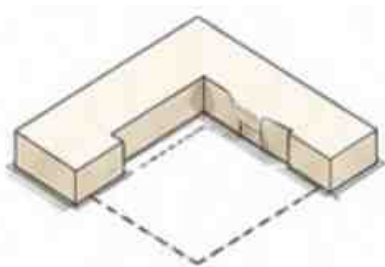
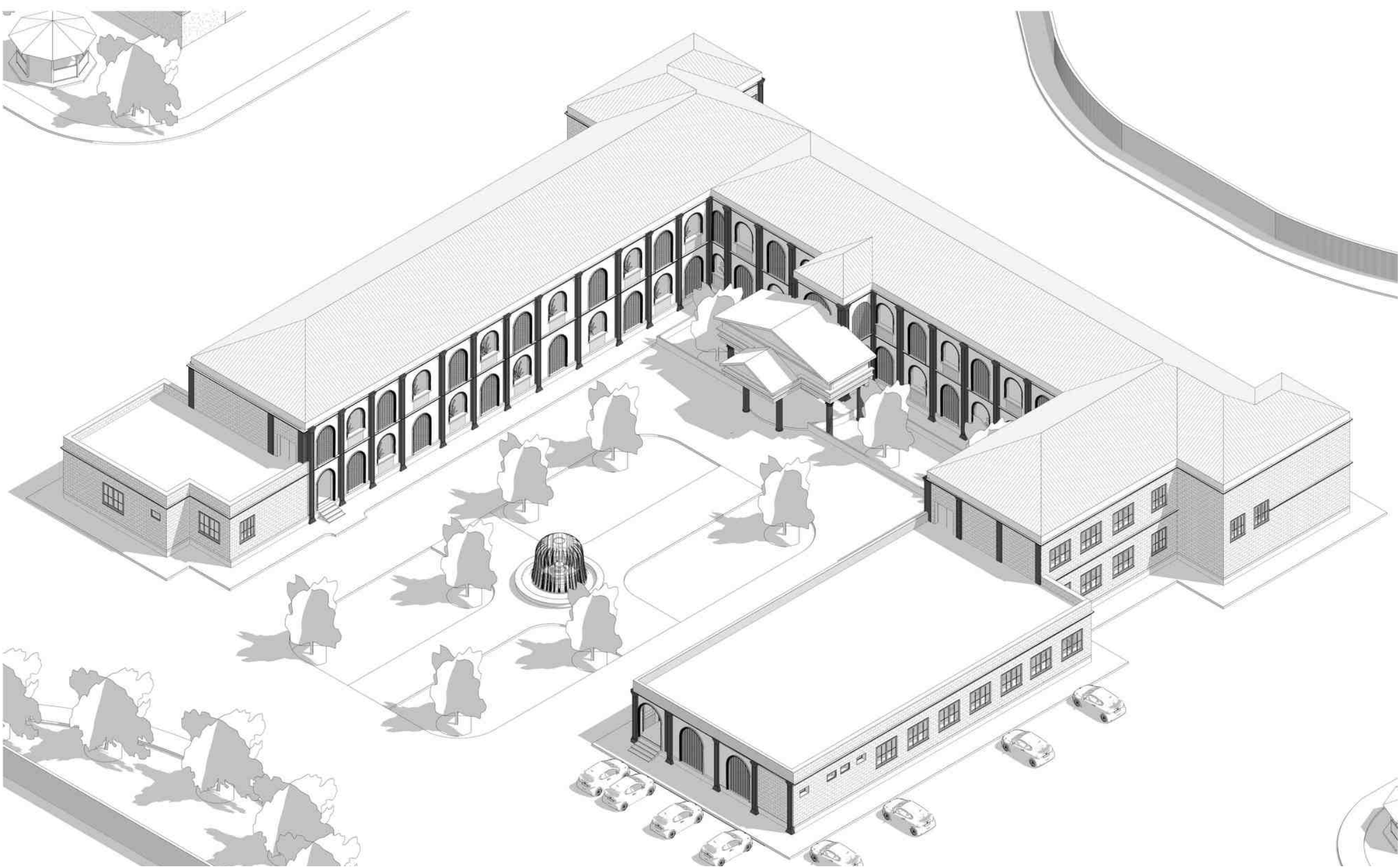
Ensure safety, privacy and a sense of ownership.



Integrate nature and open spaces for well-being.

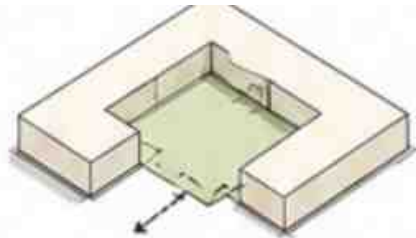


Design a timeless and functional institutional architecture.



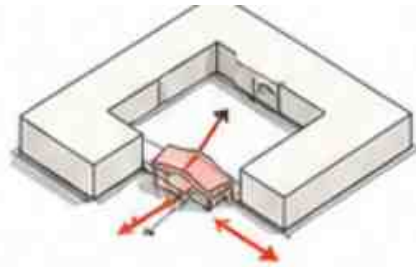
**STEP 01
BASIC MASS**

A simple U-shaped mass placed on site to create a central court.



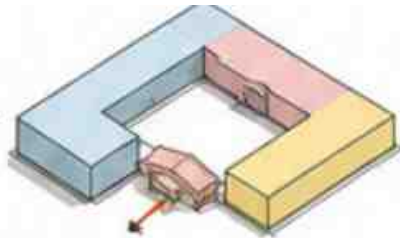
**STEP 02
COURTYARD CREATION**

The form is arranged around a central courtyard to promote interaction, light and ventilation.



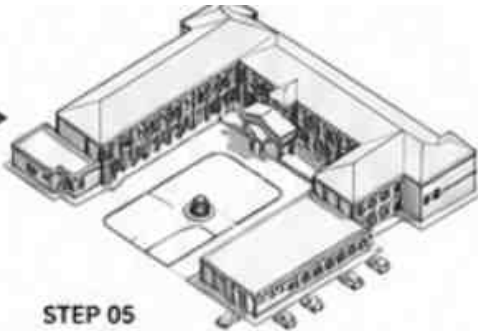
**STEP 03
AXIS & ENTRANCE**

A strong central axis is created with a main entrance to emphasize arrival and orientation.



**STEP 04
FUNCTIONAL WINGS**

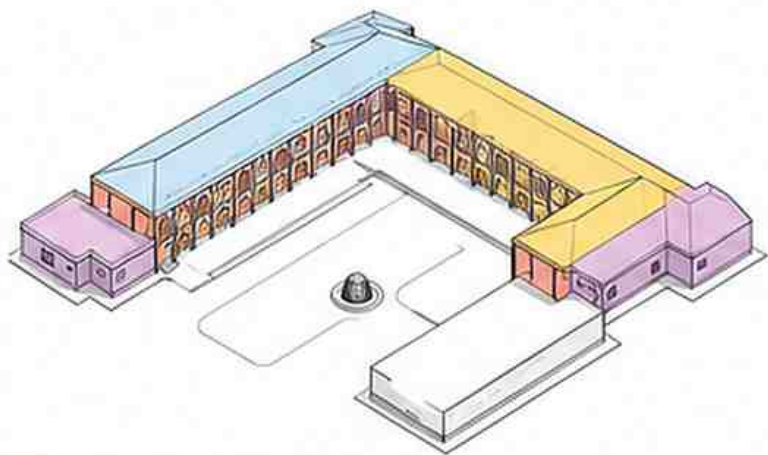
Different functions are allocated in wings while maintaining connection with the central court.



**STEP 05
FINAL FORM**

The form is refined with architectural elements, roof forms and facade treatment.

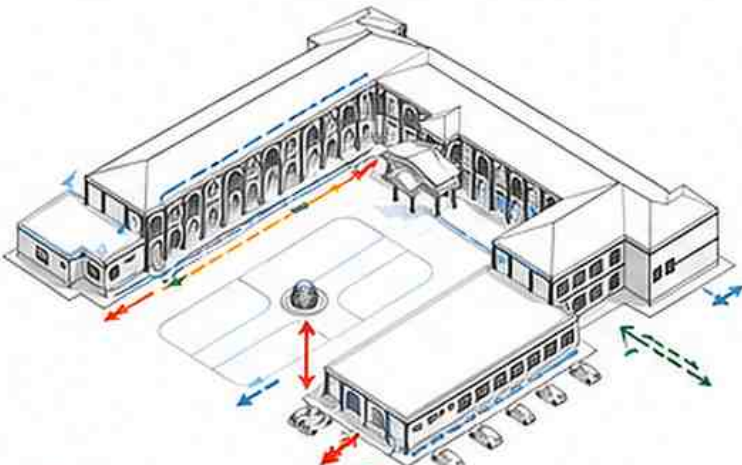
FUNCTIONAL ZONING



- ACADEMIC WING (CLASSROOMS)
- ADMINISTRATION
- LIBRARY / RESOURCE CENTER
- MULTI-PURPOSE HALL
- SUPPORT SPACES
- CIRCULATION

The planning organizes quiet, semi-public and public zones for smooth functionality and privacy.

CIRCULATION DIAGRAM



- MAIN ENTRY
- SECONDARY ENTRY
- PEDESTRIAN MOVEMENT
- SERVICE MOVEMENT

Clear circulation enhances wayfinding and separates pedestrian, vehicular and service movement.

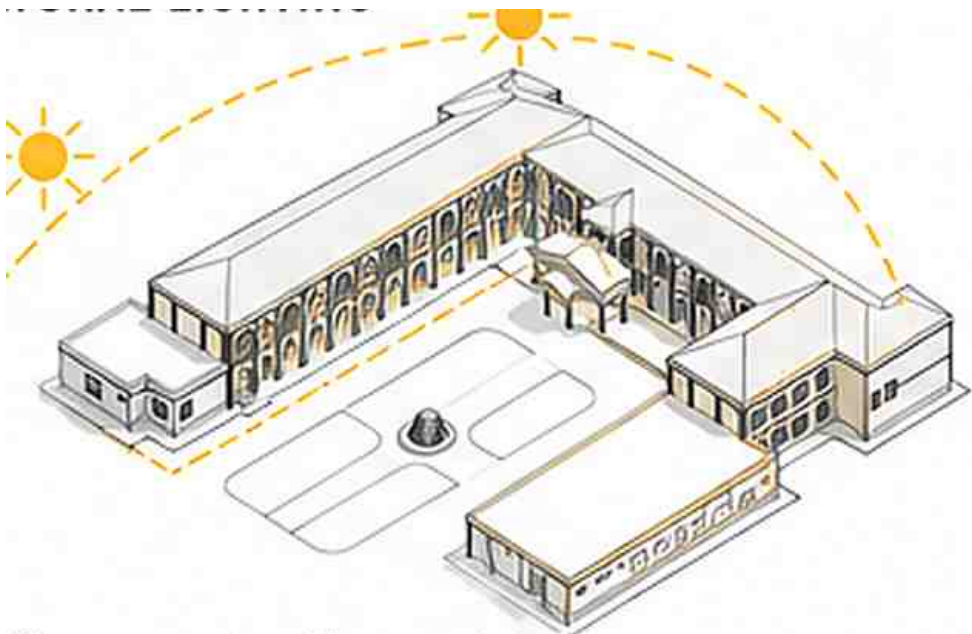
OPEN SPACES & LANDSCAPE



- CENTRAL COURTYARD
- LANDSCAPED GREEN AREAS
- RECREATIONAL ZONE

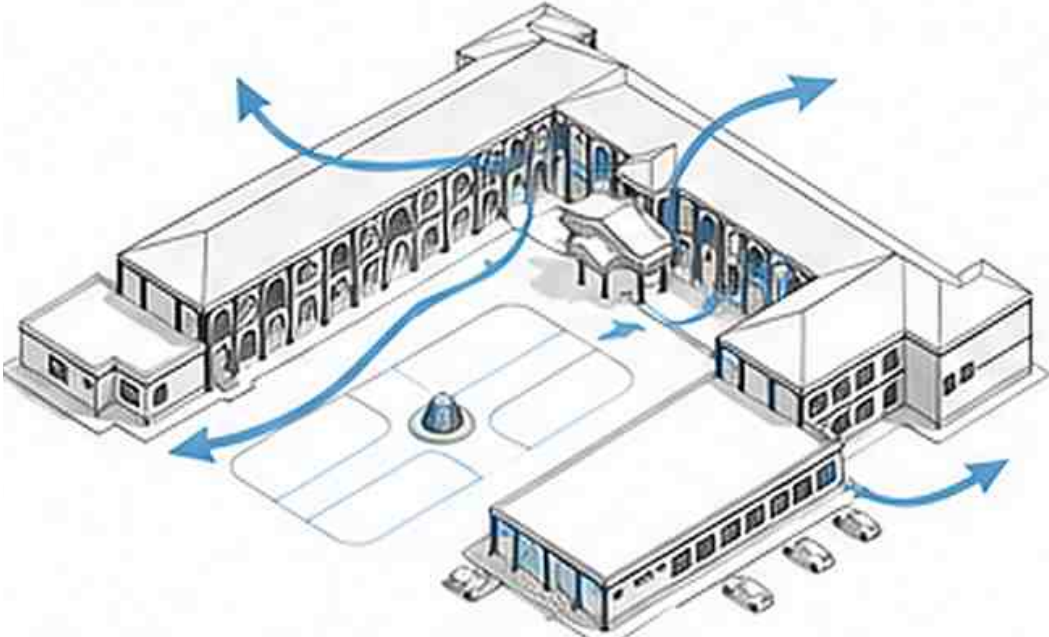
Open spaces provide recreational opportunities and create a healthy environment for learning.

NATURAL LIGHTING



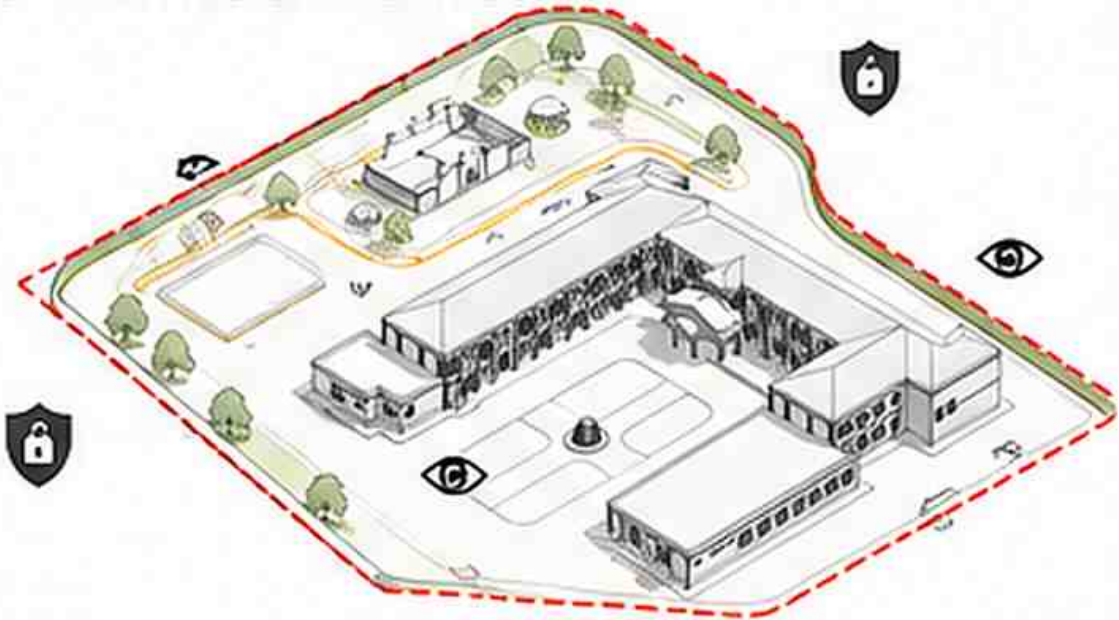
Classrooms and corridors are oriented to maximize daylight. Arched openings and large windows bring soft natural light into the interiors.

VENTILATION STRATEGY



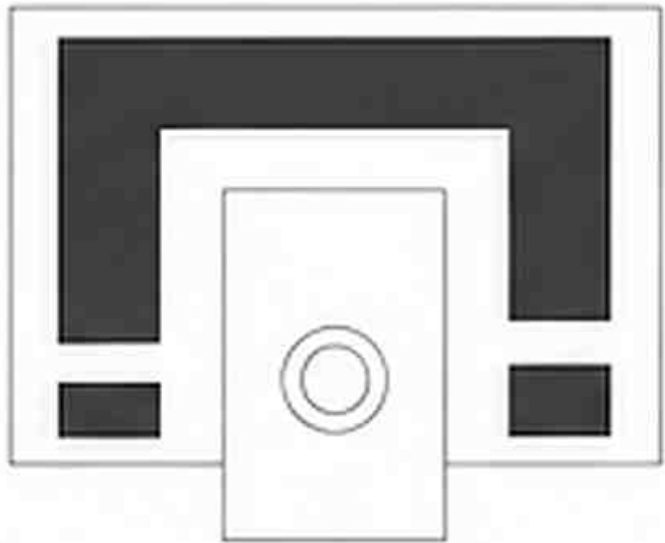
The courtyard acts as a breathing space, allowing cross ventilation in all wings for a comfortable indoor environment.

PRIVACY & SECURITY



The architectural layout ensures controlled access from main entrance while the perimeter wall and building orientation ensure privacy and safety.

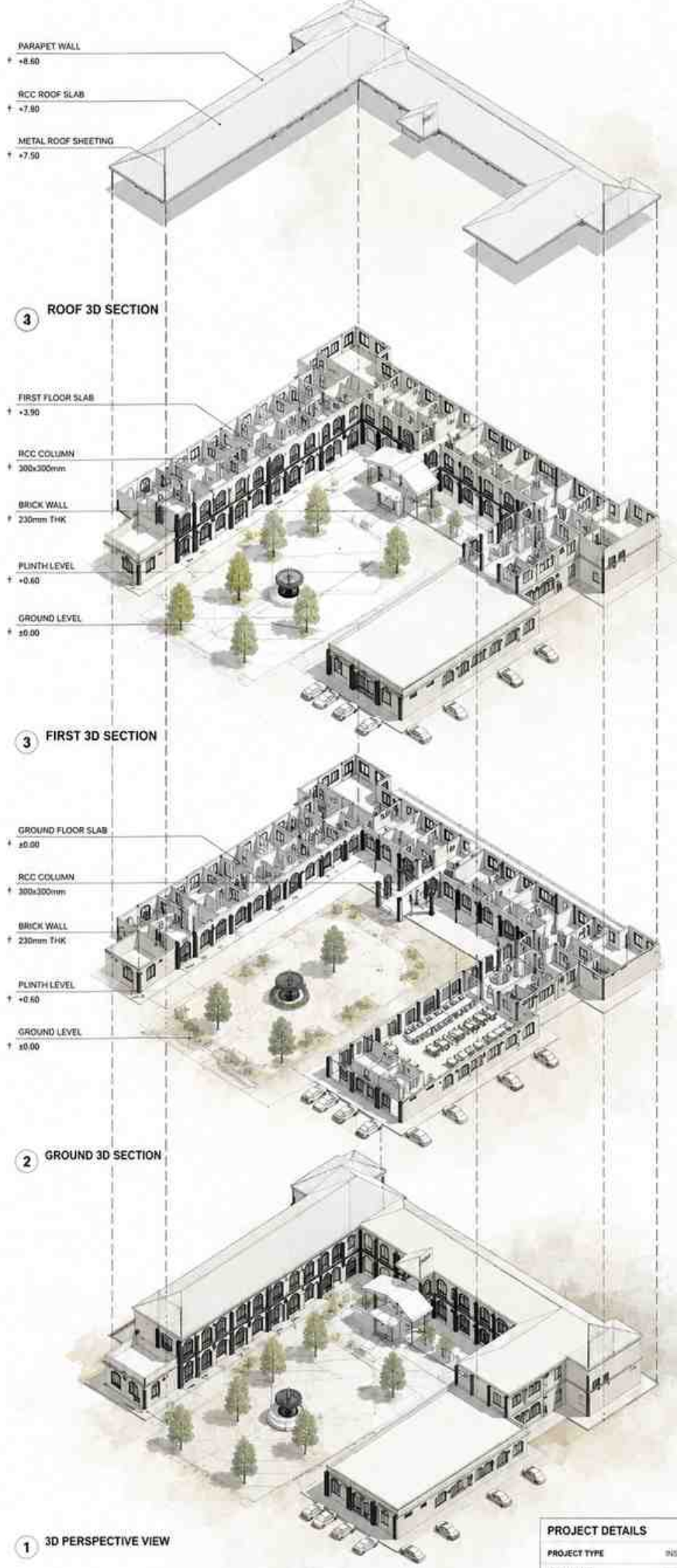
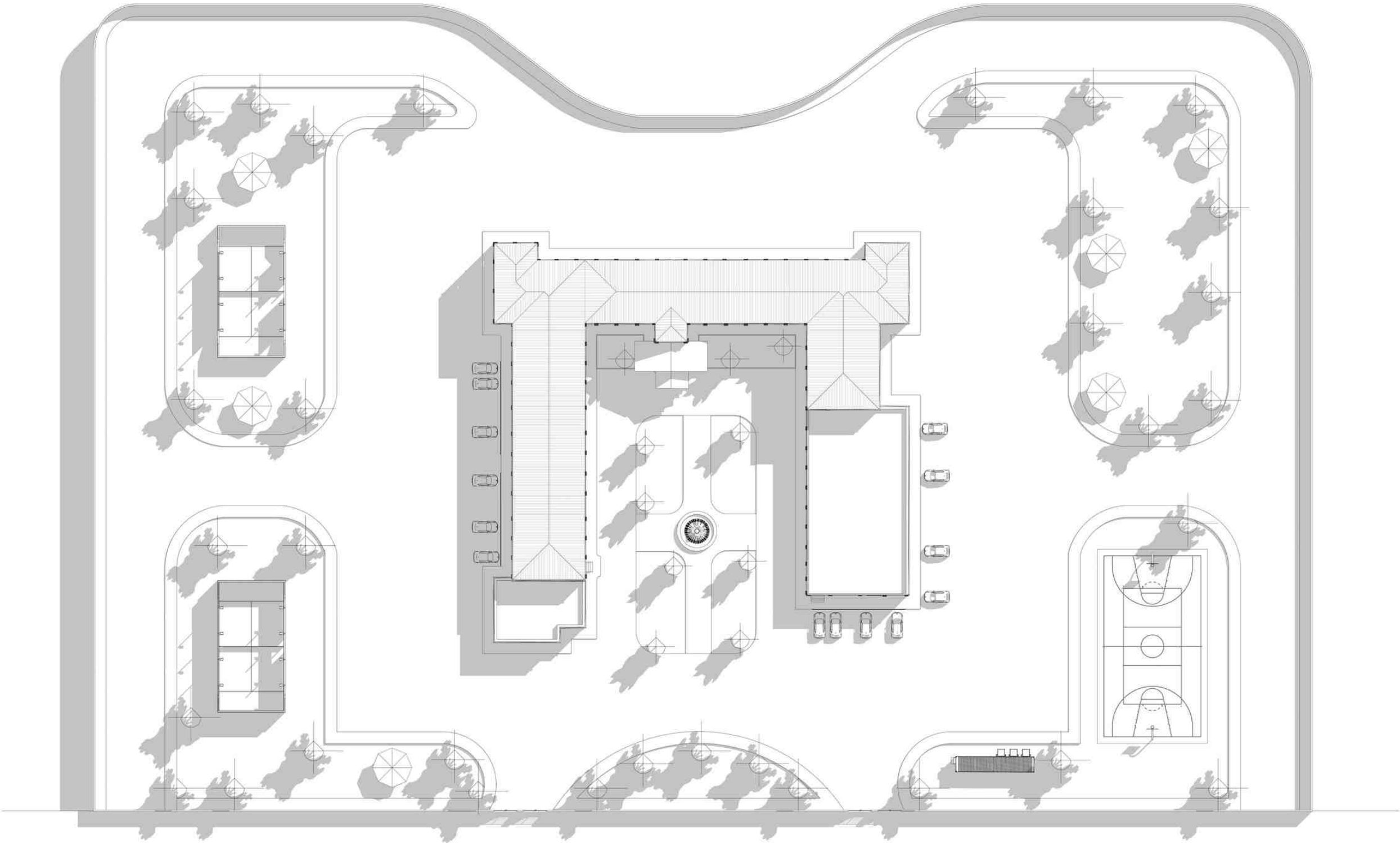
SOLID AND VOID



- SOLID (BUILT MASS)
- VOID (OPEN SPACES)

The interplay of solid and void creates balance, hierarchy and visual relief.

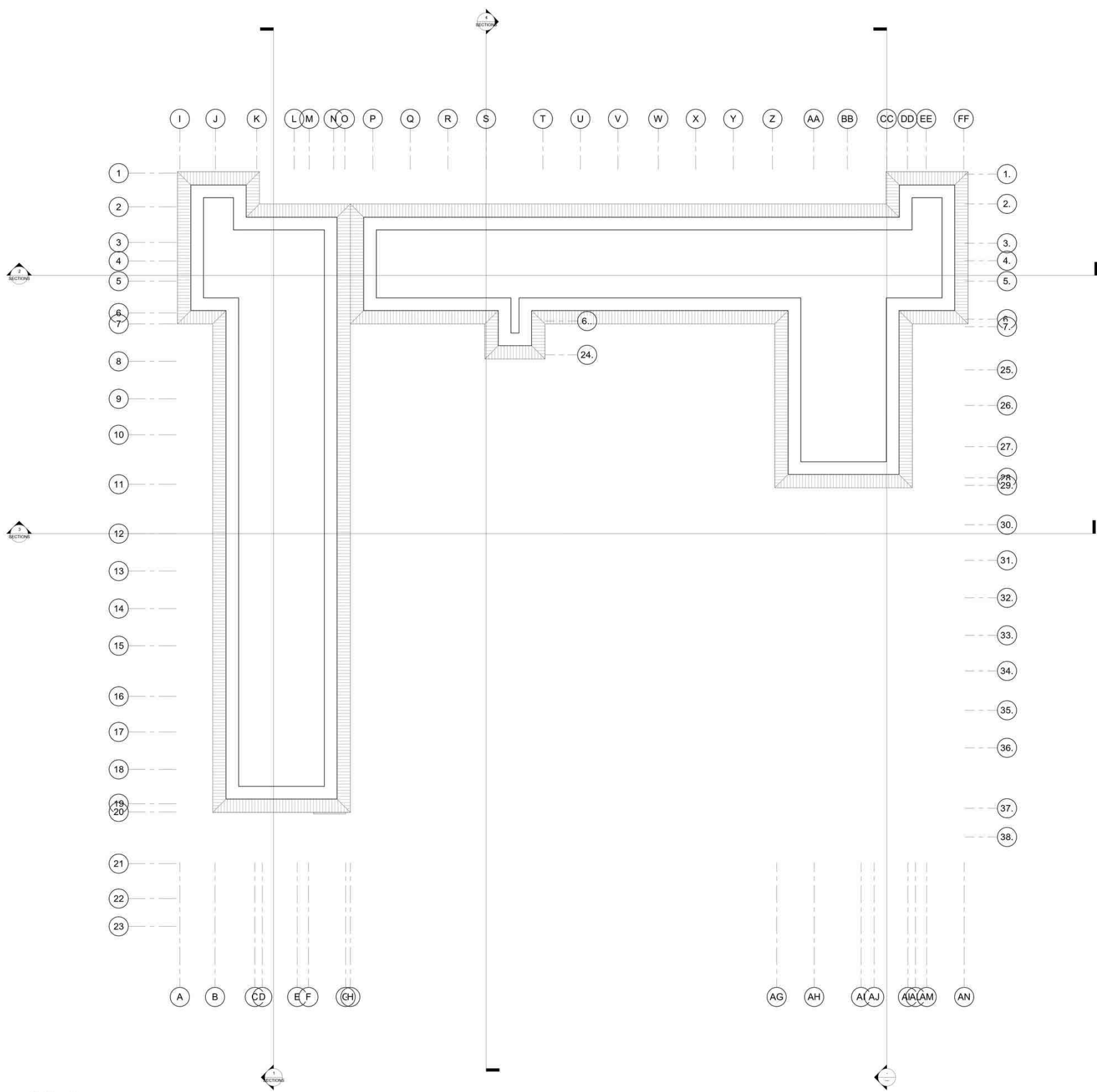
MASTER PLAN & EXPLODED VIEW



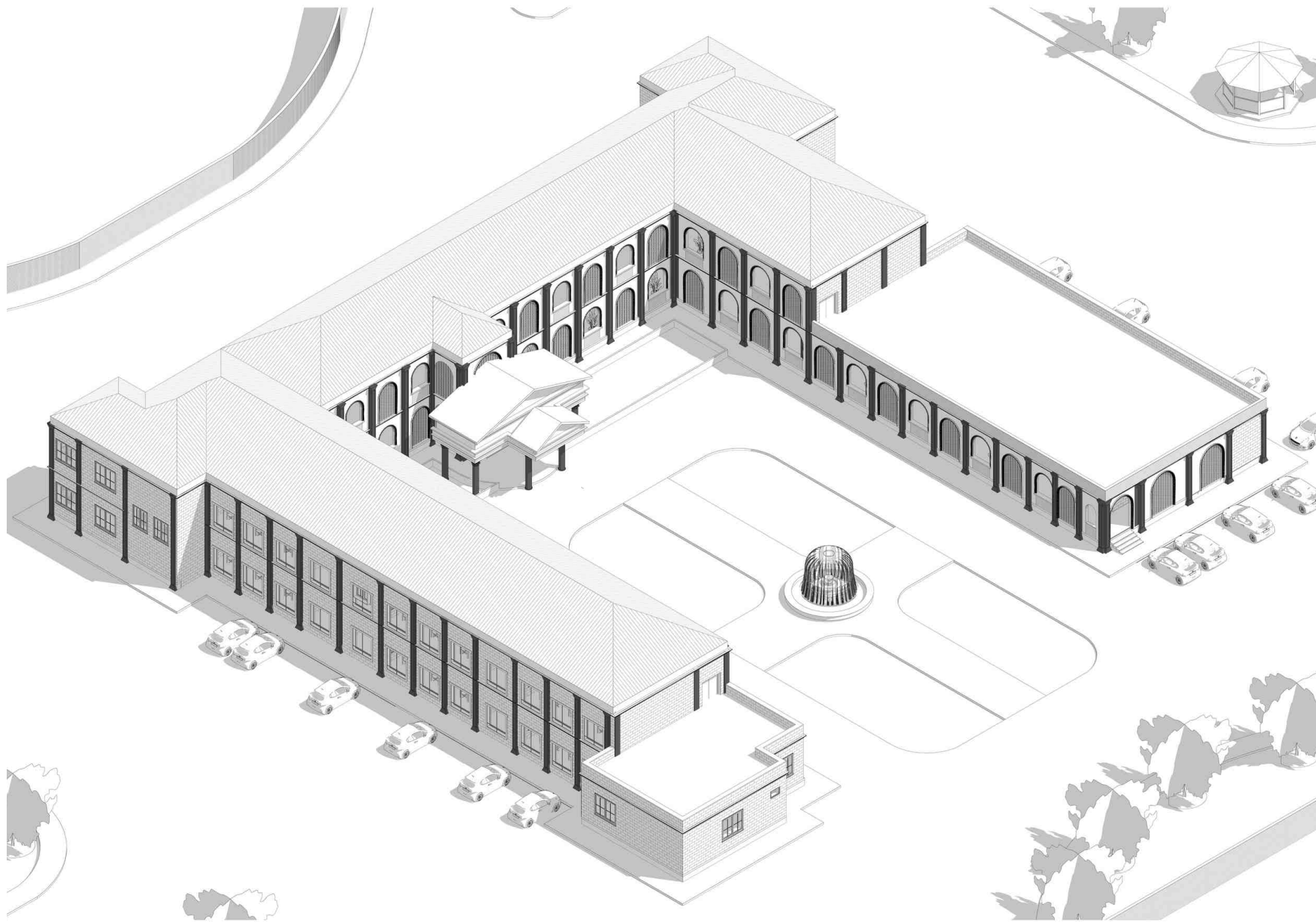
PROJECT DETAILS	
PROJECT TYPE	INSTITUTIONAL
STRUCTURE	RCC FRAME STRUCTURE
WALL THICKNESS	230mm
FLOOR TO FLOOR HEIGHT	3.60m
TOTAL HEIGHT	9.60m



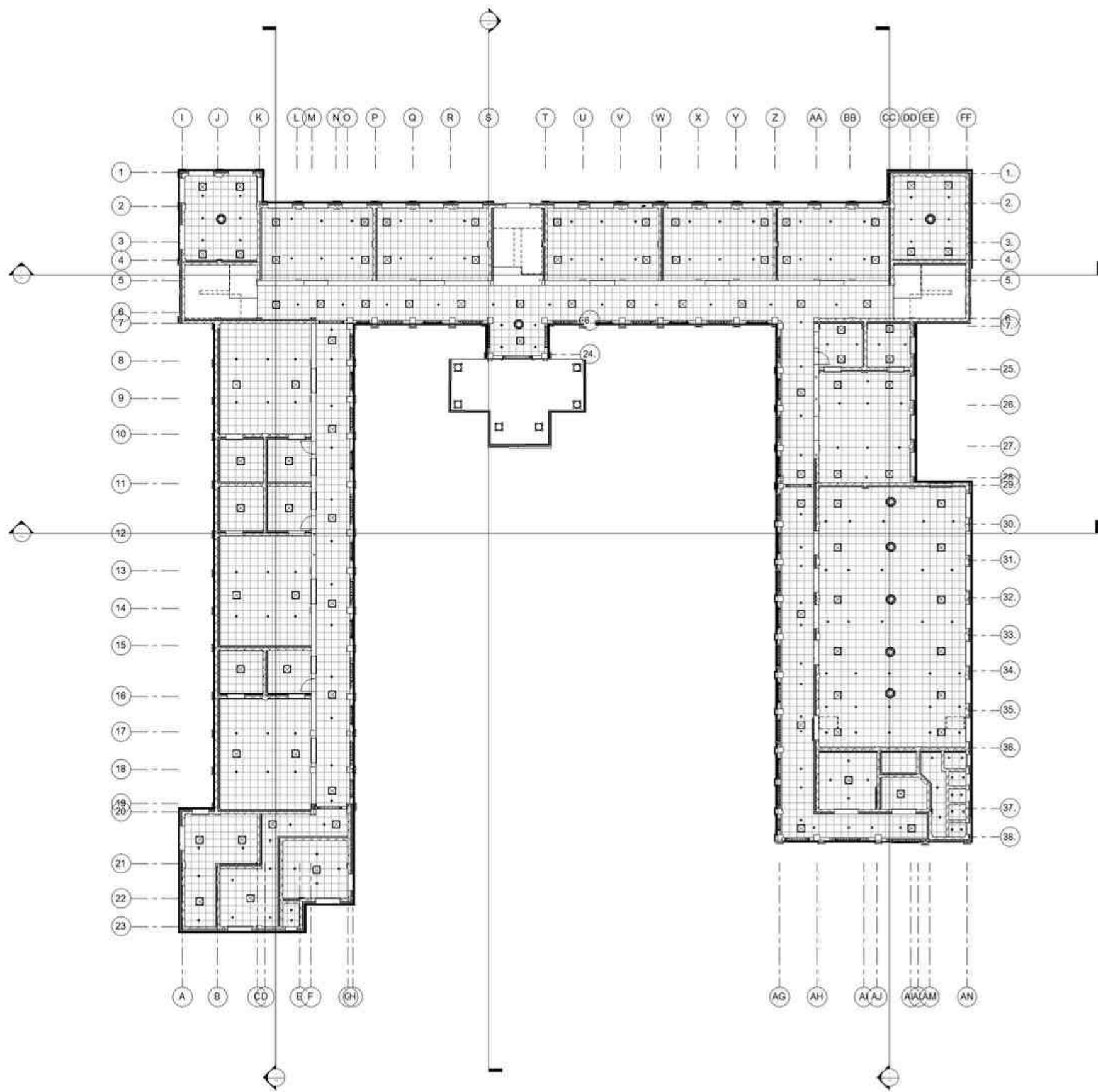
ROOF PLAN



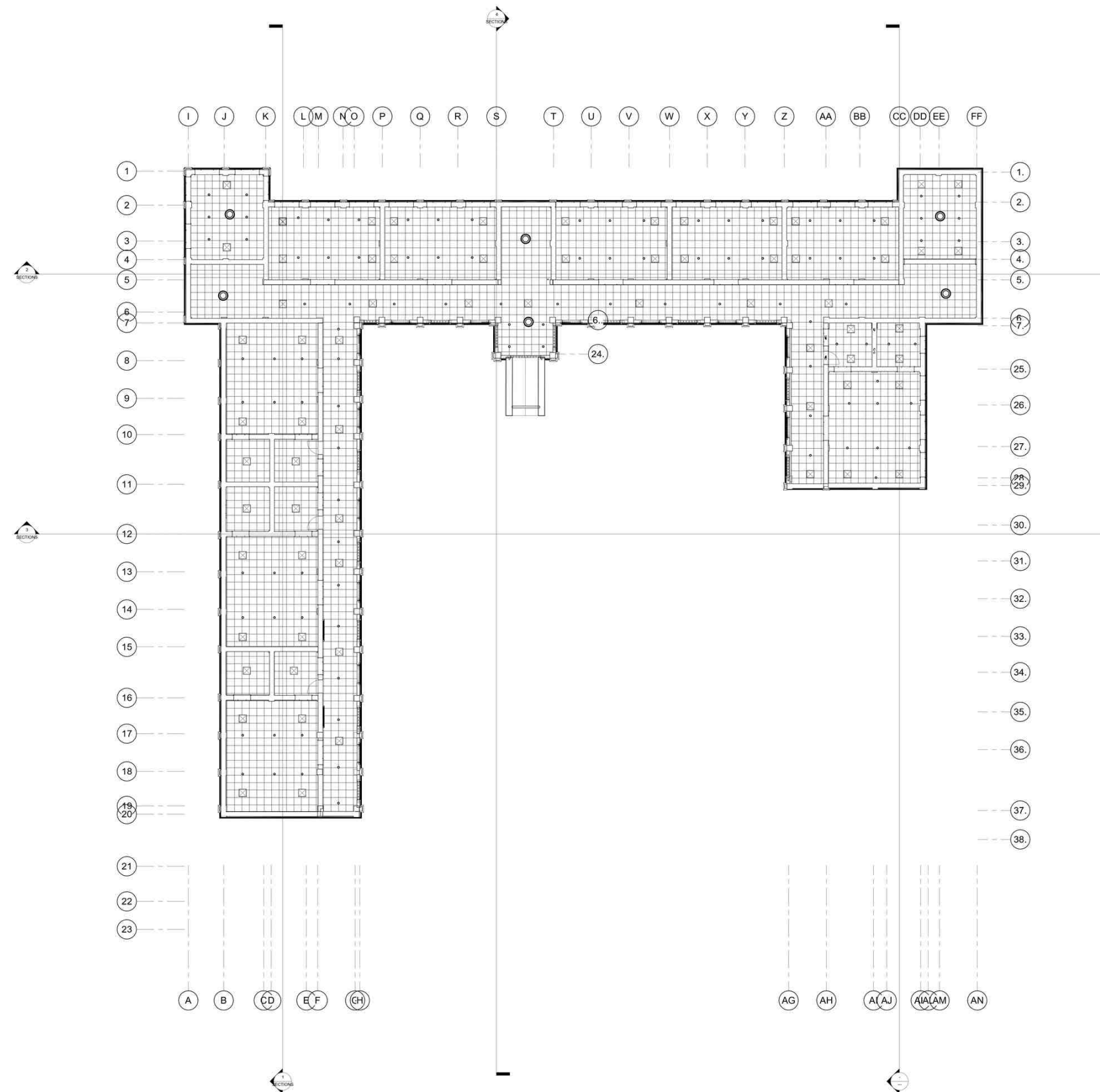
3D VIEW



CEILING PLANS

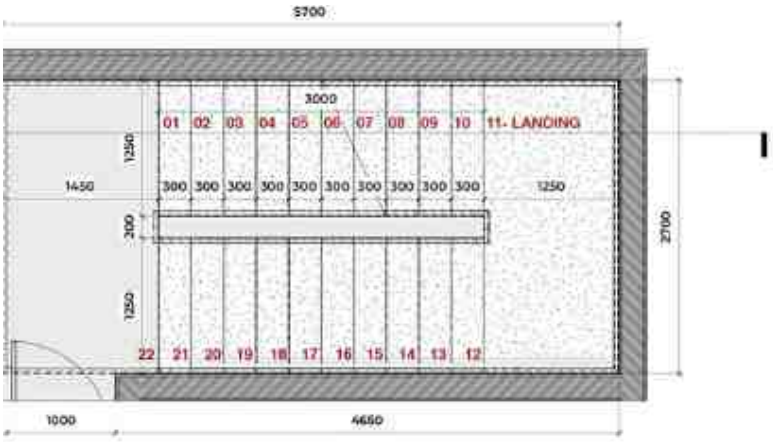


GROUND FLOOR PLAN

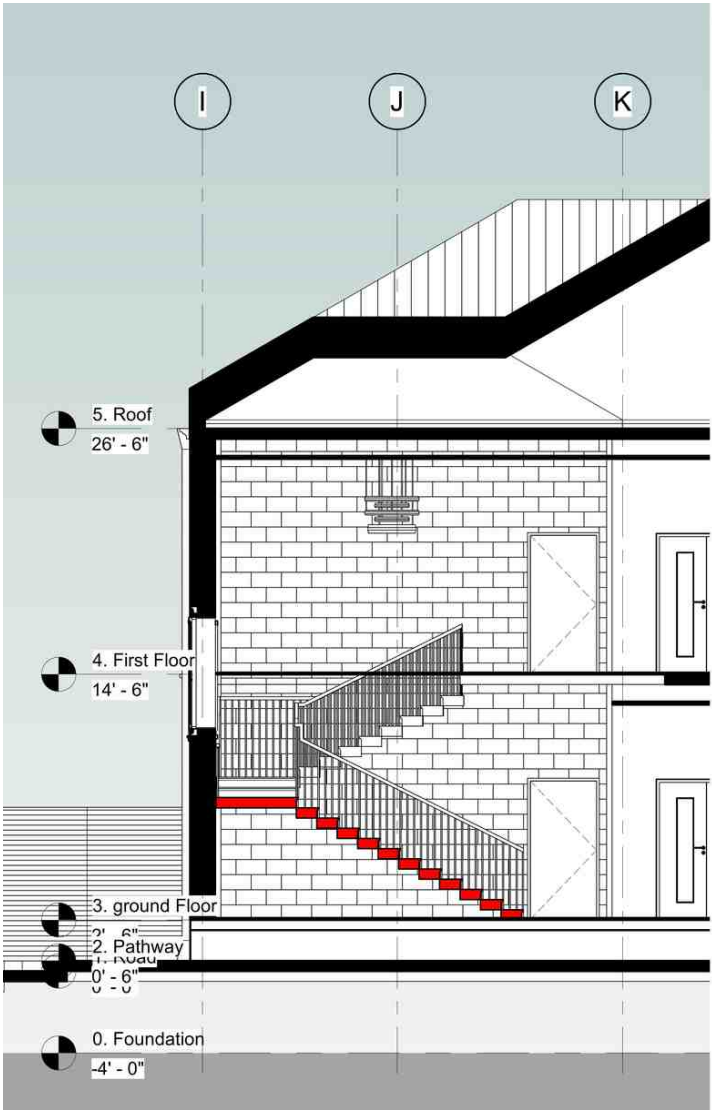


FIRST FLOOR PLAN

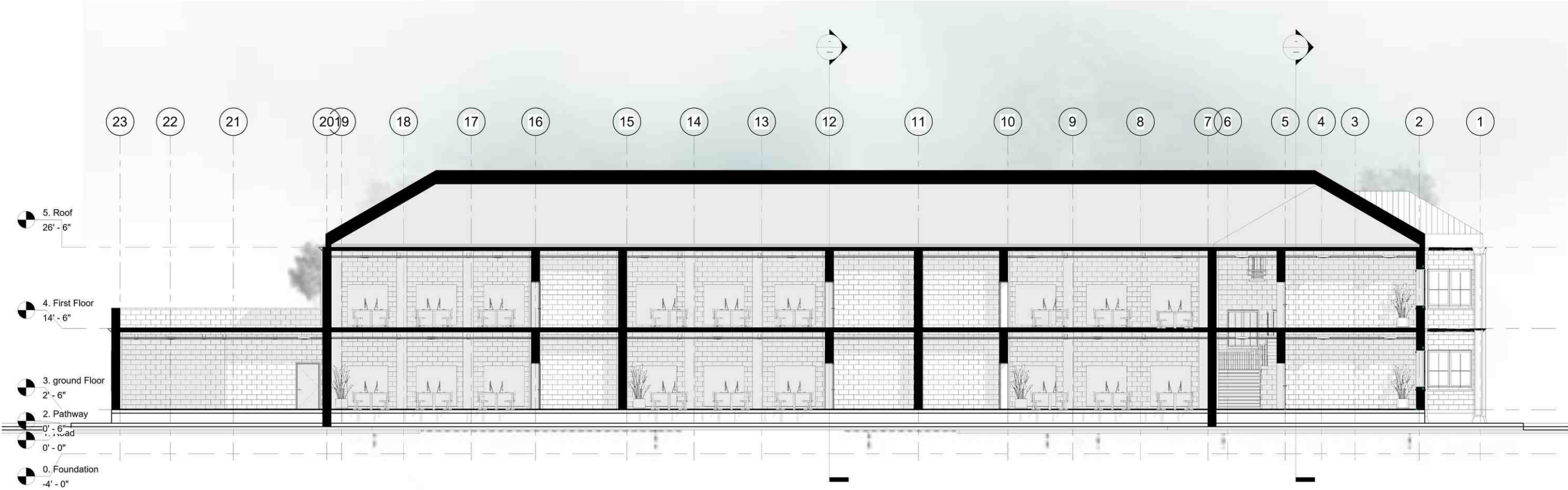
SECTIONS



STAIRCASE



STAIRCASE

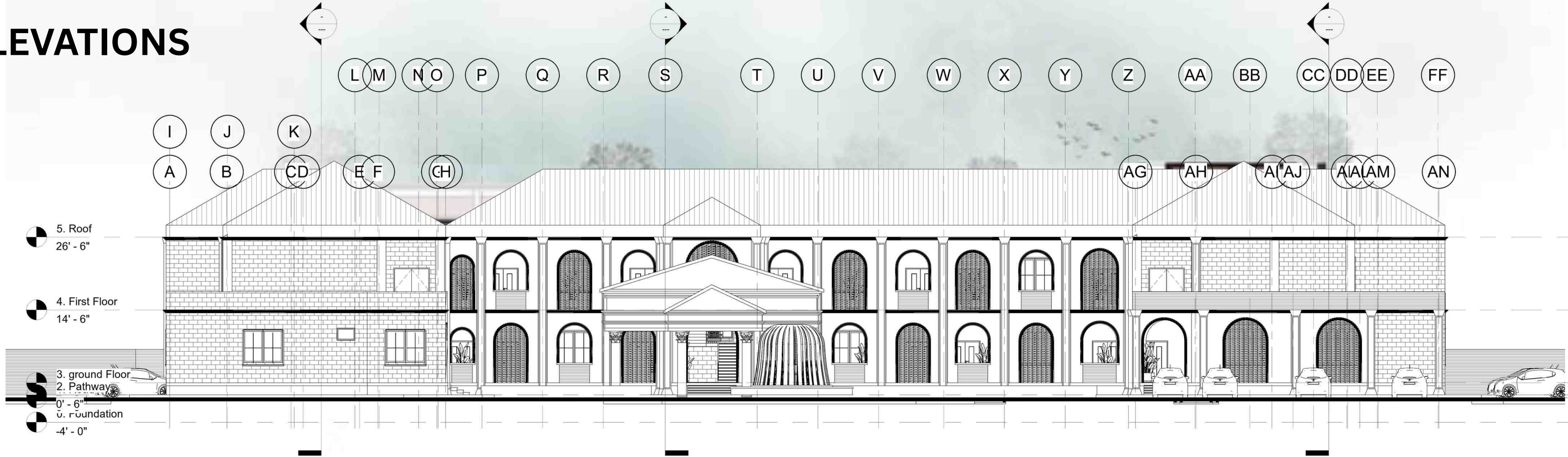


SECTION AA

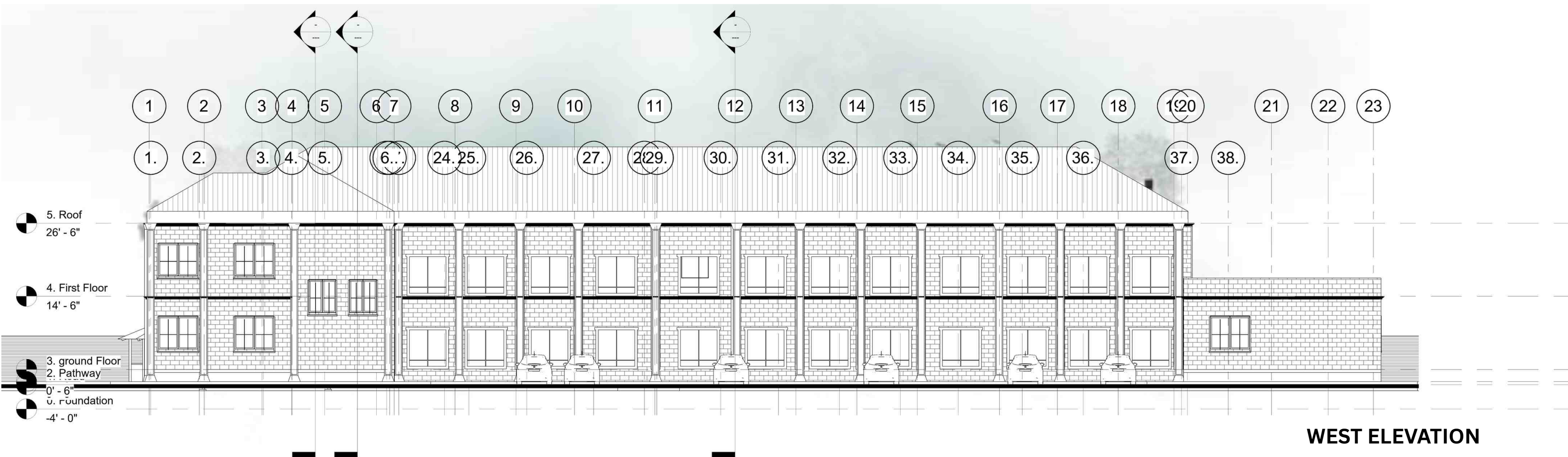


SECTION BB

ELEVATIONS

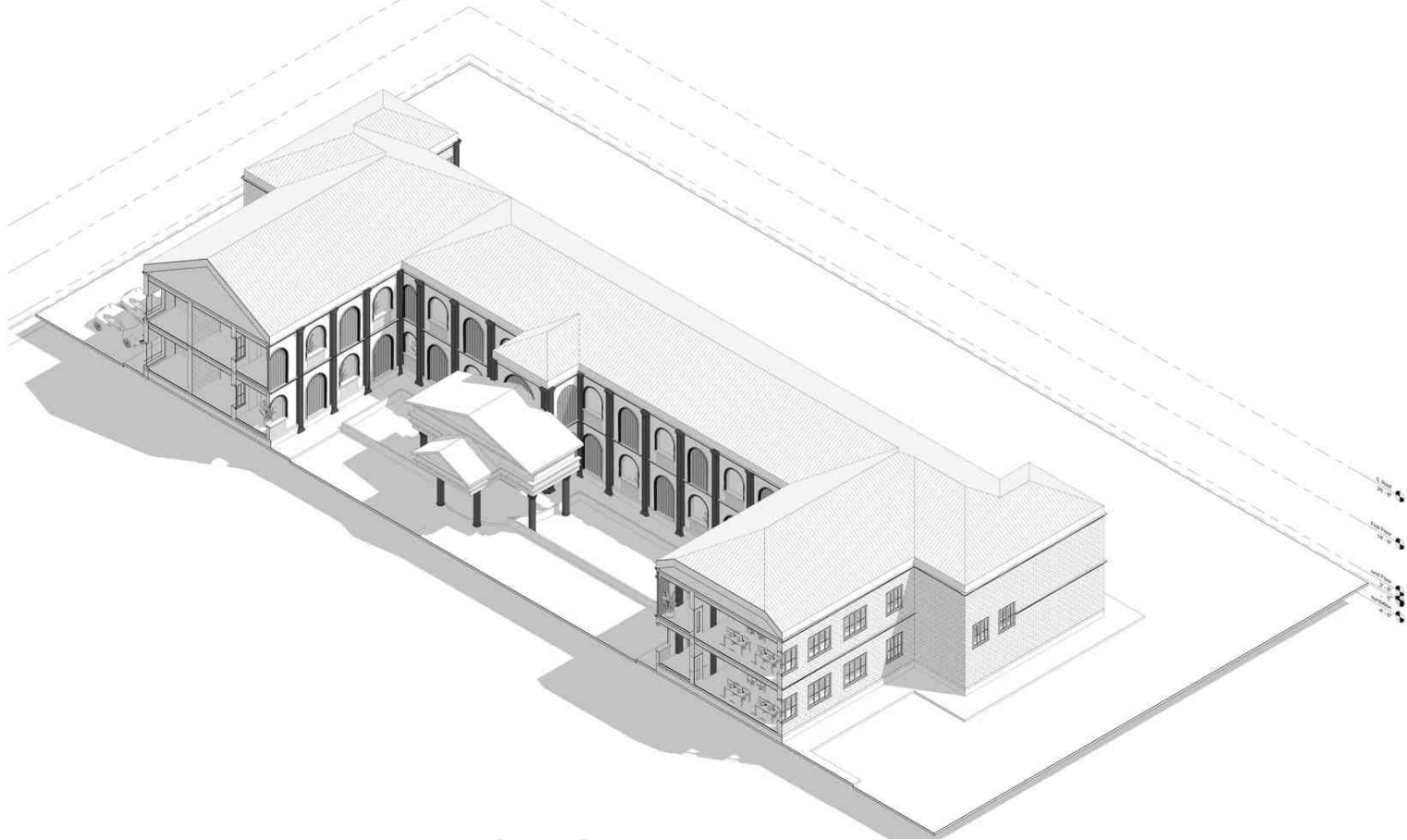


SOUTH ELEVATION

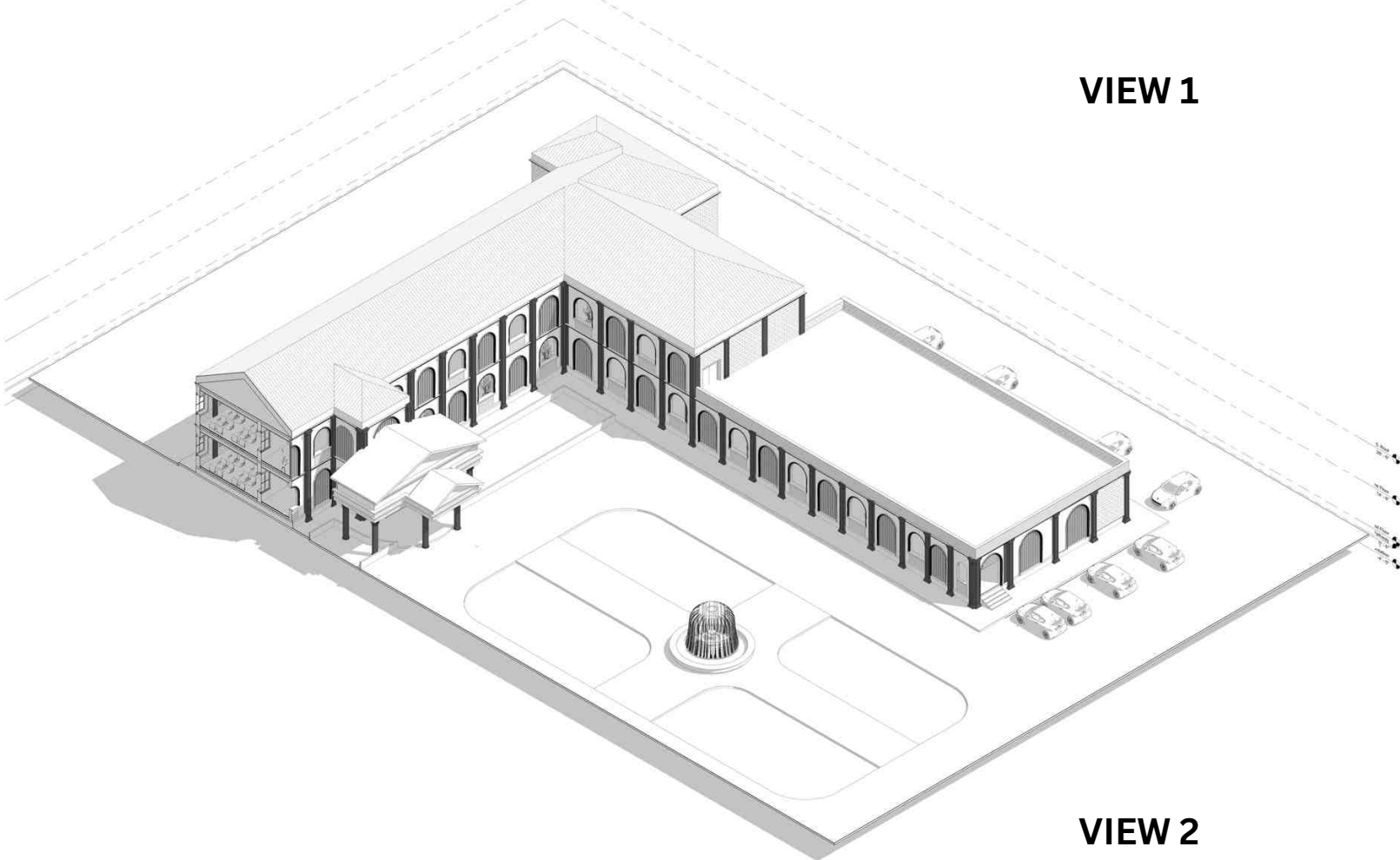


WEST ELEVATION

3D SECTIONS

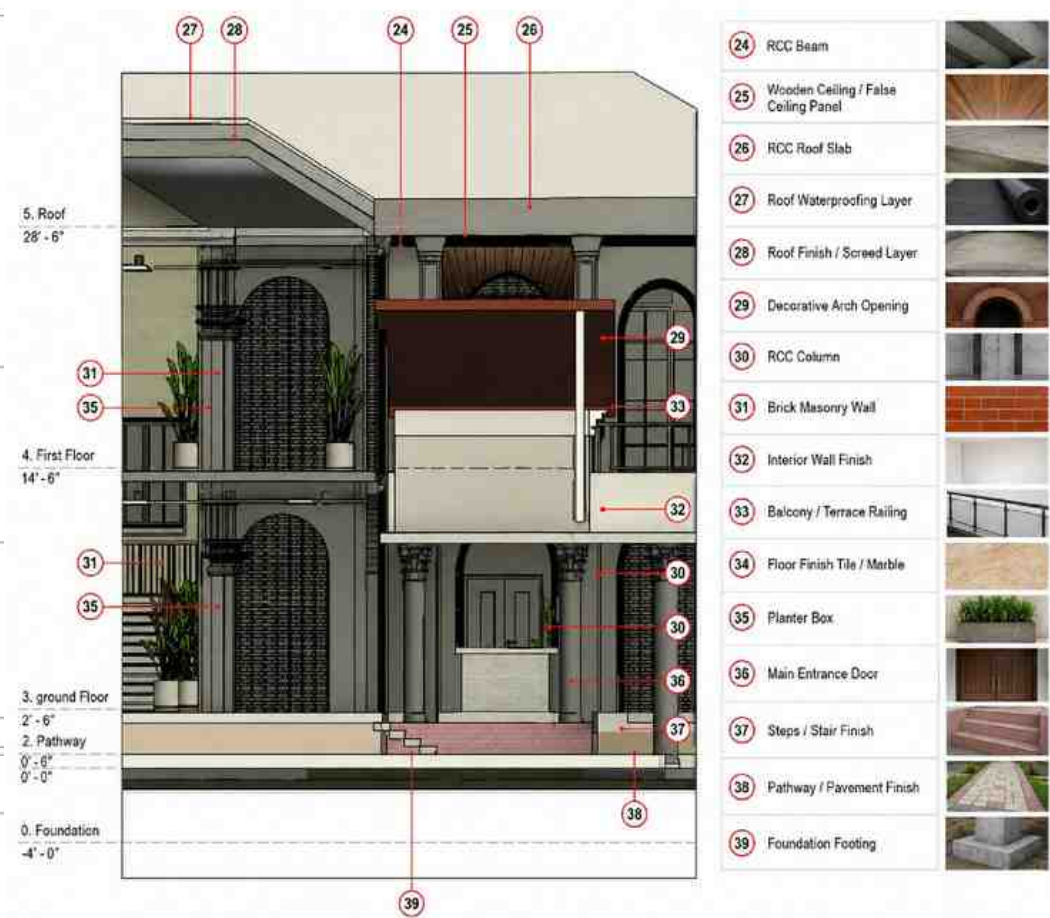
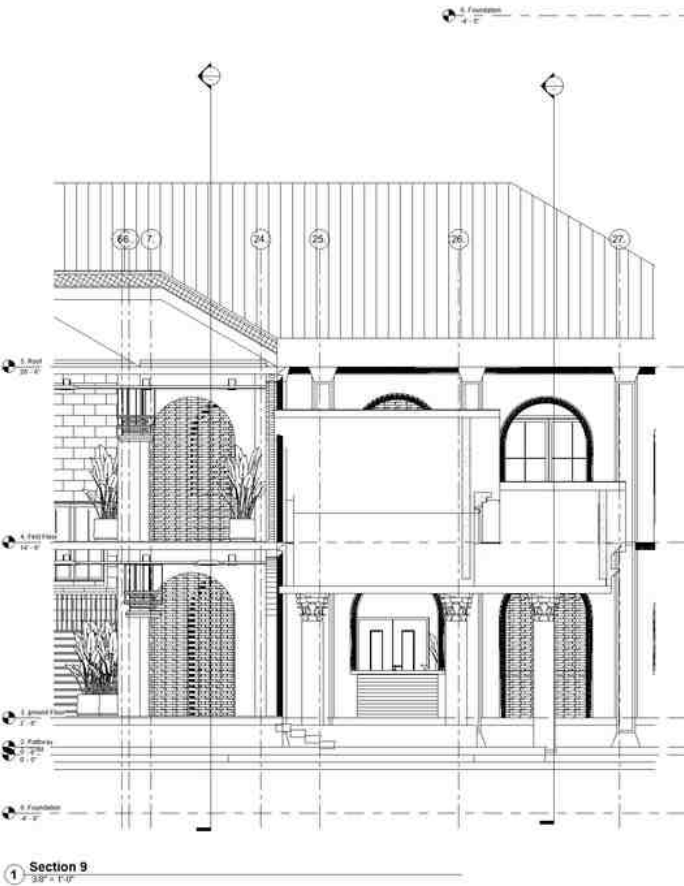
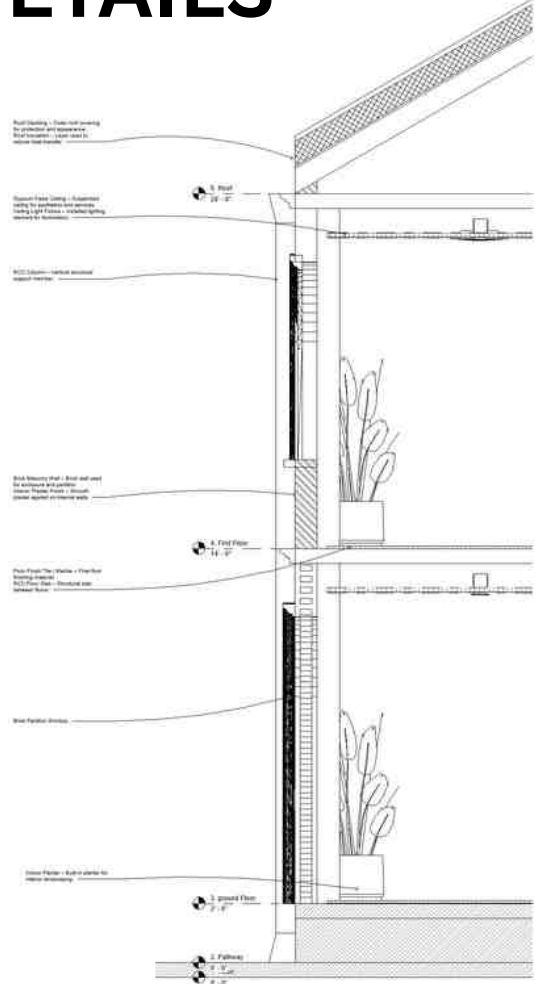


VIEW 1

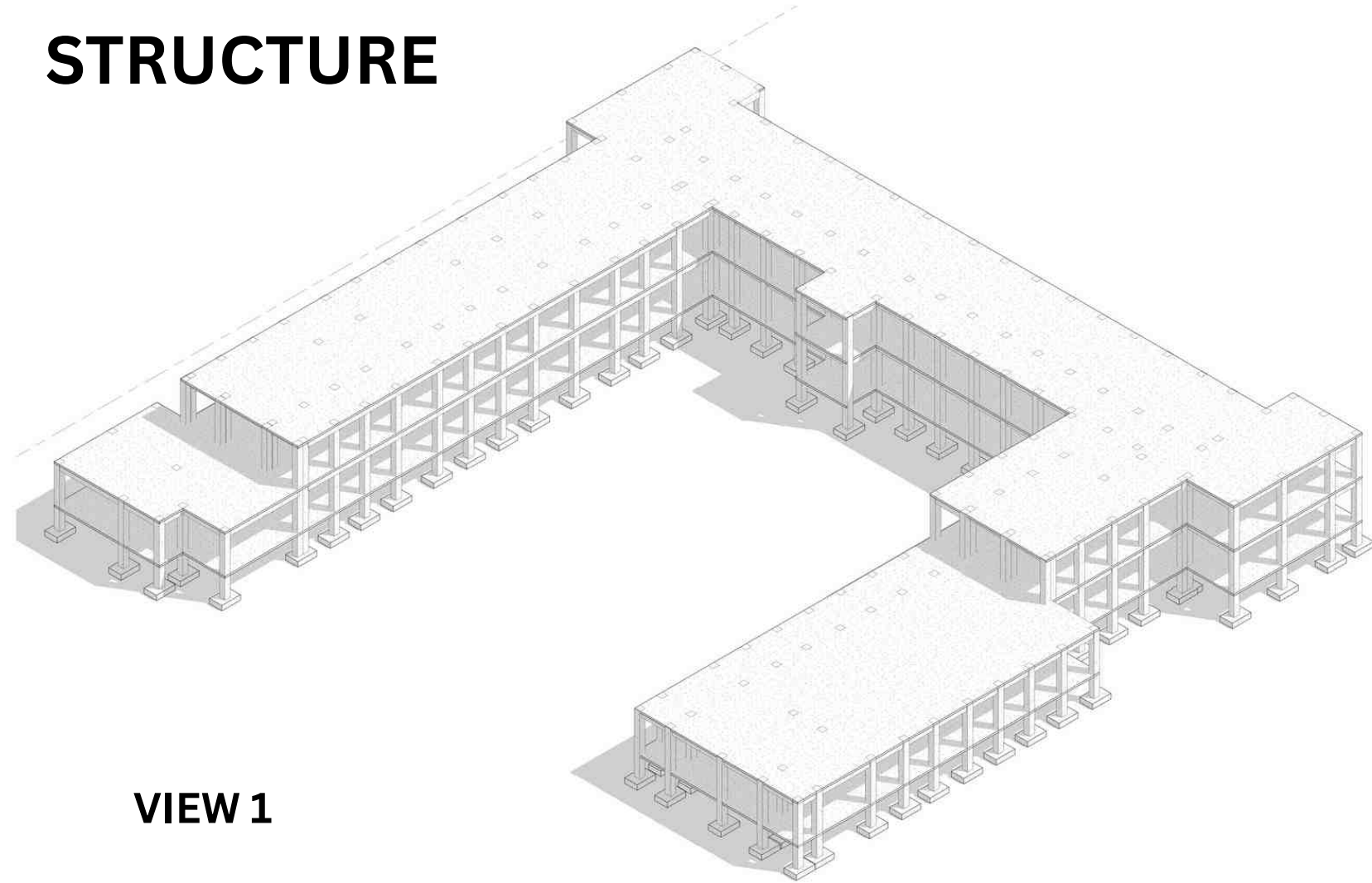


VIEW 2

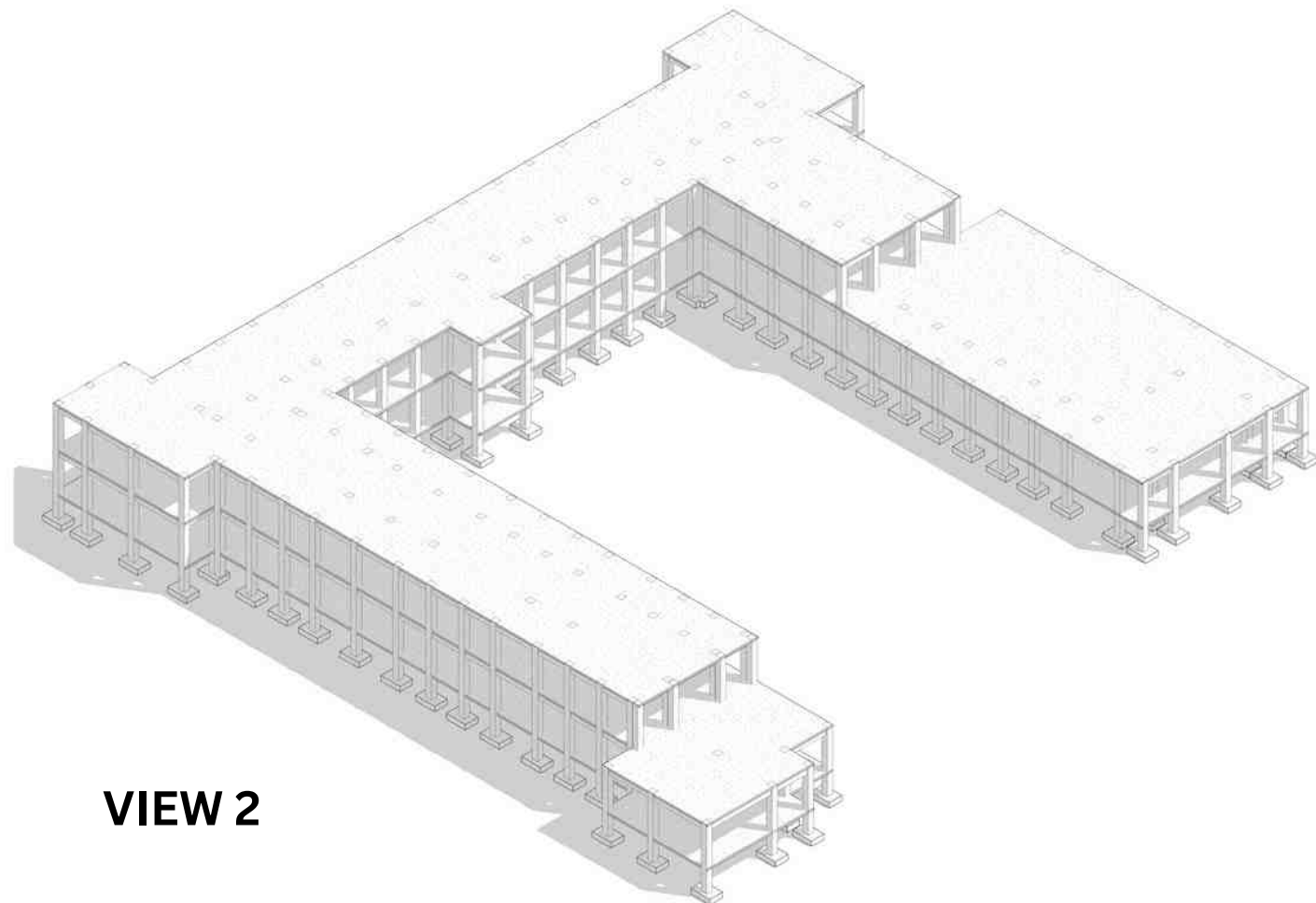
SECTIONS DETAILS



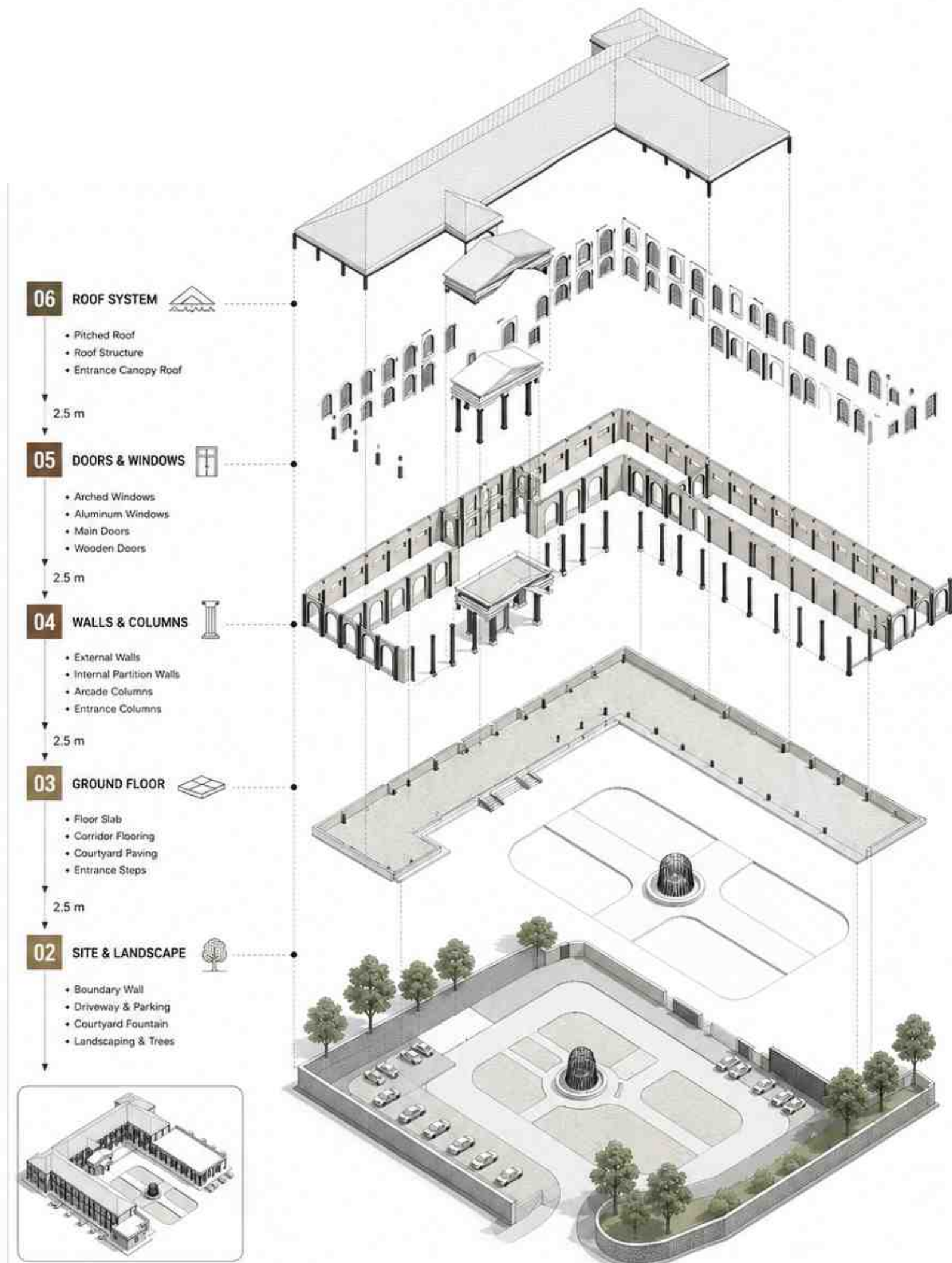
STRUCTURE



VIEW 1



VIEW 2



STRUCTURE SCHEDULE

DEVELOPMENT LENGTH (ACI 21.5.4)

BAR DIA	F _c ' = 3,000 psi		F _c ' = 3,000 psi	
	STRAIGHT BARS $l_d = f_y/25\sqrt{f_c'}$ (upto # 6 bars) $l_d = f_y/20\sqrt{f_c'}$ (#7 And above)	WITH STANDARD HOOK $L_{dh} = 0.8 \times 0.02 f_y / \sqrt{f_c'}$	STRAIGHT BARS $l_d = f_y/25\sqrt{f_c'}$ (upto # 6 bars) $l_d = f_y/20\sqrt{f_c'}$ (#7 And above)	WITH STANDARD HOOK $L_{dh} = 0.8 \times 0.02 f_y / \sqrt{f_c'}$
1/4"	8"	4"	12"	5"
3/8"	12"	5"	18"	7"
1/2"	16"	6.5"	24"	10"
5/8"	20"	8"	30"	12"
3/4"	24"	10"	36"	15"
1"	40"	13"	60"	19"

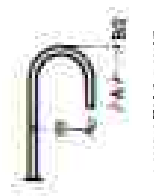
LAP SPLICE (ACI 21.3.2.4) CLASS B SPLICE = 1.3 l_d

BAR DIA	F _c ' = 3,000 psi		F _c ' = 3,000 psi	
	LENGTH OF LAP SPLICE		LENGTH OF LAP SPLICE	
1/4"	10"		15"	
3/8"	15"		24"	
1/2"	20"		30"	
5/8"	26"		40"	
3/4"	30"		46"	
1"	52"		78"	

MINIMUM CLEAR COVER

S.No	LOCATION	MINIMUM COVER
1.	FOOTING	3"
2.	COLUMN	1-1/2"
3.	BEAM	1-1/2"
4.	SLAB	3/4"
5.	WATER RETAINING STRUCTURES	3"

STANDARD HOOKS (ACI 7.1- 7.2)

BAR DIA	A	B	C	 180° HOOK
1/4"	2.5"	1.5"	3"	
3/8"	2.5"	2.25"	4.5"	 90° HOOK
1/2"	2.5"	3"	6"	
5/8"	2.5"	3.75"	7.5"	
3/4"	3"	4.5"	9"	
1"	4"	6"	12"	

1. Introduction

This document contains information about the structural design of the above stated scheme. The gravity and lateral load resisting system of the structure consists of a Reinforced Concrete Frame as detailed in the structural drawings. The 3-D Finite Element based software ETABS has been used for the analysis and design of the structural members. ACI code has been used for the design of all reinforced concrete members.

2. Design Specifications

- 2.1. ACI (American Concrete Institute) Building code 318-99
- 2.2. UBC (Uniform Building Code) 1997
- 2.3. AISC LRFD Specifications 1999
- 2.4. ASCE
- 2.5. Engineering Judgment is used where code does not satisfy local requirements

3. Design Philosophy

- 3.1. ACI Ultimate Strength Design (USD) method has been used for design of all reinforced concrete members

4. Unit Weights

- 4.1. Unit weight of Reinforced Cement Concrete = 150.00 PCF
- 4.2. Unit weight of Plain Cement Concrete = 140.00 PCF
- 4.3. Unit weight of brick masonry = 120.00 PCF
- 4.4. Unit weight of Steel = 490.00 PCF

5. Loads

- 5.1. Dead Loads: Dead load consists of self load of structure & a superimposed dead load of 25 PSF for roofing, 4.5 inch thick partition walls etc.
- 5.2. Live Loads: The live load of 40 PSF is considered on the floors.
- 5.3. Earthquake Loads: Following Earthquake analysis have been carried out for the design of buildings:
 - a. **Equivalent Static Load Analysis:** UBC-97 code with the parameters given in table 1 has been used for equivalent static analysis of the school building.

6. Structural System and Elements used in ETABS Modeling:

The gravity and lateral load resisting system of the structure consists of a Reinforced Concrete Frame as detailed in the structural drawings. The 3-D Finite Element based software ETABS has been used for the analysis and design of the structural members. Beams, Columns were modeled as follows.

- 6.1. Beams, Columns Modeled as Frame element
- 6.2. Slab Modeled as Shell element

7. General Information about Design Parameters:

- 7.1. Resistance factors, Φ ,
 - Flexure and tension for concrete Members = 0.90
 - Shear and torsion for concrete Members = 0.85
 - Axial Compression for concrete Members = 0.70
 - Tension (yield) in steel members = 0.90
- 7.2. Load Combinations: Load combinations used in the design consist of the following:
 - U = 1.4 DL (UBC 1909.2.1)
 - U = 1.4 DL + 1.7 LL (UBC 1909.2.1)
 - U = 0.9 DL ± 1.3 WL (UBC 1909.2.2)
 - U = 0.75 (1.4 DL + 1.7 LL ± 1.7 WL) (UBC 1909.2.2)
 - U = 0.9 DL ± 1.0 EL (UBC 1909.2.3, 1612.2.1)
 - U = 1.2 DL + 0.5 LL ± 1.0 EL (UBC 1909.2.3, 1612.2.1)

8. Material Properties

Following material properties have been used in the analysis and design of structures:

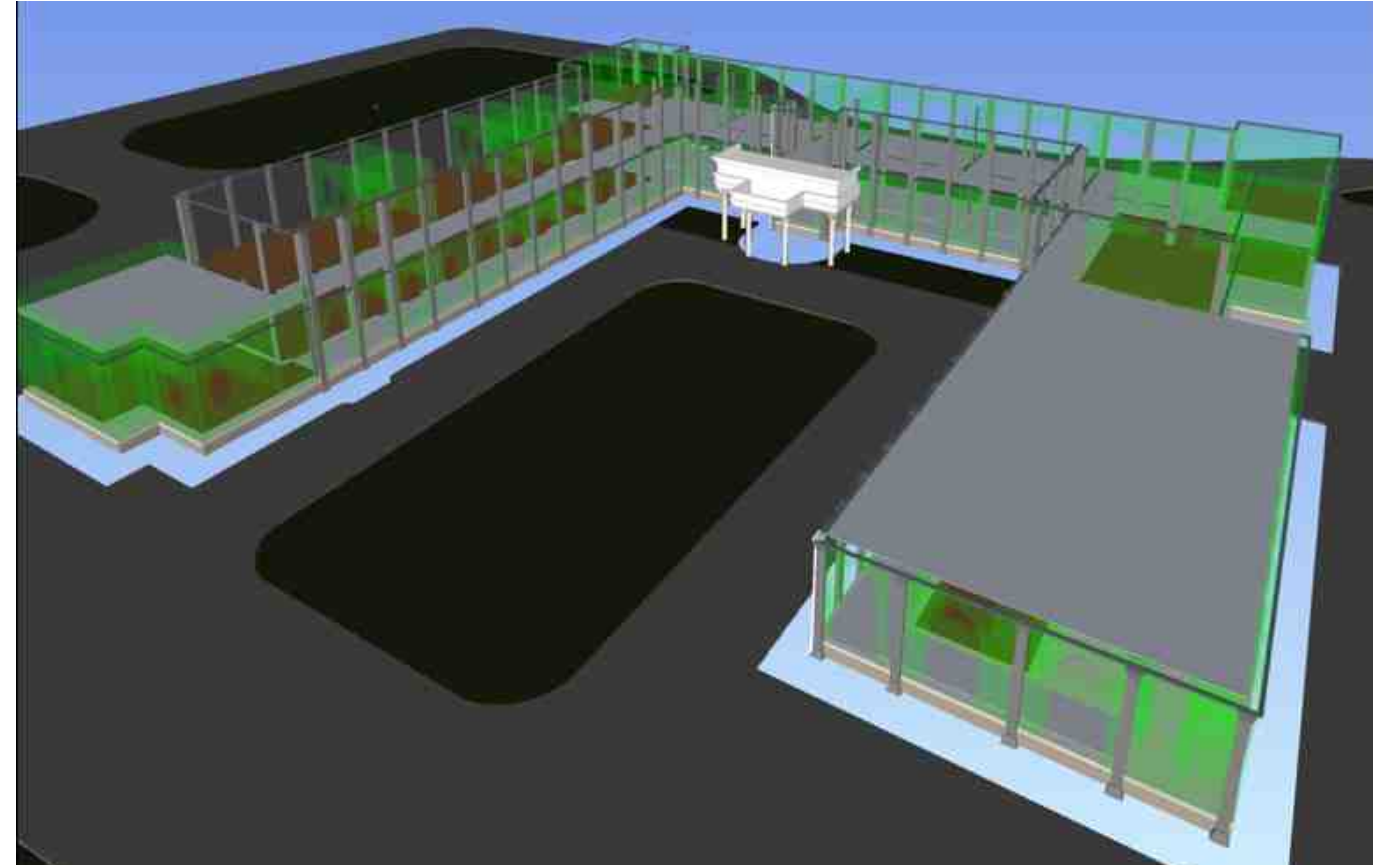
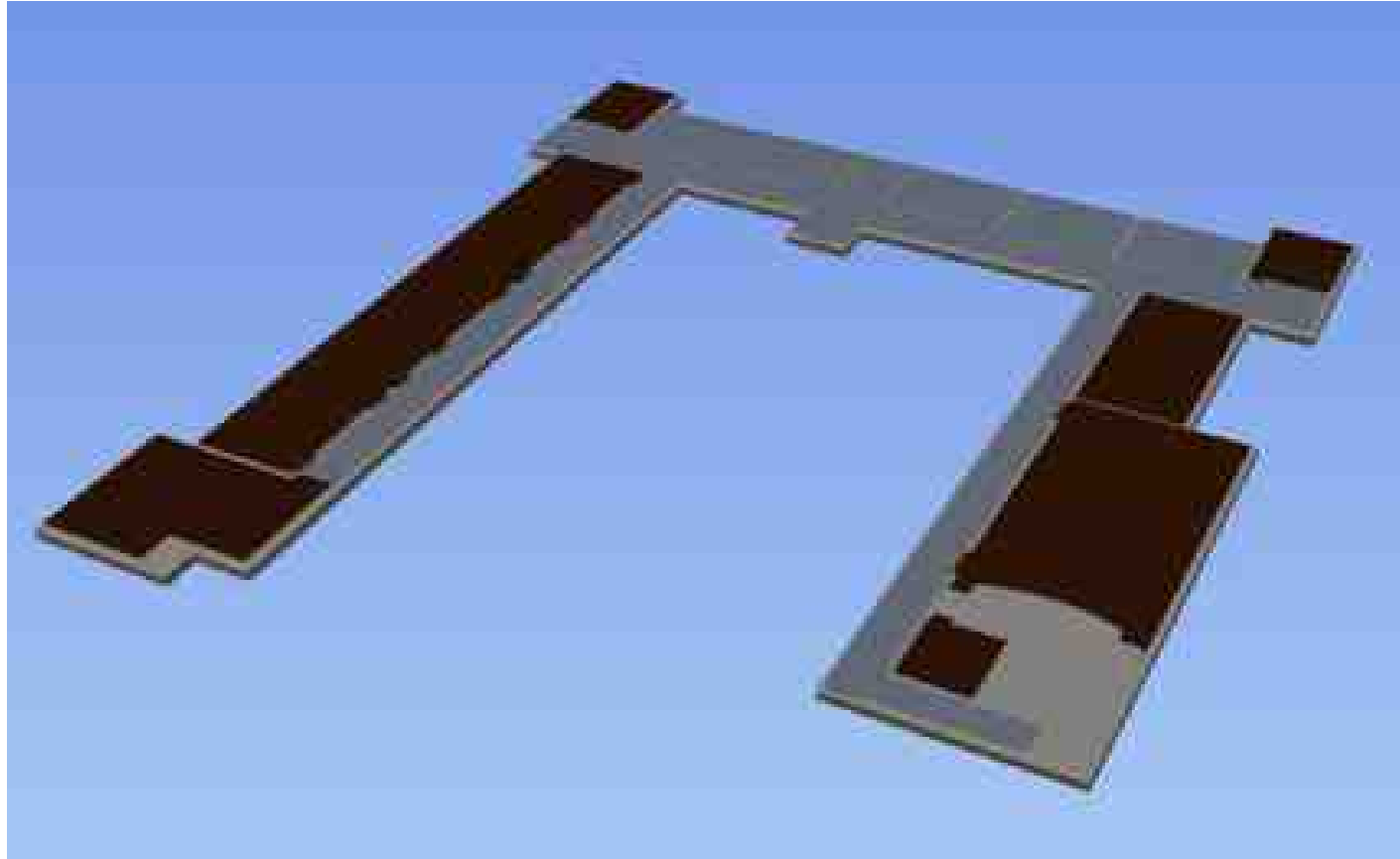
[illegible]

1001

A	B	C	D
Mark	Level	Width	Sill Height
4	3. ground Floor		3' - 0"
5	3. ground Floor		3' - 0"
6	3. ground Floor		3' - 0"
7	3. ground Floor		3' - 0"
8	3. ground Floor		3' - 0"
11	3. ground Floor		3' - 0"
12	3. ground Floor		3' - 0"
13	3. ground Floor		3' - 0"
14	3. ground Floor		3' - 0"
16	3. ground Floor		3' - 0"
17	3. ground Floor		3' - 0"
18	3. ground Floor		3' - 0"
19	3. ground Floor		3' - 0"
20	3. ground Floor		3' - 0"
21	3. ground Floor		3' - 0"
22	3. ground Floor		3' - 0"
23	3. ground Floor		3' - 0"
24	3. ground Floor		3' - 0"
26	3. ground Floor		3' - 0"
27	3. ground Floor		3' - 0"
28	3. ground Floor		3' - 0"
29	3. ground Floor		3' - 0"
30	3. ground Floor		3' - 0"
31	3. ground Floor		3' - 0"
36	3. ground Floor		3' - 0"
37	3. ground Floor		3' - 0"
38	3. ground Floor		3' - 0"
39	3. ground Floor		3' - 0"
40	3. ground Floor		3' - 0"
41	3. ground Floor		3' - 0"
42	3. ground Floor		3' - 0"
43	3. ground Floor		3' - 0"
44	3. ground Floor		3' - 0"
57	3. ground Floor		3' - 0"
58	3. ground Floor		3' - 0"
59	3. ground Floor		3' - 0"
60	3. ground Floor		3' - 0"
61	3. ground Floor		3' - 0"
64	3. ground Floor		3' - 0"
65	3. ground Floor		3' - 0"
68	3. ground Floor		3' - 0"
69	3. ground Floor		3' - 0"
72	3. ground Floor		3' - 0"
73	3. ground Floor		3' - 0"
74	3. ground Floor		3' - 0"

75	3. ground Floor		3' - 0"	154	3. ground Floor	6' - 0"	0' - 6"	6' - 0"
76	3. ground Floor		3' - 0"	155	3. ground Floor	6' - 0"	0' - 6"	6' - 0"
77	3. ground Floor		3' - 0"	156	3. ground Floor	6' - 0"	0' - 6"	6' - 0"
78	3. ground Floor		3' - 0"	157	3. ground Floor	6' - 0"	0' - 6"	6' - 0"
79	3. ground Floor		9' - 0"	158	3. ground Floor	6' - 0"	0' - 6"	6' - 0"
80	3. ground Floor		9' - 0"	159	3. ground Floor	6' - 0"	0' - 6"	6' - 0"
81	3. ground Floor		9' - 0"	160	3. ground Floor	6' - 0"	0' - 6"	6' - 0"
82	3. ground Floor		9' - 0"	161	3. ground Floor	6' - 0"	0' - 6"	6' - 0"
83	3. ground Floor		9' - 0"	162	3. ground Floor	6' - 0"	0' - 6"	6' - 0"
84	3. ground Floor		7' - 0"	163	3. ground Floor	6' - 0"	0' - 6"	6' - 0"
85	3. ground Floor		7' - 0"	164	1. Road	6' - 0"	4' - 0"	6' - 0"
86	3. ground Floor		7' - 0"	165	4. First Floor	6' - 0"	0' - 6"	6' - 0"
87	3. ground Floor		7' - 0"	166	4. First Floor	6' - 0"	0' - 6"	6' - 0"
88	3. ground Floor		7' - 0"	167	4. First Floor	6' - 0"	0' - 6"	6' - 0"
91	4. First Floor		3' - 0"	168	4. First Floor	6' - 0"	0' - 6"	6' - 0"
92	4. First Floor		3' - 0"	169	4. First Floor	6' - 0"	0' - 6"	6' - 0"
93	4. First Floor		3' - 0"	170	4. First Floor	6' - 0"	0' - 6"	6' - 0"
94	4. First Floor		3' - 0"	171	4. First Floor	6' - 0"	0' - 6"	6' - 0"
96	4. First Floor		3' - 0"	172	4. First Floor	6' - 0"	0' - 6"	6' - 0"
97	4. First Floor		3' - 0"	173	4. First Floor	6' - 0"	0' - 6"	6' - 0"
98	4. First Floor		3' - 0"	174	4. First Floor	6' - 0"	0' - 6"	6' - 0"
99	4. First Floor		3' - 0"	175	4. First Floor	6' - 0"	0' - 6"	6' - 0"
100	4. First Floor		3' - 0"	176	4. First Floor	6' - 0"	0' - 6"	6' - 0"
101	4. First Floor		3' - 0"					
106	4. First Floor		3' - 0"					
107	4. First Floor		3' - 0"					
108	4. First Floor		3' - 0"					
109	4. First Floor		3' - 0"					
110	4. First Floor		3' - 0"					
111	4. First Floor		3' - 0"					
112	4. First Floor		3' - 0"					
113	4. First Floor		3' - 0"					
114	4. First Floor		3' - 0"					
127	4. First Floor		3' - 0"					
128	4. First Floor		3' - 0"					
129	4. First Floor		3' - 0"					
130	4. First Floor		3' - 0"					
131	4. First Floor		3' - 0"					
134	4. First Floor		3' - 0"					
135	4. First Floor		3' - 0"					
138	4. First Floor		3' - 0"					
139	4. First Floor		3' - 0"					
142	4. First Floor		3' - 0"					
143	4. First Floor		3' - 0"					
144	3. ground Floor		3' - 0"					
145	4. First Floor		3' - 0"					
146	3. ground Floor		3' - 0"					
147	4. First Floor		3' - 0"					
148	1. Road	6' - 0"	3' - 0"					
153	3. ground Floor	6' - 0"	0' - 6"					

10



Activities, Dependencies & Timeline Creation



RENDERS



YEAR:
2023

LOCATION : HUNZA

TYPE : RESIDENTIAL

02

INTERACTIVE RETREAT

SITE LOCATIO: ABBOTTABAD AREA: 9,058.095QFT

PROJECT YEAR: 2024

PROJECT TYPE:RECERETIONAL PARK

Interactive Architecture fosters a dynamic relationship between people and their surroundings through active user engagement. It focuses on improving usability, efficiency, and versatility.

A retreat offers a tranquil environment that encourages relaxation and harmony with nature—promoting better health, refreshing routines, and inner peace.



VISION

The concept of the Interactive Retreat is to create a vacation home that connects people with Hunza's natural landscape, mountain views, and outdoor environment. The building is organized through layered masses, terraces, courtyards, and open spaces, allowing different levels of interaction with nature. Large openings frame panoramic views, while the stepped form responds to the site and creates a balance between privacy, social interaction, natural light, and ventilation.

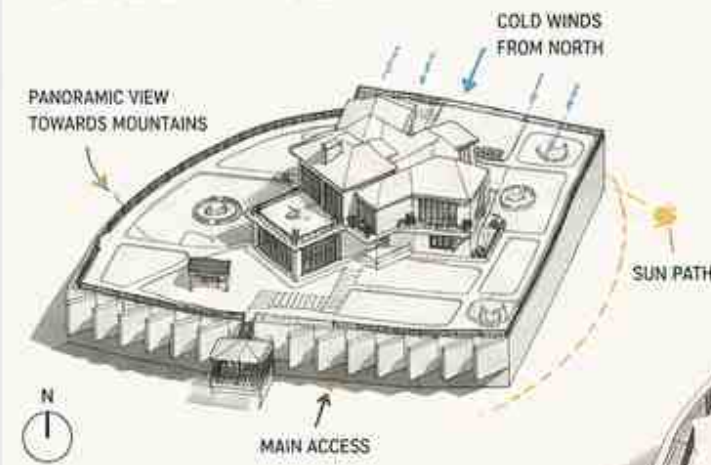


INTERACTIVE RETREAT

HUNZA, PAKISTAN

A VACATION HOME DESIGNED AS A RETREAT THAT CONNECTS WITH NATURE, VIEWS AND THE LOCAL LANDSCAPE.

1. SITE CONTEXT & RESPONSE



2. DESIGN CONCEPT



EMBRACE THE VIEWS
Open up towards the mountains and landscape.



HARMONY WITH NATURE
Use natural materials, greenery and outdoor spaces.



PRIVATE RETREAT
Create peaceful, interactive spaces for family and guests.



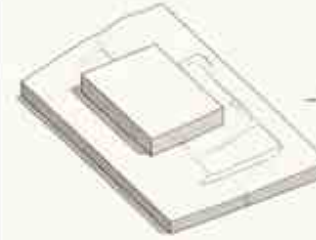
LIGHT & VENTILATION
Maximize natural light and cross ventilation.

7. LIGHT & VENTILATION STRATEGY

- Large openings towards mountains for day lighting.
- Cross ventilation through strategically placed windows and louvers.
- Deep overhangs and shading for summer comfort.

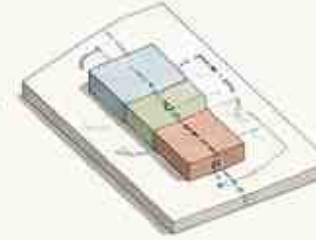


3. FORM DEVELOPMENT



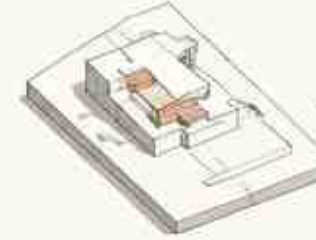
01. SITE MASS

Basic mass placed on site considering setbacks.



02. ZONING

Public spaces towards view, private spaces at back.



03. SUBTRACTION

Carving terraces and openings for views.



04. ADDITION

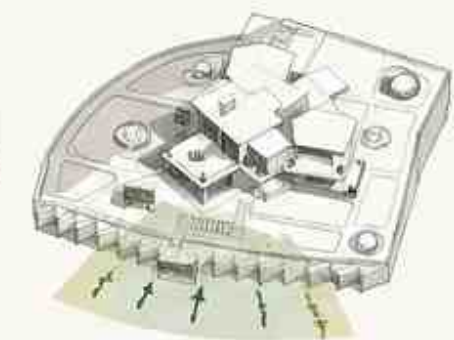
Adding volumes, terraces and connection spaces.



05. FINAL FORM

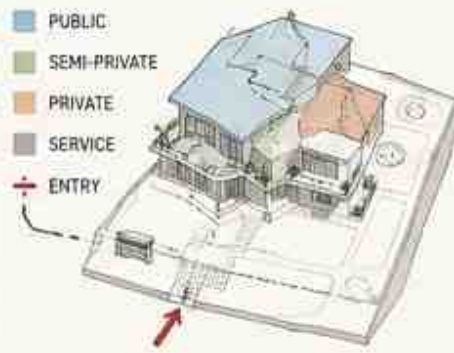
Final form responding to site, views and climate.

4. ORIENTATION & VIEWS



- BEST VIEWS
- SECONDARY VIEWS
- SERVICE / BACK

5. CIRCULATION CONCEPT



6. INDOOR - OUTDOOR CONNECTION



8. MATERIAL & CHARACTER



STONE CLADDING
Local stone for walls and base.



WOOD FINISH
Warm texture for roofs and details.



GLASS
Large glazing for views & light.



METAL RAILING
Lightweight railing for safety.

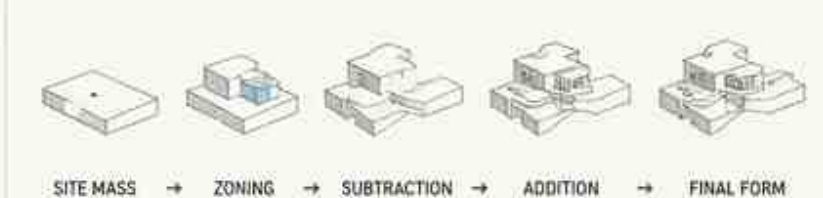


LANDSCAPE
Greenery to blend with nature.

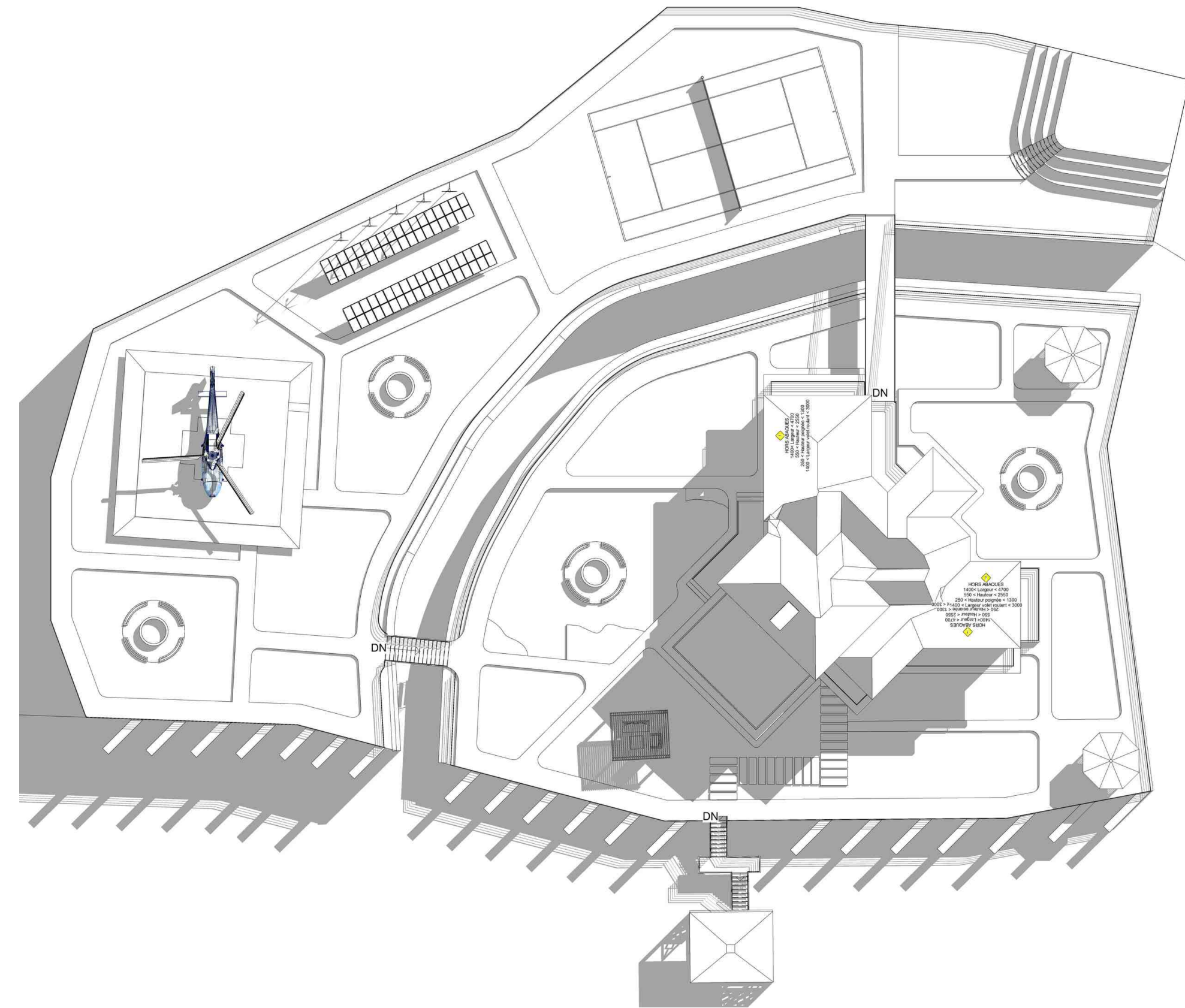
9. KEY FEATURES

- Panoramic mountain views
- Spacious terraces & decks
- Gazebos & outdoor seating
- Landscaped gardens & paths
- Privacy with interaction
- Blend of modern & local character

10. FORM EVOLUTION SUMMARY



MASTER PLAN



REVIT



PHOTOSHOP RENDER

PLANS



Room Legend

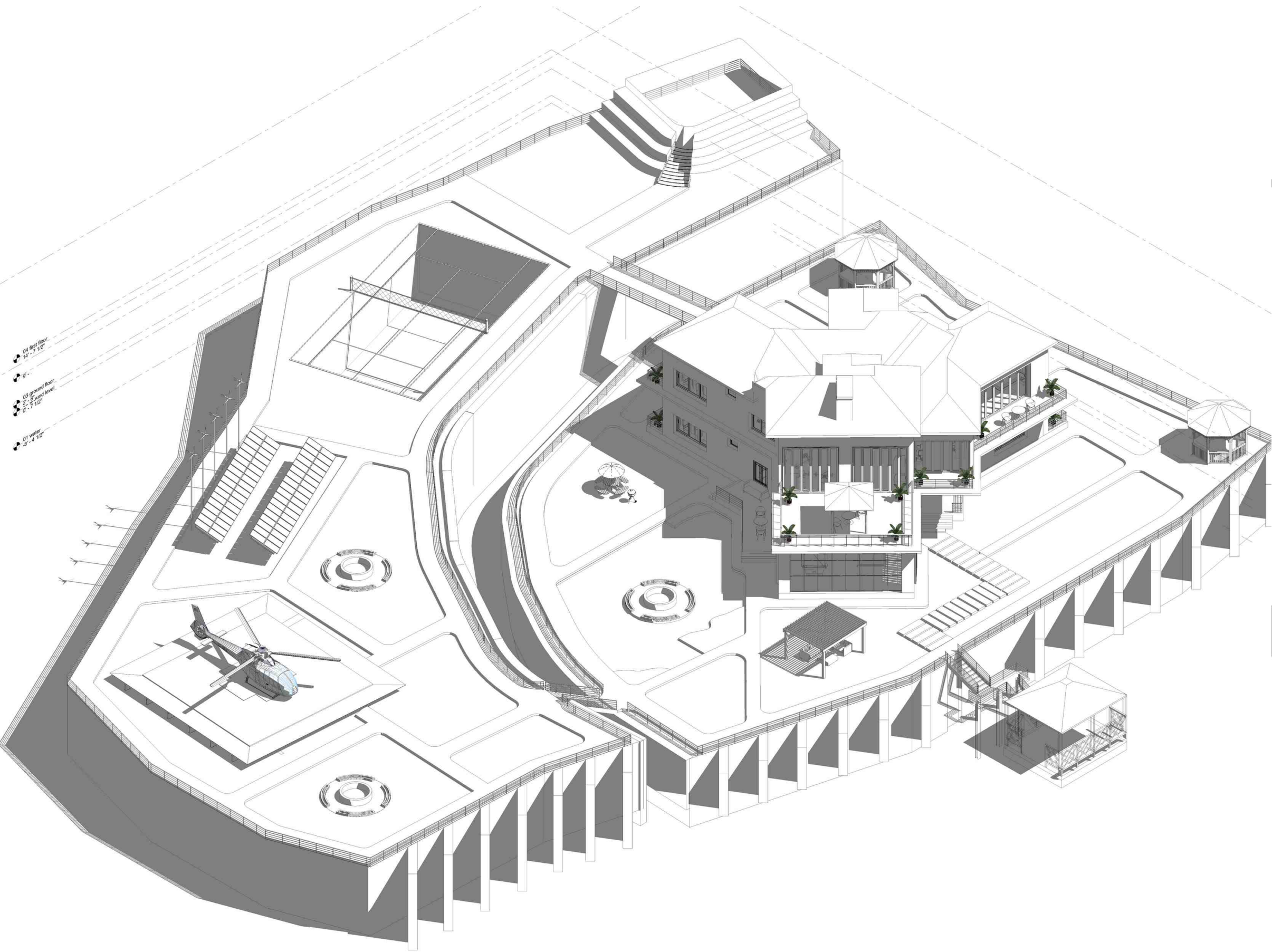
- BALCONY
- BED ROOM
- COURTYARD
- DINNING
- DRESSER
- FOYER
- LIVING AREA
- LOBBY
- OPEN KITCHEN
- STAIRS
- STORE
- STUDIO
- WASHROOM



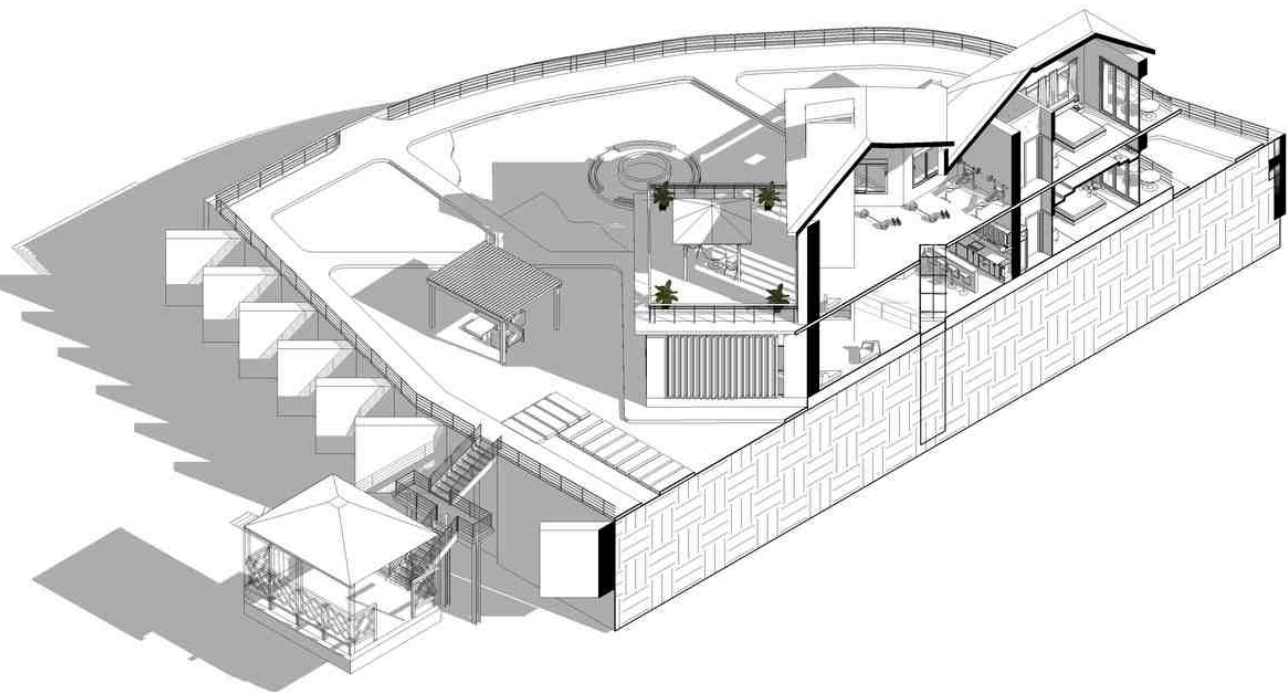
Room Legend

- BALCONY
- BED ROOM
- COURTYARD
- DRESSER
- GAMING ROOM
- GYM
- LIVING AREA
- LOBBY
- STAIRS
- TERRACE
- WASHROOM

3D VIEW



3D SECTIONS VIEWS



1 Isometric Section AA 'Arch'



2 Isometric Sectional BB' Arch

EXPLODED VIEW DETAILS

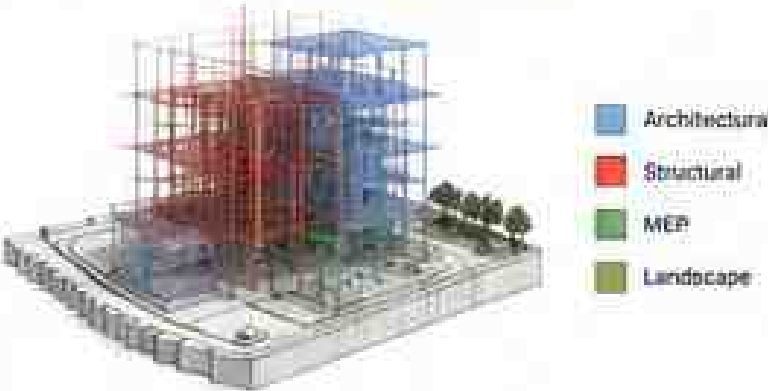
LEVEL ORGANIZATION



MODEL STATISTICS

Total Floors	4 (Including Roof)
Total Rooms	28
Total Area	1,248.50 m ²
Total Volume	4,325.60 m ³
Total Components	1,742
Families Used	312

DISCIPLINE INTEGRATION



BIM STANDARDS

- ISO 19650 Workflow Compliant
- BIM Execution Plan Implemented
- Information Management in Accordance
- Data Rich Model Enabled

MATERIAL INFORMATION

EXTERIOR FINISH

Concrete Tiles	Glass Panels
Glass Panels	Aluminum Frames

INTERIOR FINISH

Gypsum Walls	Ceramic Tiles
Wooden Flooring	Paint Finish

LANDSCAPE

Concrete Pavers	Plantation
Lawn	Decorative Elements

BUILDING COMPONENTS

10 ROOF SYSTEM

- Sloped Roof Geometry
- Roof Finish
- Structural Roof Framing
- Roof Edge Detail
- Waterproof Layer



09 UPPER FLOORS

- Bedrooms
- Study Area
- Balcony
- Circulation
- Void Space



08 INTERMEDIATE FLOORS

- Living Room
- Family Lounge
- Dining Area
- Double Height Space
- Stair Connection



05 GROUND FLOOR

- Main Entrance
- Reception
- Lounge
- Kitchen
- Dining
- Service Area



04 STRUCTURAL SYSTEM

- RCC Columns
- RCC Beams
- Floor Slabs
- Shear Walls
- Structural Core



03 VERTICAL CIRCULATION

- Main Staircase
- Landings
- Internal Connectivity
- Vertical Core



02 BUILDING ENVELOPE

- Exterior Walls
- Curtain Walls
- Glazing Panels
- Window System
- Door Assemblies



01 LANDSCAPE ELEMENTS

- Gazebo
- Walkways
- Green Spaces
- Plantation
- Outdoor Sitting
- Entry Plaza



BUILDING COMPONENT

- 01 ROOF SYSTEM
 - Sloped Roof Geometry
 - Roof Finish
 - Structural Roof Framing
 - Roof Edge Detail
 - Waterproof Layer

- 02 UPPER FLOORS
 - Bedrooms
 - Study Area
 - Balcony
 - Circulation
 - Void Space

- 03 TERRACES & BALCONIE
 - Open Terraces
 - Planter Boxes
 - Outdoor Furniture
 - Guard Rails

- 04 INTERMEDIATE FLOORS
 - Living Room
 - Family Lounge
 - Dining Area
 - Double Height Space
 - Stair Connection

- 05 VERTICAL CIRCULATION
 - Main Staircase
 - Landings
 - Internal Connectivity
 - Vertical Core

- 06 GROUND FLOOR
 - Main Entrance
 - Reception
 - Lounge
 - Kitchen
 - Dining
 - Service Area

- 07 BUILDING ENVELOPE
 - Exterior Walls
 - Curtain Walls
 - Glazing Panels
 - Window System
 - Door Assemblies

- 08 STRUCTURAL SYSTEM
 - RCC Columns
 - RCC Beams
 - Floor Slabs
 - Shear Walls
 - Structural Core

- 09 LANDSCAPE ELEMENT
 - Green Spaces
 - Plantation
 - Walkways
 - Water Feature
 - Outdoor Lighting

- 10 OUTDOOR SITOUTS
 - Sitout Area
 - Paved Flooring
 - Pergola / Shades

- 11 SITE ACCESS
 - Entry Gate
 - Driveway
 - Parking Area
 - Pedestrian Path

- 12 SITE DEVELOPMENT
 - Retaining Walls
 - Boundary Wall
 - Site Grading
 - Drainage System

- 13 FOUNDATION SYSTEM
 - Foundation Footings
 - Plinth Beam
 - Damp Proof Course

- 14 BASEMENT (OPTIONAL)
 - Storage
 - Services
 - Mechanical Room

- 15 SERVICE AREAS
 - Laundry
 - Utility Room
 - Electrical Room

- 16 MEP SYSTEMS (COORD.)
 - HVAC Ducts
 - Plumbing Lines
 - Electrical Conduits

- 17 EXTERIOR FINISHES
 - Stone Cladding
 - Paint Finish
 - Texture Finish

- 18 INTERIOR FINISHES
 - Gypsum Walls
 - Ceramic Tiles
 - Wooden Flooring

- 19 FURNITURE & FIXTURES
 - Furniture
 - Sanitary Fixtures
 - Joinery

- 20 SITE FURNITURE
 - Benches
 - Lighting Poles
 - Planters



ELEVATIONS DESIGN



INTERACTIVE RETREAT – HUNZA

EXTERIOR ELEVATIONS (DETAILED)

1. FRONT ELEVATION (NORTH)

SCALE: 1/8" = 1'-0"

08 ROOF PEAK
38' - 3 29/32"

09 ROOF EAVE
34' - 7 1/2"

06 TOP
26' - 2 13/32"

04 FIRST FLOOR
14' - 7 1/2"

03 GROUND FLOOR
0' - 0"

02 GROUND (EXTERNAL)
-2' - 8"

01 WATER LEVEL
-8' - 4 1/2"



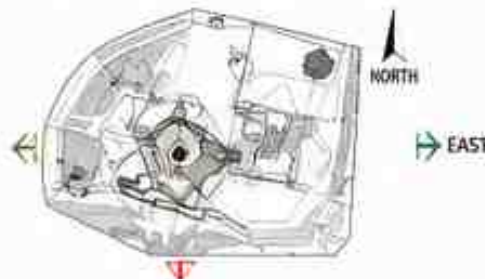
ELEVATION MATERIAL LEGEND

- 1 STANDING SEAM METAL ROOF (DARK GREY)
- 2 WOOD SOFFIT / FASCIA (TEAK FINISH)
- 3 EXTERIOR PLASTER FINISH (OFF WHITE)
- 4 STONE CLADDING (LOCAL STONE)
- 5 ALUMINUM WINDOW FRAME (DARK GREY)
- 6 ALUMINUM / GLASS SLIDING DOOR
- 7 MS RAILING (POWDER COATED BLACK)
- 8 VERTICAL WOODEN LOUVERS
- 9 EXTERIOR WALL LIGHT
- 10 STONE PLINTH / BASE

ELEVATION NOTES

- All external plaster to be weather resistant.
- Stone cladding to be natural local stone with proper waterproofing behind.
- All railings min. height 3'-6".
- Window sill height 2'-6" from FFL (typ.).
- RWP (rain water pipe) concealed in wall.

KEY PLAN



2. SIDE ELEVATION (EAST)

SCALE: 1/8" = 1'-0"

08 ROOF PEAK
38' - 3 29/32"

09 ROOF EAVE
34' - 7 1/2"

06 TOP
14' - 2 13/32"

04 FIRST FLOOR
14' - 7 1/2"

05 2ND COUNTER
12' - 7 1/2"

03 GROUND FLOOR
0' - 0"

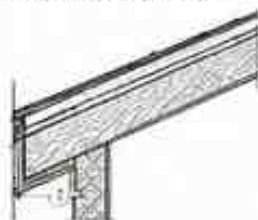
02 GROUND (EXTERNAL)
-2' - 8"

01 WATER LEVEL
-8' - 4 1/2"



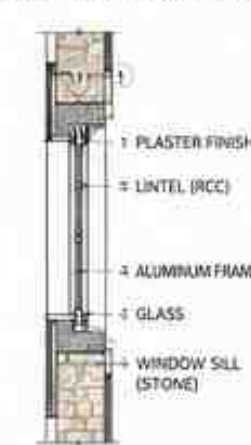
TYPICAL ELEVATION DETAILS

E-01 ROOF EDGE DETAIL

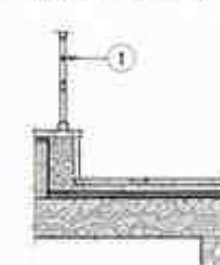


- 1 METAL FLASHING
- 2 PLYWOOD SHEATHING
- 3 WATER PROOF MEMBRANE
- 4 ROOF INSULATION
- 5 ROOF STRUCTURE
- 6 SOFFIT WOOD FINISH

E-02 WINDOW HEAD / SILL DETAIL



E-03 PARAPET / RAILING DETAIL



- 1 MS RAILING
- 2 STONE COPING
- 3 PARAPET WALL
- 4 WATER PROOFING
- 5 FLOOR SLAB

DOOR SCHEDULE

TYPE	SIZE	DESCRIPTION	QTY.
D1	4'-0" x 7'-0"	MAIN ENTRANCE DOOR (WOODEN)	01
D2	3'-0" x 7'-0"	BEDROOM DOOR (FLUSH)	06
D3	3'-0" x 7'-0"	TOILET DOOR (PVC / WPC)	04
D4	6'-0" x 7'-0"	SLIDING GLASS DOOR	02

WINDOW SCHEDULE

TYPE	SIZE	DESCRIPTION	SILL LVL.	QTY.
W1	4'-0" x 5'-0"	FIXED WINDOW	2'-6"	04
W2	4'-0" x 4'-0"	SLIDING WINDOW	2'-6"	08
W3	2'-0" x 2'-0"	FIXED VENTILATOR	7'-0"	02
W4	8'-0" x 7'-0"	FOLDING / SLIDING WINDOW	0'-0"	02

FINISH / MATERIAL PALETTE



GENERAL NOTES

- All dimensions are in feet and inches.
- All levels are in reference to 0'-0" (Ground Floor FFL).
- All external finishes to be as per specifications.
- Provide proper slope on all horizontal surfaces.
- All glass to be firm clear / as per specification.
- All drawings to be read in conjunction with structural drawings.

ABBREVIATIONS

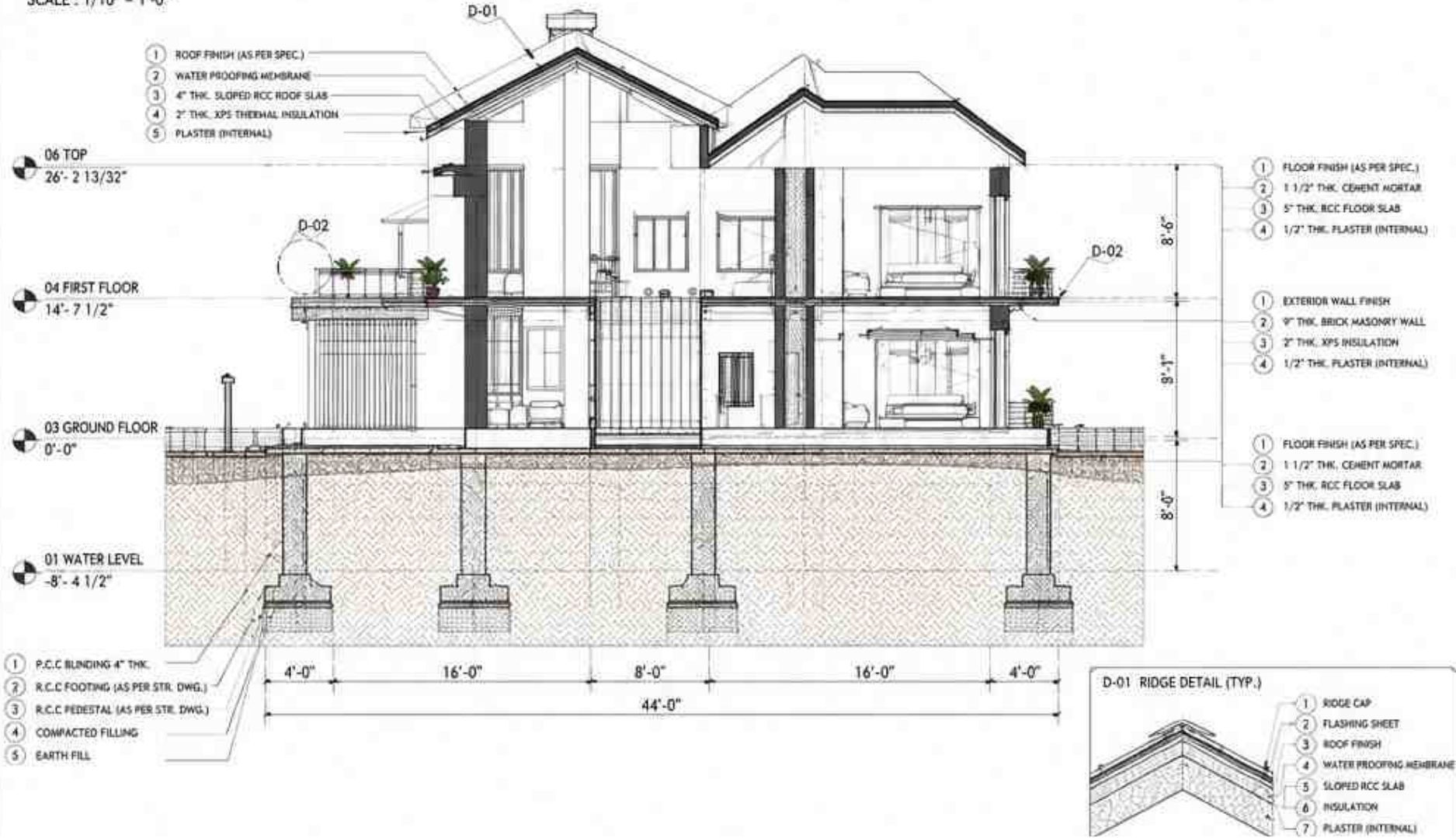
- FFL = FINISHED FLOOR LEVEL
- RCC = REINFORCED CEMENT CONCRETE
- RWP = RAIN WATER PIPE
- TYP. = TYPICAL
- CL = CENTER LINE
- EXT. = EXTERIOR
- INT. = INTERIOR

SECTIONS



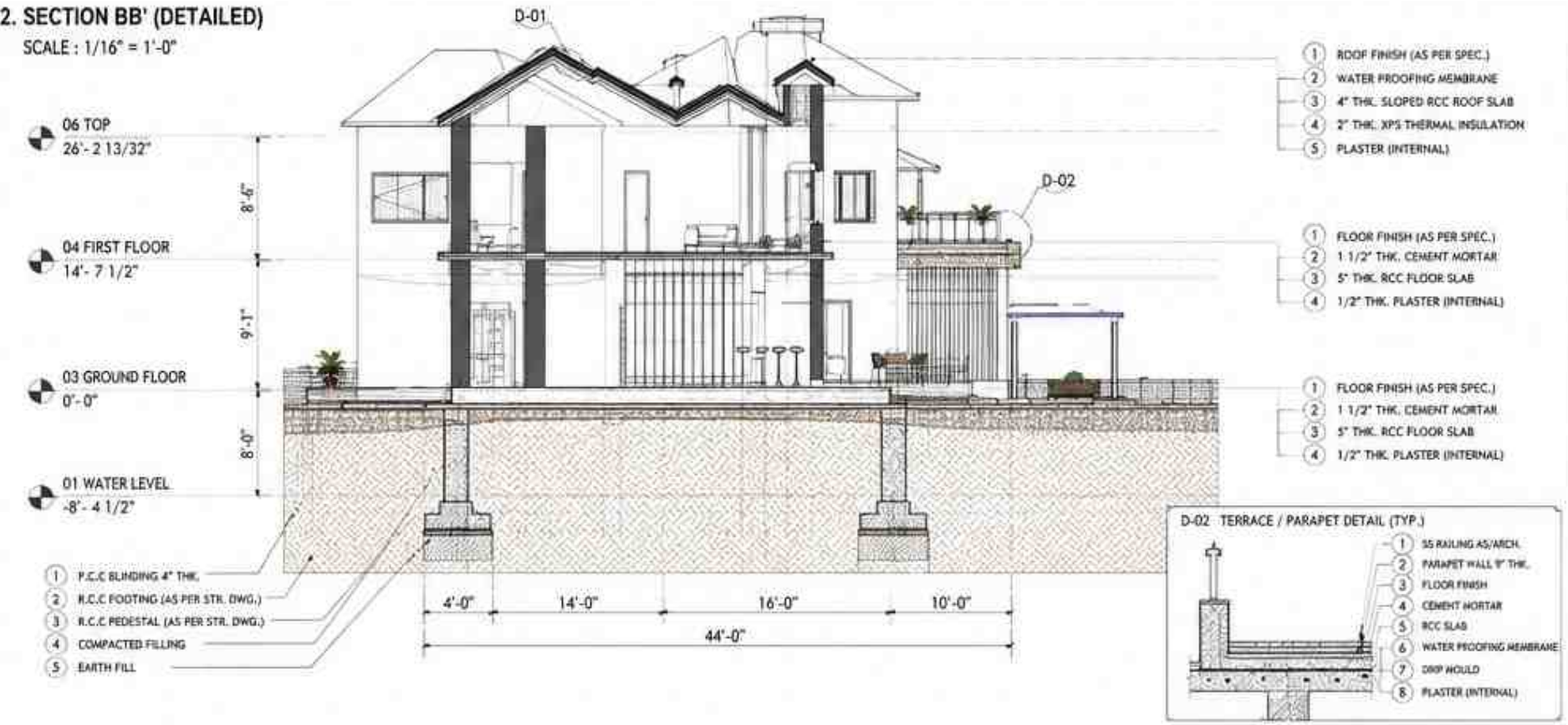
1. SECTION AA' (DETAILED)

SCALE : 1/16" = 1'-0"

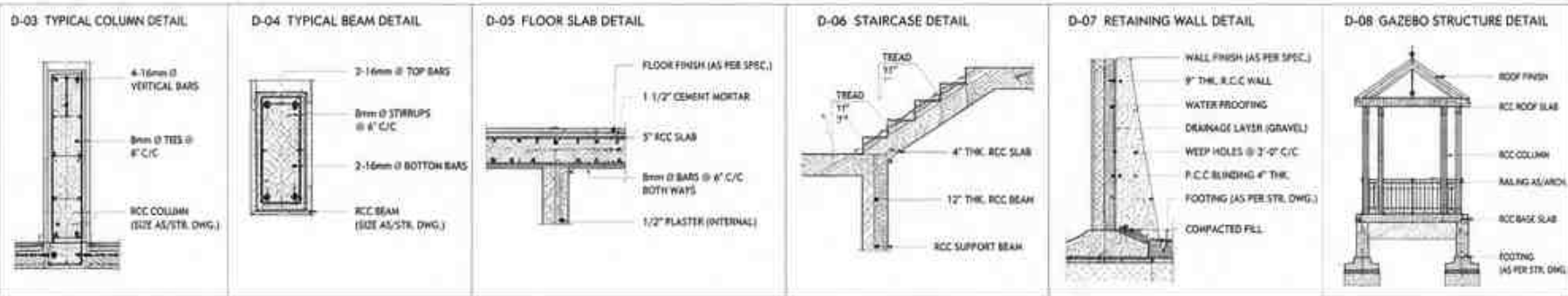


2. SECTION BB' (DETAILED)

SCALE : 1/16" = 1'-0"



3. TYPICAL STRUCTURAL DETAILS



4. MATERIAL / FINISH LEGEND



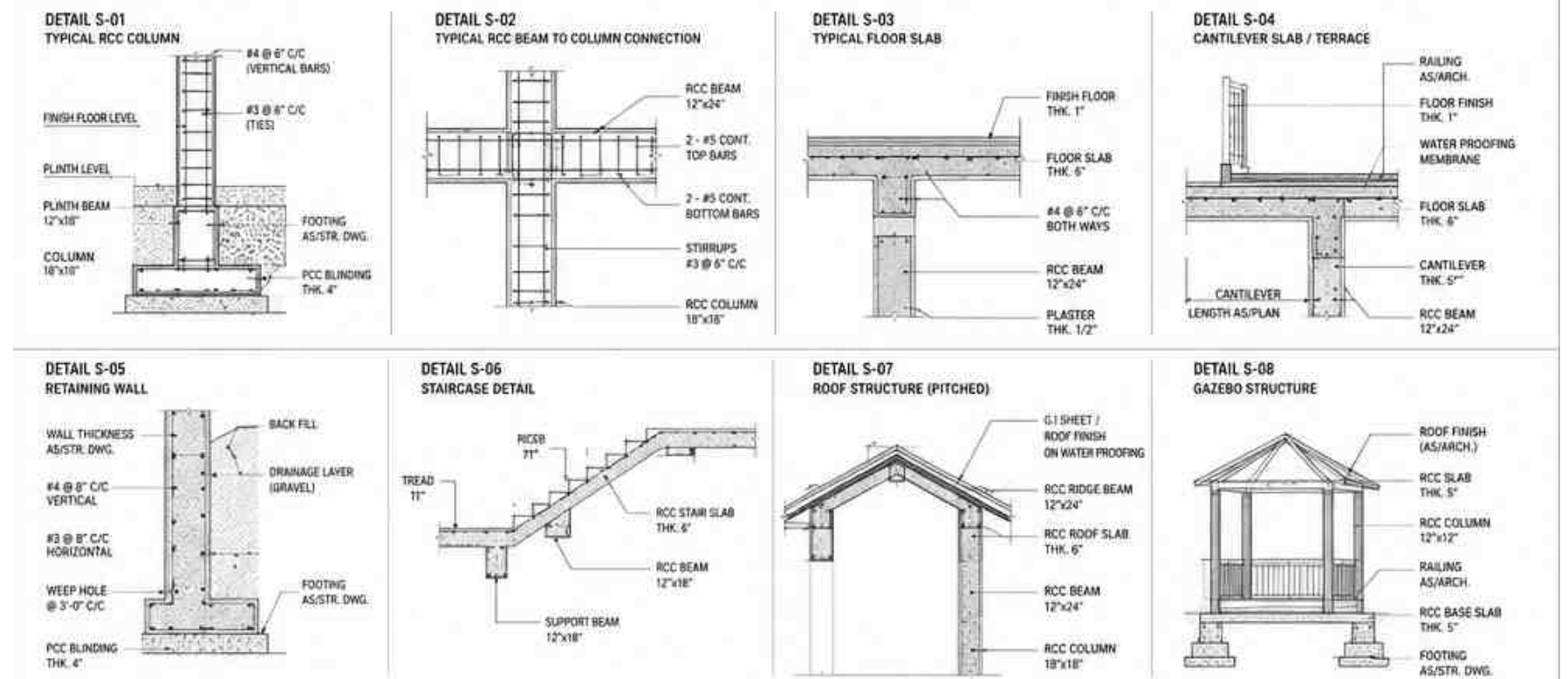
5. STRUCTURAL NOTES

- All RCC work to be as per structural drawings and engineer's specifications.
- Concrete grade : M25
- Steel grade : Fe 500
- Cover to reinforcement :
 - Beams/Columns = 1.5"
 - Slabs = 0.75"
 - Footings = 2.0"
- Provide adequate curing for all concrete elements.
- All dimensions are in feet and inches.

6. LOAD PATH DIAGRAM (TYP.)



4. TYPICAL STRUCTURAL DETAILS

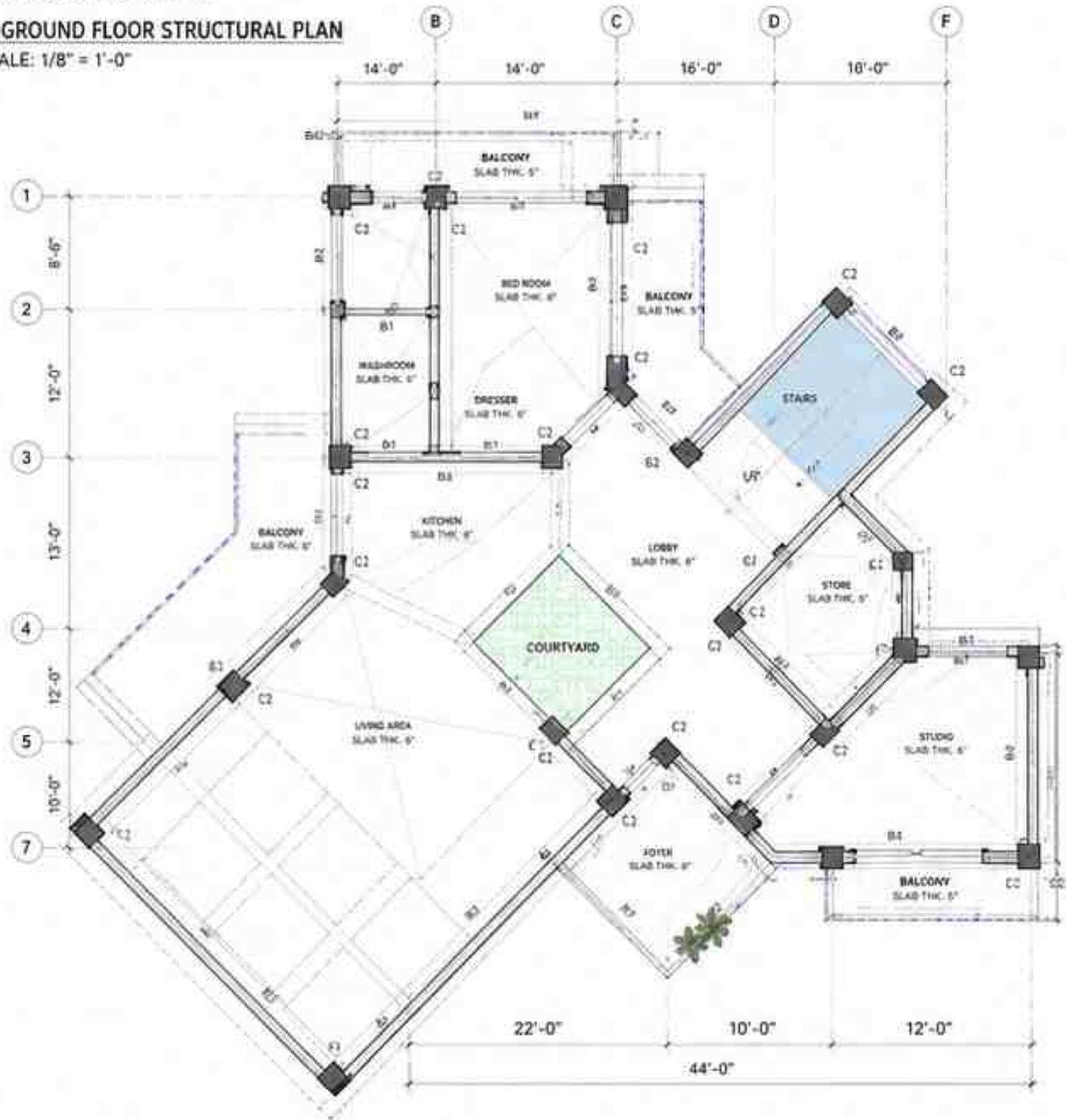


INTERACTIVE RETREAT – HUNZA

STRUCTURAL PLANS

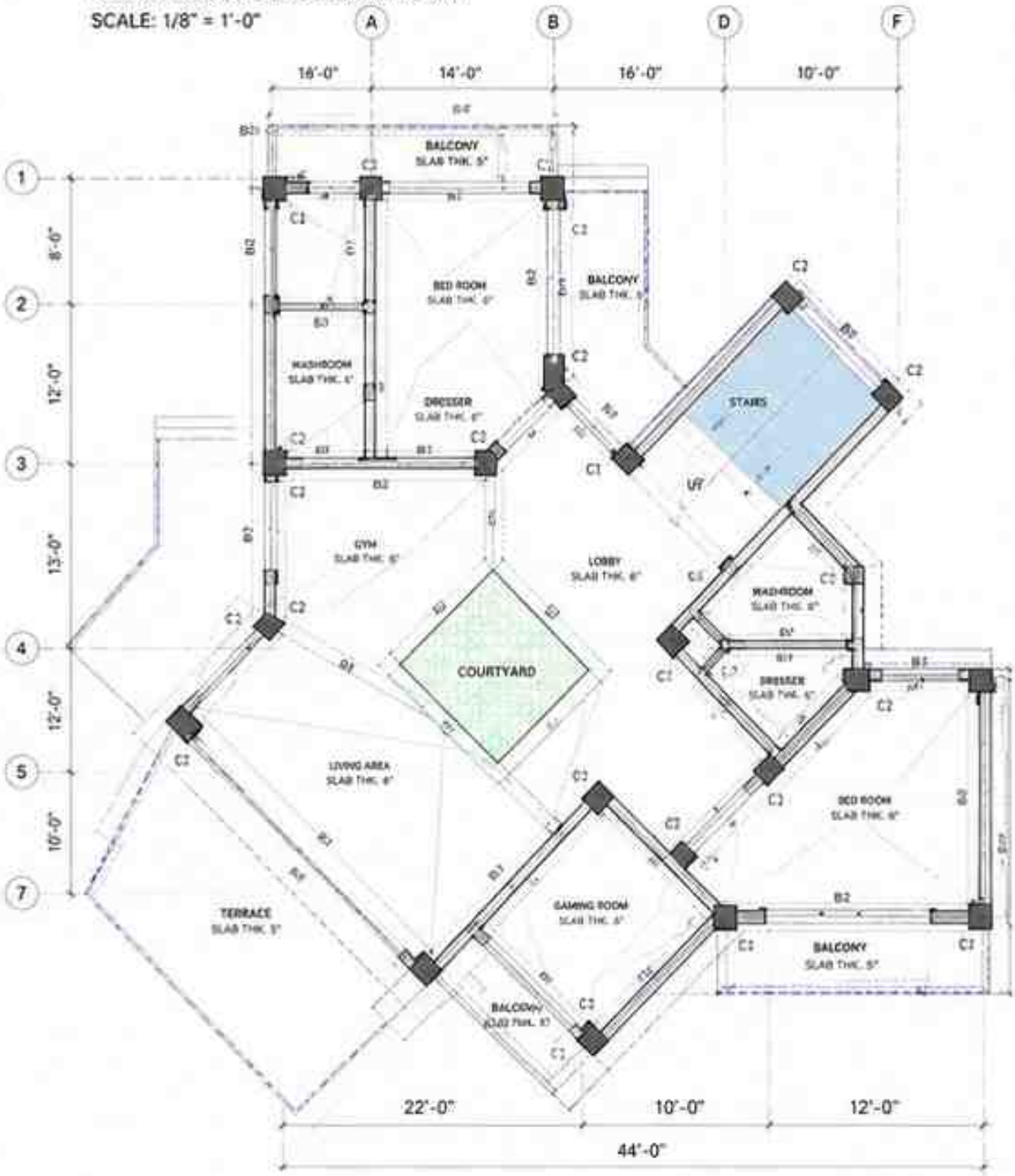
1. GROUND FLOOR STRUCTURAL PLAN

SCALE: 1/8" = 1'-0"



2. FIRST FLOOR STRUCTURAL PLAN

SCALE: 1/8" = 1'-0"



STRUCTURAL LEGEND

- C1 18"x18" RCC COLUMN
- C2 12"x18" RCC COLUMN
- B1 12"x24" RCC BEAM
- B2 12"x18" RCC BEAM
- B3 9"x18" RCC BEAM
- B4 9"x15" RCC BEAM
- PLINTH / GROUND BEAM
- CANTILEVER BEAM
- SLAB DIRECTION
- STAIRCASE OPENING
- COURTYARD OPEN TO SKY

NOTES:

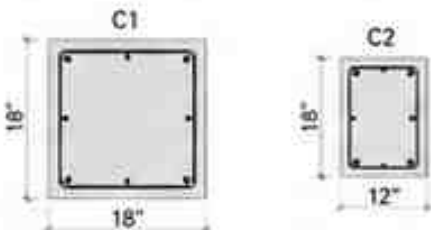
- ALL DIMENSIONS ARE IN FEET AND INCHES.
- ALL RCC WORK TO BE AS PER STRUCTURAL DRAWINGS AND ENGINEER'S SPECIFICATIONS.
- CONCRETE GRADE: M25
- STEEL GRADE: Fe 500
- COVER TO REINFORCEMENT:
 - BEAMS/COLUMNS = 1.5"
 - SLABS = 0.75"
 - FOOTINGS = 2.0"
- PROVIDE ADEQUATE CURING FOR ALL CONCRETE ELEMENTS.
- ALL SLAB THICKNESS: 6" (U.N.O.)
- STAIR SLAB THICKNESS: 6"
- TERRACE / BALCONY SLAB THICKNESS: 5"

KEY PLAN



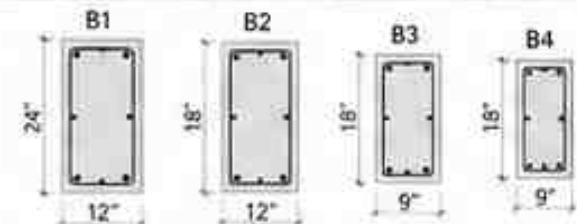
COLUMN SCHEDULE

MARK	SIZE	REINFORCEMENT	STIRRUPS
C1	18" X 18"	8 - 16 Ø	8mm @ 6" C/C
C2	12" X 18"	6 - 16 Ø	8mm @ 6" C/C

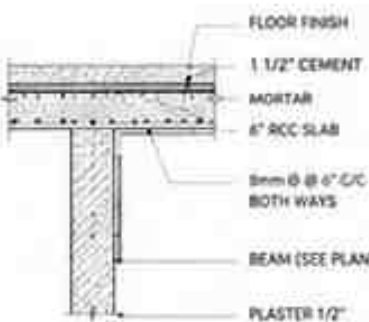


BEAM SCHEDULE

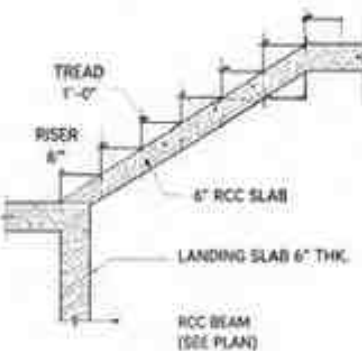
MARK	SIZE	TOP STEEL	BOTTOM STEEL	STIRRUPS
B1	12" X 24"	3 - 16 Ø	3 - 16 Ø	8mm @ 6" C/C
B2	12" X 18"	2 - 16 Ø	2 - 16 Ø	8mm @ 6" C/C
B3	9" X 18"	2 - 16 Ø	2 - 16 Ø	8mm @ 6" C/C
B4	9" X 15"	2 - 12 Ø	2 - 12 Ø	8mm @ 6" C/C



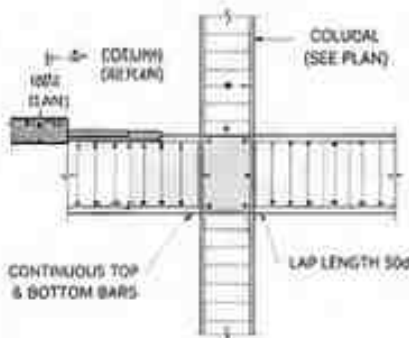
SLAB SECTION (TYP.)



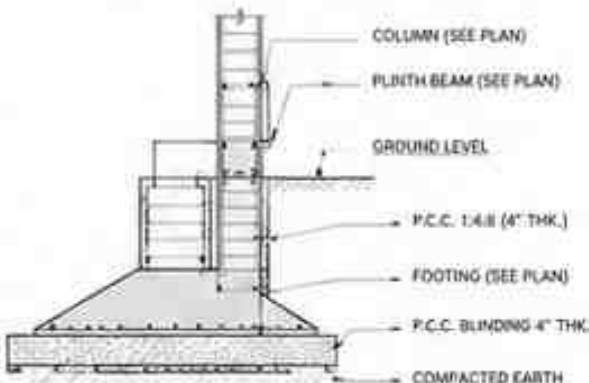
STAIRCASE SECTION (TYP.)



TYPICAL BEAM-COLUMN JOINT



TYPICAL FOUNDATION (ISOLATED FOOTING)

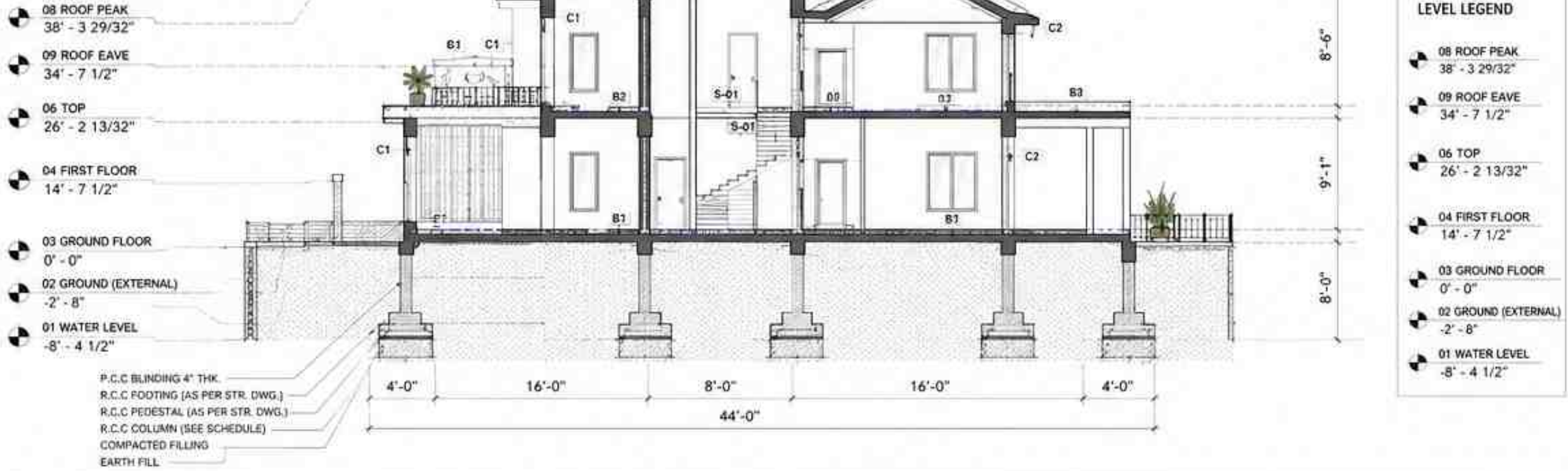


INTERACTIVE RETREAT – HUNZA

STRUCTURAL SECTIONS (DETAILED)

1. SECTION AA' (STRUCTURAL)

SCALE: 1/8" = 1'-0"



2. SECTION BB' (STRUCTURAL)

SCALE: 1/8" = 1'-0"



COLUMN SCHEDULE			
MARK	SIZE	REINFORCEMENT	STIRRUPS
C1	18"x18"	8 - 16 Ø	8mm @ 6" C/C
C2	12"x12"	6 - 16 Ø	8mm @ 6" C/C

BEAM SCHEDULE				
MARK	SIZE	TOP STEEL	BOTTOM STEEL	STIRRUPS
B1	12"x24"	3 - 16 Ø	3 - 16 Ø	8mm @ 6" C/C
B2	12"x18"	2 - 16 Ø	2 - 16 Ø	8mm @ 6" C/C
B3	9"x18"	2 - 16 Ø	2 - 16 Ø	8mm @ 6" C/C

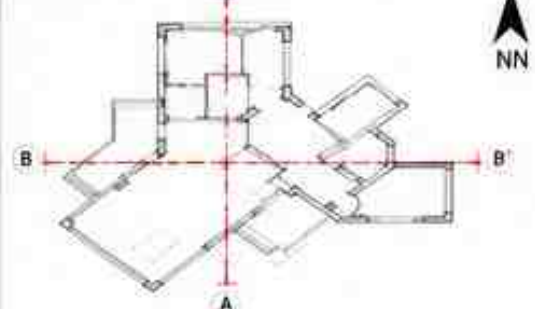
NOTES:

- ALL DIMENSIONS ARE IN FEET AND INCHES.
- ALL RCC WORK TO BE AS PER STRUCTURAL DRAWINGS AND ENGINEER'S SPECIFICATIONS.
- CONCRETE GRADE: M25
- STEEL GRADE: Fe 500
- COVER TO REINFORCEMENT:
 - BEAMS / COLUMNS = 1.5"
 - SLABS = 0.75"
 - FOOTINGS = 2.0"
- PROVIDE ADEQUATE CURING FOR ALL CONCRETE ELEMENTS.

LEGEND

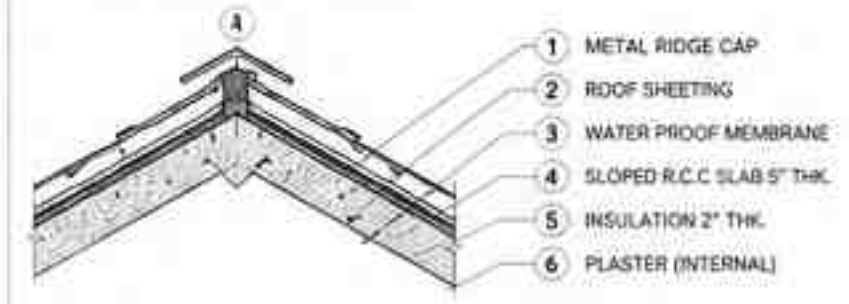
- R.C.C COLUMN
- R.C.C BEAM
- PLINTH / GROUND BEAM
- CANTILEVER BEAM
- SLAB OPENING
- SLAB DIRECTION

KEY PLAN

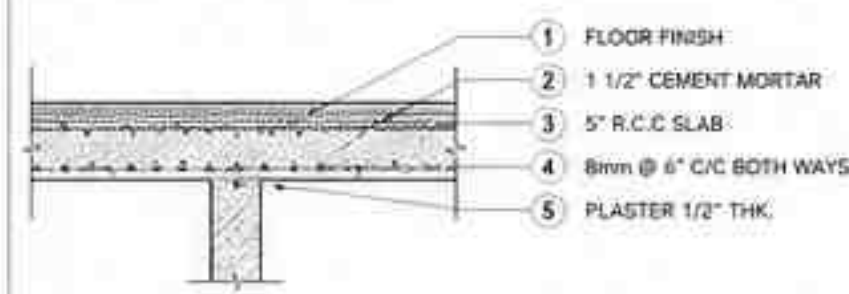


TYPICAL STRUCTURAL DETAILS

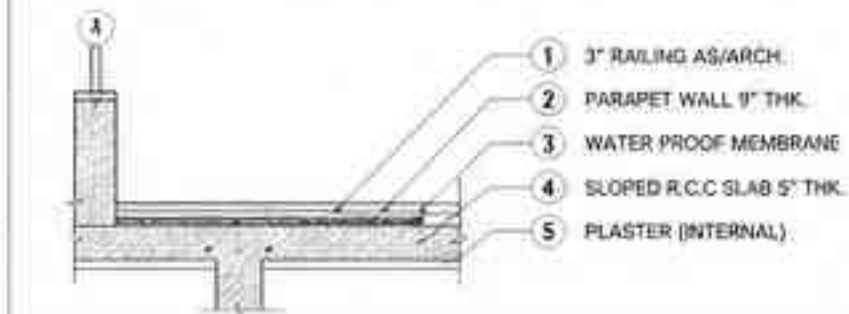
R-01 ROOF / RIDGE DETAIL (TYP.)



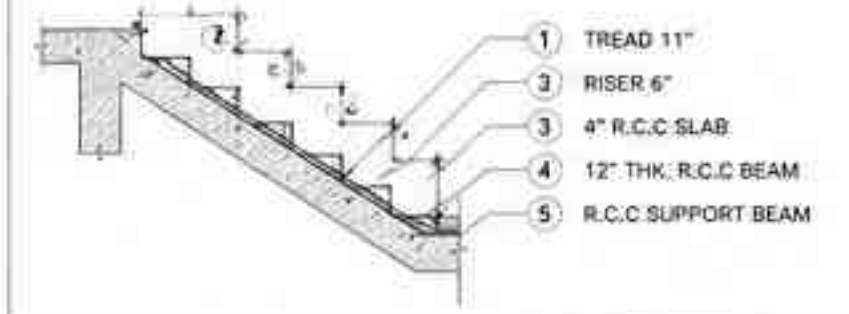
D-01 FLOOR SLAB DETAIL (TYP.)



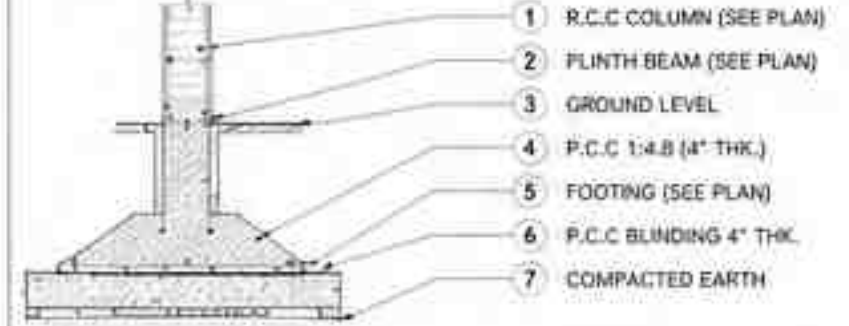
D-02 TERRACE / PARAPET DETAIL (TYP.)



S-01 STAIRCASE DETAIL (TYP.)



D-03 FOUNDATION DETAIL (TYP.)



INTERACTIVE RETREAT – HUNZA

MEP SERVICES DRAWINGS (DETAILED)

1. GROUND FLOOR MEP PLAN

SCALE: 1/8" = 1'-0"



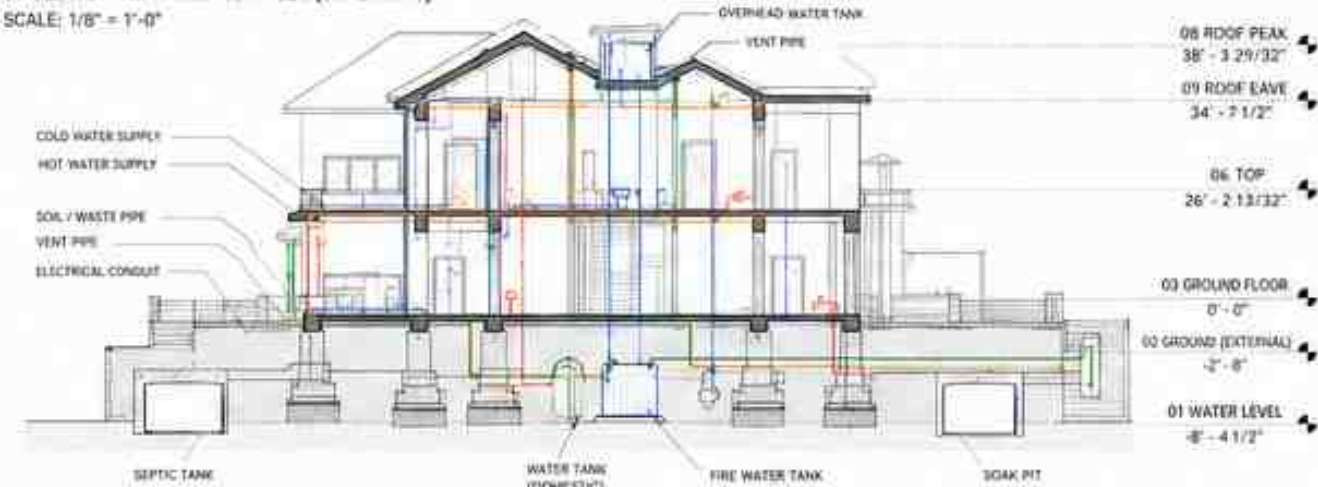
2. FIRST FLOOR MEP PLAN

SCALE: 1/8" = 1'-0"



6. SECTION AA' – MEP SERVICES (SCHEMATIC)

SCALE: 1/8" = 1'-0"



7. SECTION BB' – MEP SERVICES (SCHEMATIC)

SCALE: 1/8" = 1'-0"



8. PLUMBING FIXTURE SCHEDULE

S.NO.	FIXTURE	SYMBOL	QTY.	REMARKS
1	WATER CLOSET		06	FLUSH TANK
2	WASH BASIN		06	HOT & COLD
3	SHOWER		04	HOT & COLD
4	KITCHEN SINK		02	HOT & COLD
5	FLOOR DRAIN		07	100mm DIA.
6	HOSE BIB		03	

9. DOOR / WINDOW SCHEDULE (REFERENCE)

TYPE	DESCRIPTION	WIDTH	HEIGHT	QTY.
D1	MAIN ENTRANCE DOOR (WOODEN)	4'-0"	7'-0"	01
D2	BEDROOM DOOR (FLUSH)	3'-0"	7'-0"	06
D3	TOILET DOOR (PVC / WPC)	3'-0"	7'-0"	04
D4	SLIDING GLASS DOOR	8'-0"	7'-0"	02

10. GENERAL NOTES (MEP)

- ALL MEP DRAWINGS ARE SCHEMATIC AND TO BE READ IN CONJUNCTION WITH ARCHITECTURAL & STRUCTURAL DRAWINGS.
- ALL PIPE SIZES ARE INDICATIVE. FINAL SIZES AS PER SITE CONDITIONS & ENGINEER'S DESIGN.
- USE CPVC / PPR PIPES FOR HOT & COLD WATER SUPPLY.
- PROVIDE ADEQUATE SLEEVES AT ALL WALLS / SLABS FOR PIPES & CONDUITS.
- ELECTRICAL WIRING AS PER NEA / IEC STANDARDS.
- PROVIDE EARTHING FOR ALL DBs AND METALLIC ITEMS.

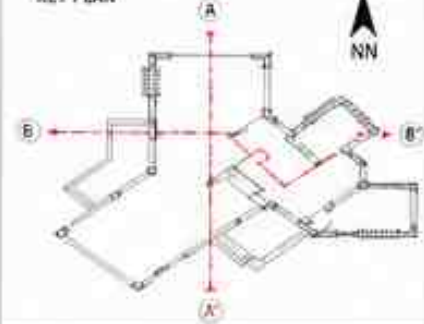
11. MEP SYSTEM OVERVIEW

- WATER SUPPLY SYSTEM
- SANITARY DRAINAGE SYSTEM
- FIRE FIGHTING SYSTEM
- ELECTRICAL SYSTEM
- LOW CURRENT / ELV SYSTEM

12. ABBREVIATIONS

- DB = DISTRIBUTION BOARD
- RWP = RAIN WATER PIPE
- SWR = SOIL WASTE PIPE
- CWS = COLD WATER SUPPLY
- HWS = HOT WATER SUPPLY
- FD = FLOOR DRAIN
- CO = CLEAN OUT
- GV = GATE VALVE
- SV = STOP VALVE
- PRV = PRESSURE REDUCING VALVE
- ELV = EXTRA LOW VOLTAGE

KEY PLAN



3. MEP LEGEND

PLUMBING (WATER SUPPLY)

- COLD WATER PIPE
- HOT WATER PIPE
- STOP VALVE
- GATE VALVE
- WATER METER
- PRESSURE REDUCING VALVE

SANITARY (SOIL & WASTE)

- SOIL / WASTE PIPE
- VENT PIPE
- FLOOR DRAIN
- CLEAN OUT

FIRE FIGHTING

- FIRE PIPE (WET RISER)
- FIRE HYDRANT
- FIRE EXTINGUISHER
- FIRE HOSE REEL

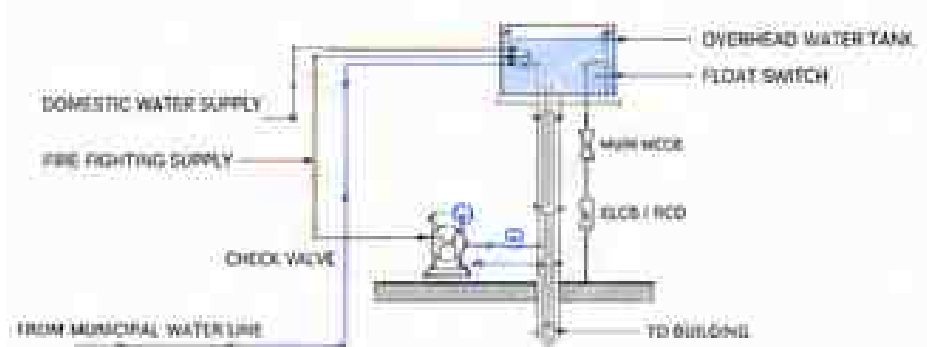
ELECTRICAL

- DISTRIBUTION BOARD (DB)
- CEILING LIGHT
- WALL LIGHT
- CEILING FAN
- EXHAUST FAN
- SINGLE SOCKET OUTLET
- DOUBLE SOCKET OUTLET
- SWITCH (SINGLE POLE)
- SWITCH (TWO WAY)

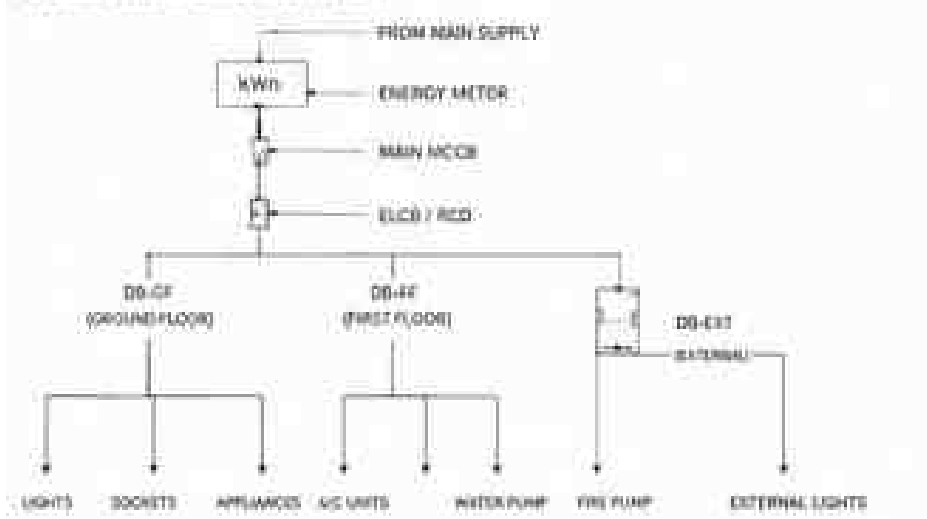
LOW CURRENT / ELV

- TV OUTLET
- TELEPHONE OUTLET
- DATA OUTLET
- DOOR PHONE / INTERCOM
- CCTV CAMERA

4. WATER TANK & PUMP LAYOUT



5. ELECTRICAL RISER DIAGRAM



INTERACTIVE RETREAT — HUNZA

COMPLETESTRUCTURAL+MEPSCHEDULESET | Revit / TechnicalDocumentationStyle

PROJECT INFO

Project	Interactive Retreat — Hunza
Use	Residential Vacation Home / Retreat
Documentation	Structural + MEP Schedule Set
Status	Conceptual / Academic Coordination
Important	Final structural sizes, reinforcement, MEP capacities and routing require engineering calculations, soil/seismic data and local code verification.

COLUMN SCHEDULE

Tag	Element	Indicative Size	Material	Use	Remarks
C-01	RCC Column	300×300 mm	RCC	Primary frame	Verify by analysis
C-02	RCC Column	300×450 mm	RCC	Major/long-span zones	Verify by analysis
C-03	RCC Column	250×300 mm	RCC	Secondary frame	Conceptual
C-04	Edge Column	250×300 mm	RCC	Perimeter/terrace	Coordinate glazing

BEAM SCHEDULE

Tag	Element	Indicative Size	Material	Use	Remarks
B-01	Primary Beam	300×450 mm	RCC	Main spans	Final depth by analysis
B-02	Secondary Beam	250×400 mm	RCC	Room/slab framing	Coordinate ceiling
B-03	Edge Beam	250×350 mm	RCC	Perimeter/balcony	Support façade
B-04	Cantilever Beam	250×450 mm	RCC	Terraces/balconies	Deflection check

SLAB SCHEDULE

Tag	Type	Thickness	System	Location	Remarks
S-01	Typical floor	150 mm	RCC solid slab	Ground/first	Verify spans
S-02	Terrace	150 mm	RCC solid slab	Outdoor terraces	Waterproof + slope
S-03	Balcony	150 mm	RCC cantilever	Balconies	Deflection check
S-04	Roof	125–150 mm	RCC/roof framing	Roof zones	Coordinate roof

FOUNDATION SCHEDULE

Tag	Element	Concept	Indicative Size	Material	Remarks
F-01	Isolated footing	RCC pad	1500×1500×450 mm*	RCC	*Indicative only; soil report required
F-02	Combined footing	RCC pad	TBD	RCC	Where geometry requires
F-03	Plinth beam	Ground tie beam	250×300 mm	RCC	Tie foundations
F-04	Retaining wall	RCC wall	250–300 mm*	RCC	Drainage + waterproofing

STAIR SCHEDULE

Tag	Element	Type	Thickness	Remarks
ST-01	Main stair	RCC waist slab	150 mm*	Final reinforcement by engineer
ST-02	Landing	RCC slab	150 mm*	Support on beam/wall
ST-03	Handrail	Metal railing	TBD	Coordinate architecture

PLUMBING FIXTURE SCHEDULE

Tag	Fixture	Connection	Indicative Qty	Location	Remarks
P-01	WC	100 mm soil	4–6*	Washrooms	Confirm final plan
P-02	Wash basin	40–50 mm waste + CW	4–6*	Washrooms	HW where required
P-03	Shower	CW + HW + 50 mm waste	3–4*	Washrooms	Floor trap
P-04	Kitchen sink	CW + HW + 50 mm waste	1	Open kitchen	Bottle trap
P-05	Floor drain	75–100 mm waste	All wet areas	Washrooms/kitchen/terraces	Trap + grate
P-06	Outdoor tap	CW	3–5*	Garden/terrace	Isolation valve

WATER SUPPLY SCHEDULE

Tag	System	Pipe 25–	Material	Equipment	Remarks
CW-01	Cold main	40 mm	PPR/PEX	Tank + pump	Size by demand
CW-02	Cold branch	20–25 mm	PPR/PEX	Fixtures	Isolation valves
HW-01	Hot main	20–25 mm	PPR/PEX	Water heater	Insulate
HW-02	Hot branch	15–20 mm	PPR/PEX	Basins/showers/kitchen	Short runs
WT-01	Storage tank	TBD	HDPE/GRP	Service/roof	Capacity by occupancy
WP-01	Booster pump	TBD	—	Pump set	Head calculation required

DRAINAGE SCHEDULE

Tag	System	Pipe	Material	Component	Remarks
SP-01	Soil stack	100 mm	uPVC/HDPE	WC discharge	Vent
WP-01	Waste stack	75 mm	uPVC/HDPE	Basins/showers	Maintain gradient
VP-01	Vent stack	50–75 mm	uPVC	Drainage vent	Above roof
FD-01	Floor drain	75–100 mm	SS/uPVC	Wet areas	Trap
CO-01	Cleanout	100 mm	uPVC	Drainage	Accessible
IC-01	Inspection chamber	TBD	RCC/brick	External drainage	Final size by design

HVAC SCHEDULE

Tag	Space	Equipment	Indicative Capacity	Qty	Remarks
AC-01	Living area	Split AC	2.0–2.5 TR* 1.0–	1	Heat-load calculation
AC-02	Bedrooms	Split AC	1.5 TR*	2	Final room loads
AC-03	Gaming room	Split AC	1.5 TR*	1	Equipment load check
AC-04	Gym	Split AC	1.5–2.0 TR* 150–	1	Fresh air recommended
EF-01	Washrooms	Exhaust fan	250 mm	Each WC	External discharge
EF-02	Kitchen	Range hood	TBD	1	Dedicated duct

ELECTRICAL FIXTURE SCHEDULE

Tag L-	Fixture	Qty	Typical Location	Remarks
01 L-	LED ceiling light	As per layout	All rooms	Energy efficient
02 L-	Wall/accent light	As required	Living/terraces	Architectural
03 L-	Outdoor bollard	As required	Landscape	Weather protected
04	Outdoor wall light	As required	Balconies/gazebos	IP-rated
SO-01	13A socket	As required	Rooms	Furniture coordination
SO-02	Weatherproof socket	As required	Outdoor	IP-rated
AC-E	Dedicated AC point	Each AC	AC rooms	Dedicated circuit

ELECTRICAL DISTRIBUTION SCHEDULE

Tag DB-	Equipment	Rating	Location	Remarks
01 DB-	Main DB	TBD	Utility/entry	Final load schedule
02 DB-	Floor DB	TBD	First floor	Separate circuits
03	Outdoor DB	TBD	Service zone	Weatherproof
GEN-01	Generator provision	TBD	Service yard	Optional
PV-01	Solar PV provision	TBD	Roof	Coordinate roof

MEP EQUIPMENT SCHEDULE

Tag	Equipment	Specification	Qty	Location	Remarks
ME-01	Water storage tank	TBD by demand	1	Service/roof	Maintenance access
ME-02	Water pump	TBD by head	1–2	Pump area	Duty/standby as required
ME-03	Water heater	TBD by demand	1	Utility	Electric/solar option
ME-04	AC outdoor units	Matched to indoor	As required	External service zone	Clearance required
ME-05	Kitchen exhaust	TBD	1	Kitchen	Safe discharge

INTERACTIVE RETREAT — HUNZA

DOOR & WINDOWSCCHEDULE | GROUND+FIRSTFLOOR | Revit / ArchitecturalDocumentation Style

DOOR SCHEDULE

Tag	Door Type	Size W×H (mm)	Operation	Frame / Material	Qty	Location	Remarks
D-01	Main Entrance Door	1200 × 2400	Double leaf / glazed	Aluminium frame + glass	1	Main foyer entry	Feature entrance; large opening
D-02	Bedroom Door	900 × 2100	Single leaf	Timber / laminate	2	Ground bedrooms	Solid-core internal door
D-03	Washroom Door	750 × 2100	Single leaf	PVC/laminate	2	Ground washrooms	Moisture resistant
D-04	Kitchen Door	900 × 2100	Single leaf / glazed	Aluminium + glass	1	Open kitchen/service	Coordinate with cabinetry
D-05	Store Door	800 × 2100	Single leaf	Flush timber	1	Ground store	Solid core
D-06	Studio Door	900 × 2100	Single leaf / glazed	Aluminium + glass	1	Ground studio	Visual connection
D-07	Terrace / Balcony Door	1800 × 2400	Double sliding	Aluminium + double glazing	3	Ground/first terraces	Large indoor-outdoor opening
D-08	Bedroom Door	900 × 2100	Single leaf	Timber / laminate	2	First-floor bedrooms	Solid-core internal door
D-09	Washroom Door	750 × 2100	Single leaf	PVC/laminate	2	First-floor washrooms	Moisture resistant
D-10	Gaming Room Door	900 × 2100	Single leaf	Timber / laminate	1	First-floor gaming room	Solid-core
D-11	Gym Door	900 × 2100	Single leaf / glazed	Aluminium + glass	1	First-floor gym	Safety glass
D-12	Service / Utility Door	800 × 2100	Single leaf	Metal / flush	1	Service zone	Coordinate with MEP

WINDOW SCHEDULE

Tag	Window Type	Size W×H (mm)	Operation	Frame / Glazing	Qty	Location	Remarks
W-01	Large Living Window	2400 × 1800	Fixed + sliding	Aluminium + DGU	2	Living area	Panoramic view
W-02	Bedroom Window	1800 × 1500	Sliding	Aluminium + DGU	4	Bedrooms	Mountain/view orientation
W-03	Bedroom Side Window	1200 × 1500	Sliding / fixed	Aluminium + DGU	2	Bedrooms	Cross ventilation
W-04	Washroom Window	600 × 600	Awning / top-hung	Aluminium + frosted glass	4	Washrooms	Privacy glazing
W-05	Kitchen Window	1200 × 1000	Sliding / fixed	Aluminium + DGU	1	Open kitchen	Above counter
W-06	Studio Window	1800 × 1500	Sliding	Aluminium + DGU	1	Studio	Natural light
W-07	Gym Window	1800 × 1500	Sliding	Aluminium + DGU	2	Gym	Ventilation + daylight
W-08	Gaming Room Window	1800 × 1500	Sliding	Aluminium + DGU	1	Gaming room	View + daylight
W-09	Lobby / Stair Window	1200 × 1800	Fixed + operable	Aluminium + DGU	2	Lobby/stair Service/wet	Daylight
W-10	High-Level Window	900 × 600	Awning	Aluminium + frosted/DGU	2	zones First-floor living	Controlled ventilation
W-11	First-Floor Living Window	2400 × 1800	Sliding/fixed	Aluminium + DGU	2	First-floor terrace/living	Panoramic view
W-12	Terrace Full-Height Glazing	3000 × 2400	Sliding/folding	Aluminium + DGU	2		Indoor-outdoor connection

DOOR & WINDOW TYPE SUMMARY

Type	Typical Size (W×H)	Frame	Glazing/Leaf	Primary Use
D-01	1200×2400	Aluminium	Double glazed / double leaf	Main entrance
D-02/D-08	900×2100	Timber	Solid single leaf	Bedrooms
D-03/D-09	750×2100	PVC/laminate	Solid single leaf	Washrooms
D-07	1800×2400	Aluminium	DGU sliding	Terraces/balconies
W-01/W-11	2400×1800	Aluminium	DGU fixed/sliding	Living
W-02/W-03	1800×1500 / 1200×1500	Aluminium	DGU sliding/fixed	Bedrooms
W-04	600×600	Aluminium	Frosted awning	Washrooms
W-12	3000×2400	Aluminium	DGU sliding/folding	Terrace glazing

GENERAL SCHEDULE NOTES

01	Schedule is coordinated conceptually with the supplied Ground Floor and First Floor architectural plans.
02	Dimensions are indicative design values where the supplied plan graphics do not provide a readable family/type dimension.
03	Exact quantity, handing, opening direction and family dimensions should be checked against the original Revit model before construction documentation.
04	Use double glazing / insulated glazing as appropriate for Hunza's cold climate; final U-value and glass specification should be based on energy analysis.
05	Use safety/toughened glass at large glazed openings, doors and areas adjacent to floor level.
06	Provide weather seals, sill flashing and proper waterproofing at all external openings.

INTERACTIVE RETREAT – HUNZA

ENERGY MODEL (REVIT)

This is an analytical energy model used for energy simulation and analysis in Autodesk Revit. The model represents the thermal envelope of the building including walls, roofs, floors, windows and doors. It is used to evaluate the energy performance, cooling & heating loads, and overall building efficiency.

ENERGY MODEL LEGEND

- Analytical Exterior Surfaces
- Analytical Interior Surfaces (Adjacent to Unconditioned Spaces)
- Glazed Surfaces (Windows / Doors)
- Model Boundary (Analytical Envelope)

ENERGY MODEL NOTES

- The energy model is simplified for simulation purposes.
- Only conditioned spaces are included in the analytical model.
- Courtyards, terraces and open spaces are considered as unconditioned / exterior.
- Window to Wall Ratio is optimized for daylight and energy performance.
- Roofs are modeled as insulated sloped surfaces.

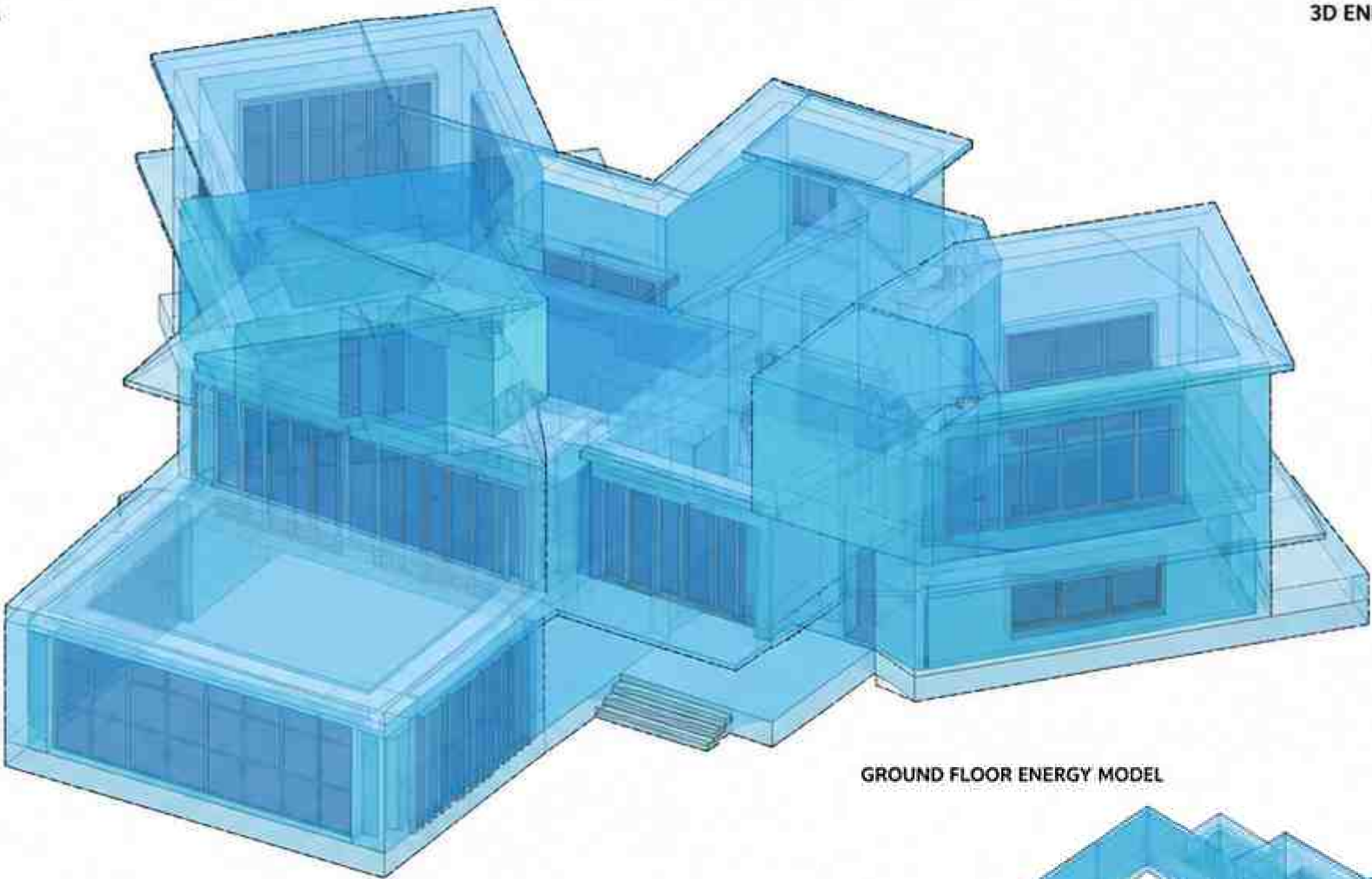
<Analytical Spaces>

A	B	C	D	E
Area	Count	Number	Name	Volume
318 SF	1	1-1	Analytical Space	1310.93 CF
162 SF	1	1-3	Analytical Space	1276.88 CF
396 SF	1	1-2	Analytical Space	2208.08 CF
45 SF	1	1-4	Analytical Space	299.87 CF
63 SF	1	1-5	Analytical Space	509.07 CF
121 SF	1	2-2	Analytical Space	705.36 CF
29 SF	1	2-1	Analytical Space	415.52 CF
1134 SF				6725.70 CF

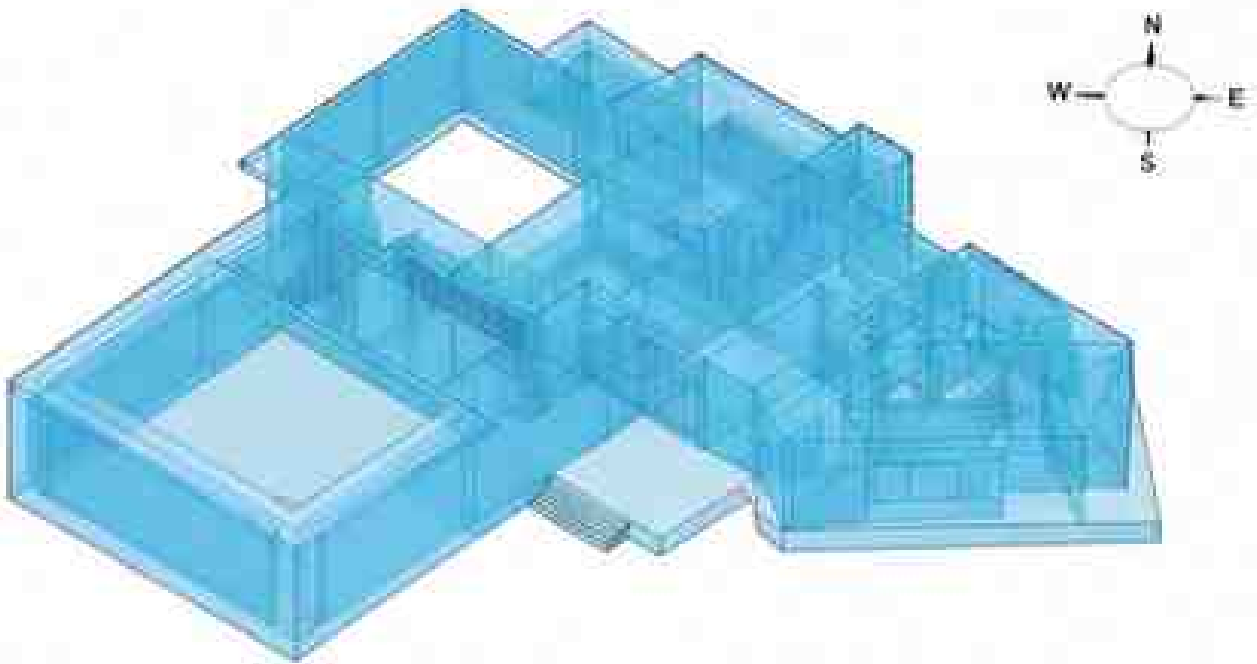
<Analytical Surfaces>

A	B	C	D
Area	Count	Opening Type	Surface Type
27 SF	2	Non-sliding Door	
310 SF	9		Interior Wall
10798 SF	114		Shade
150 SF	5		Underground Ceiling
1927 SF	12		Underground Slab
1893 SF	23		Underground Wall
15104 SF			

3D ENERGY MODEL VIEW

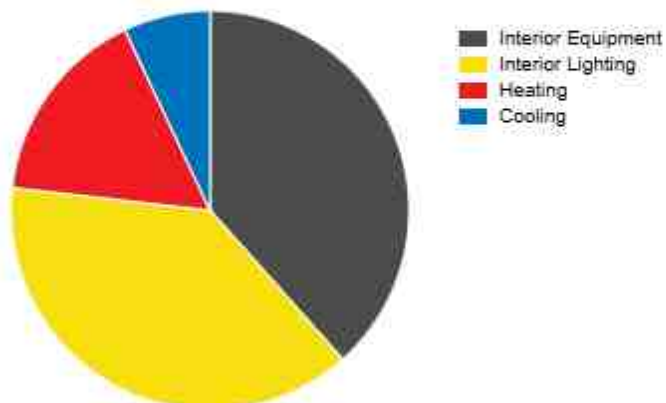


GROUND FLOOR ENERGY MODEL

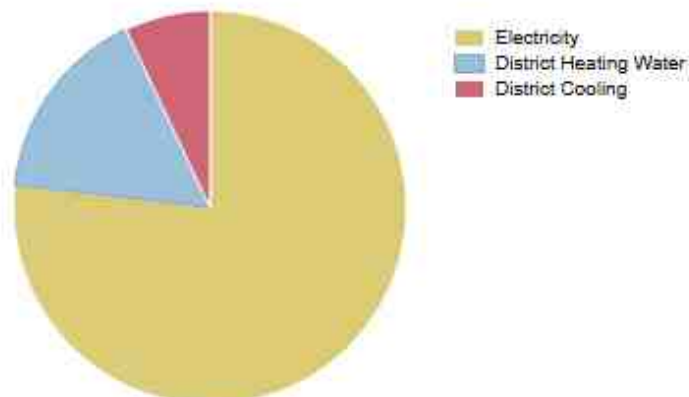


SYSTEM ANALYSIS REPORT ANNUAL REPORT

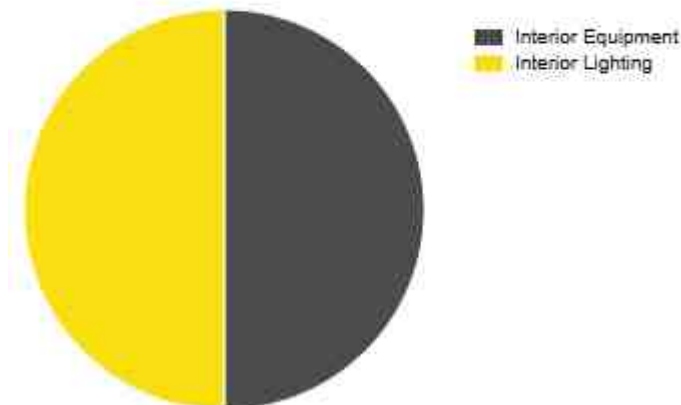
End Use - view table



Energy Use - view table



EUI - Electricity - view table



Text

About this insight

Insight scope: Envelope and Interiors subset of architectural elements based on the data included in the Energy Analytical Model
Total carbon analysis scope: manufacturing stage (A1 - A3) + Operational energy (B6)

Building Location

Base Model

Ladakh

Building Type

Base Model

Single Family

Building Area

Base Model

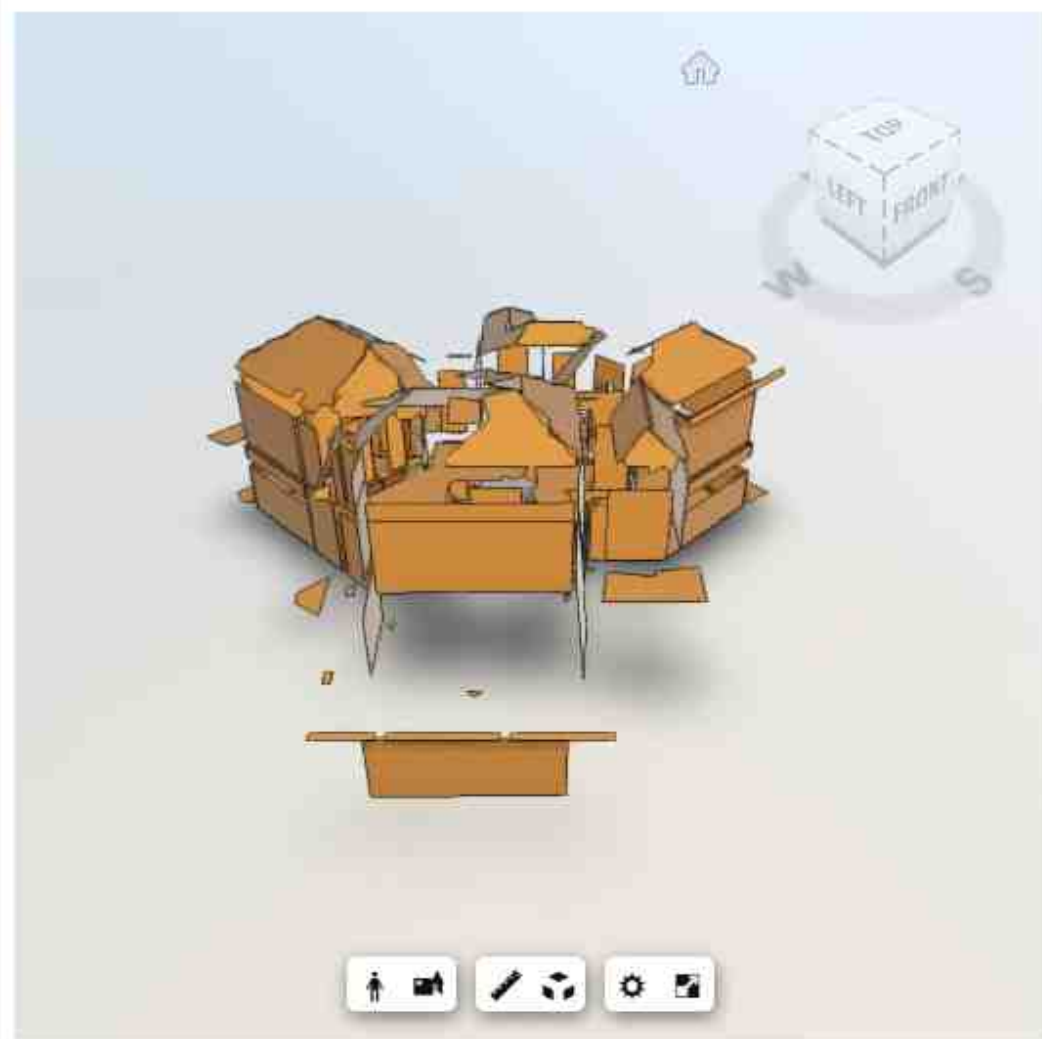
27.74 m²

Building Lifespan

10 years

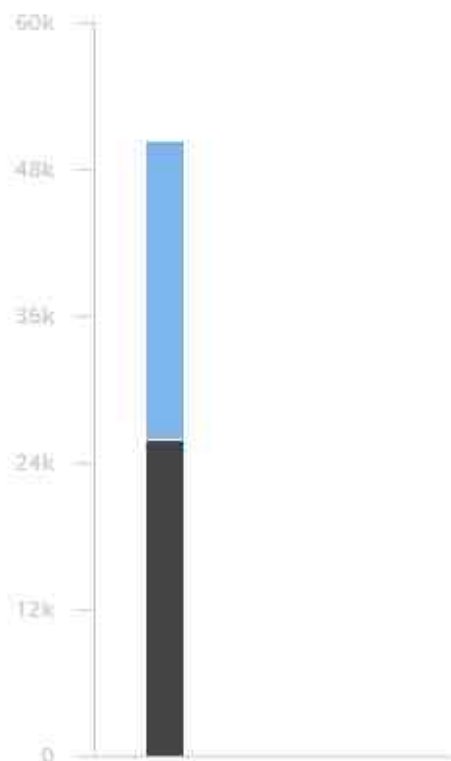
Medium: 10 years

2026 UPDATED MODEL RETREAT



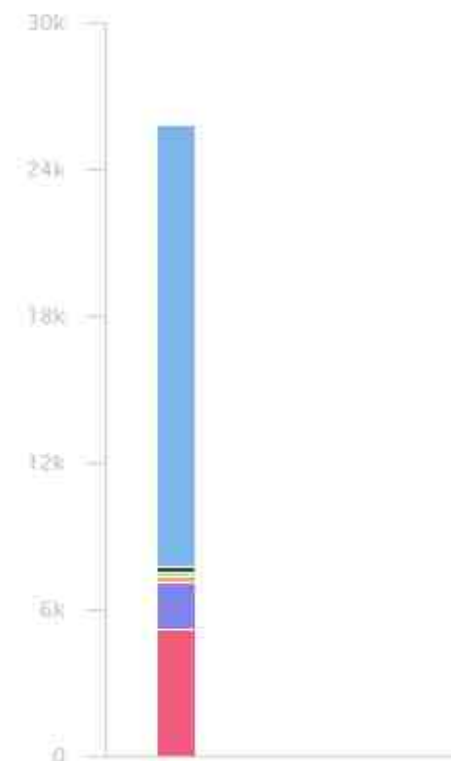
Total Carbon

50,263.9 kgCO₂e



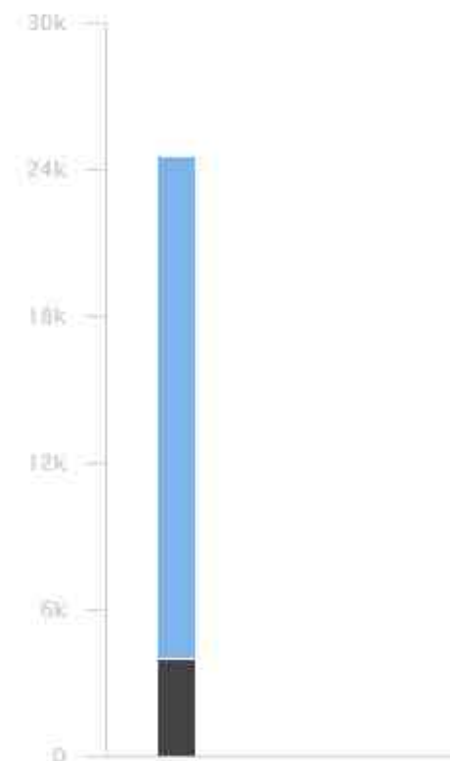
Embodied Carbon

25,764.45 kgCO₂e



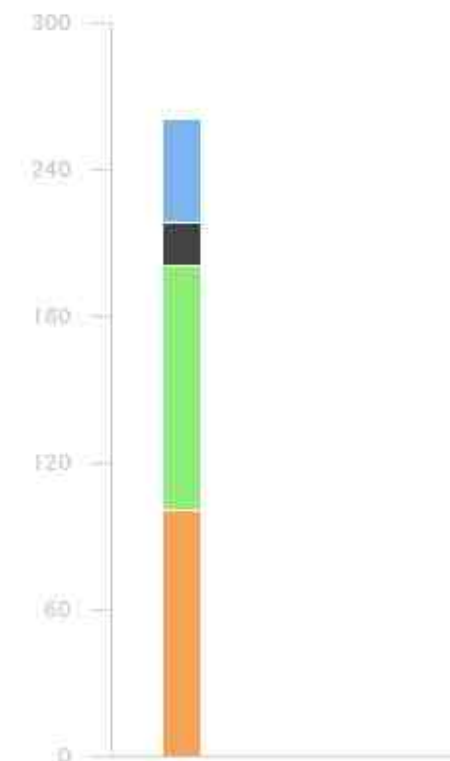
Operational Carbon

24,499.44 kgCO₂e

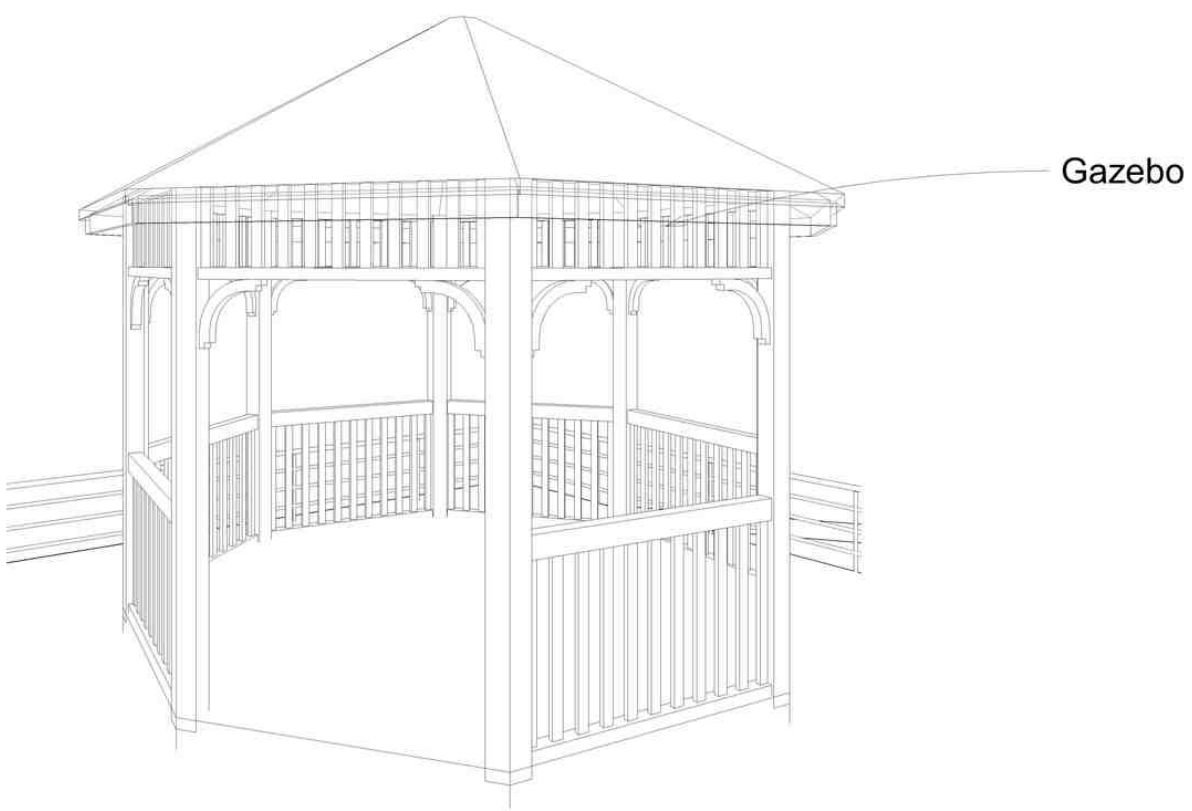
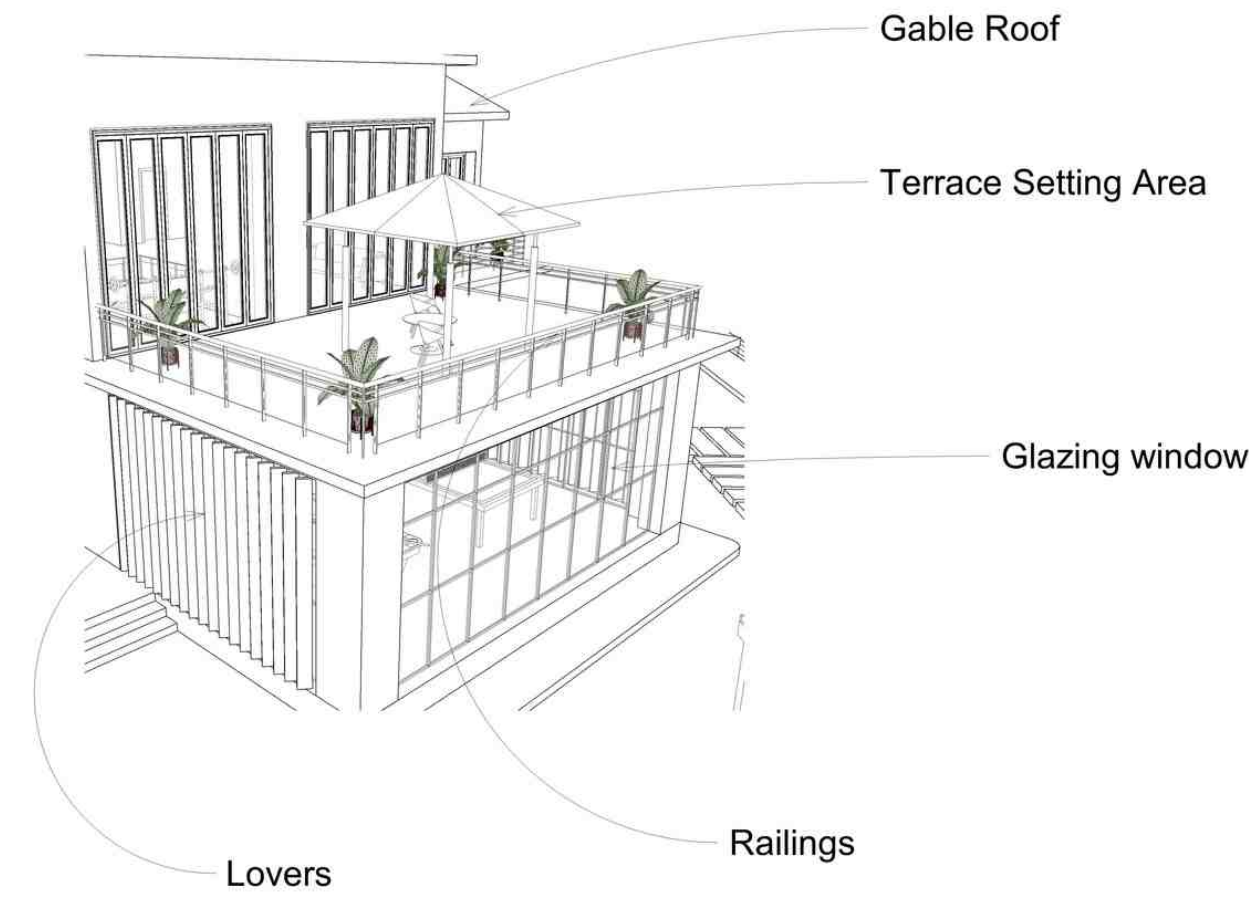


Annual Energy Use Intensity

260.44 kWh/m²



BLOWUPS



RENDERS



YEAR:
2025

LOCATION :
LAHORE

TYPE : HUB

03

RISING THE TECH

Crafting Future Hubs for AI, Creativity, and Technological Growth"

SITE LOCATION: LAHORE NEAR TO QAZAFI STADIUM

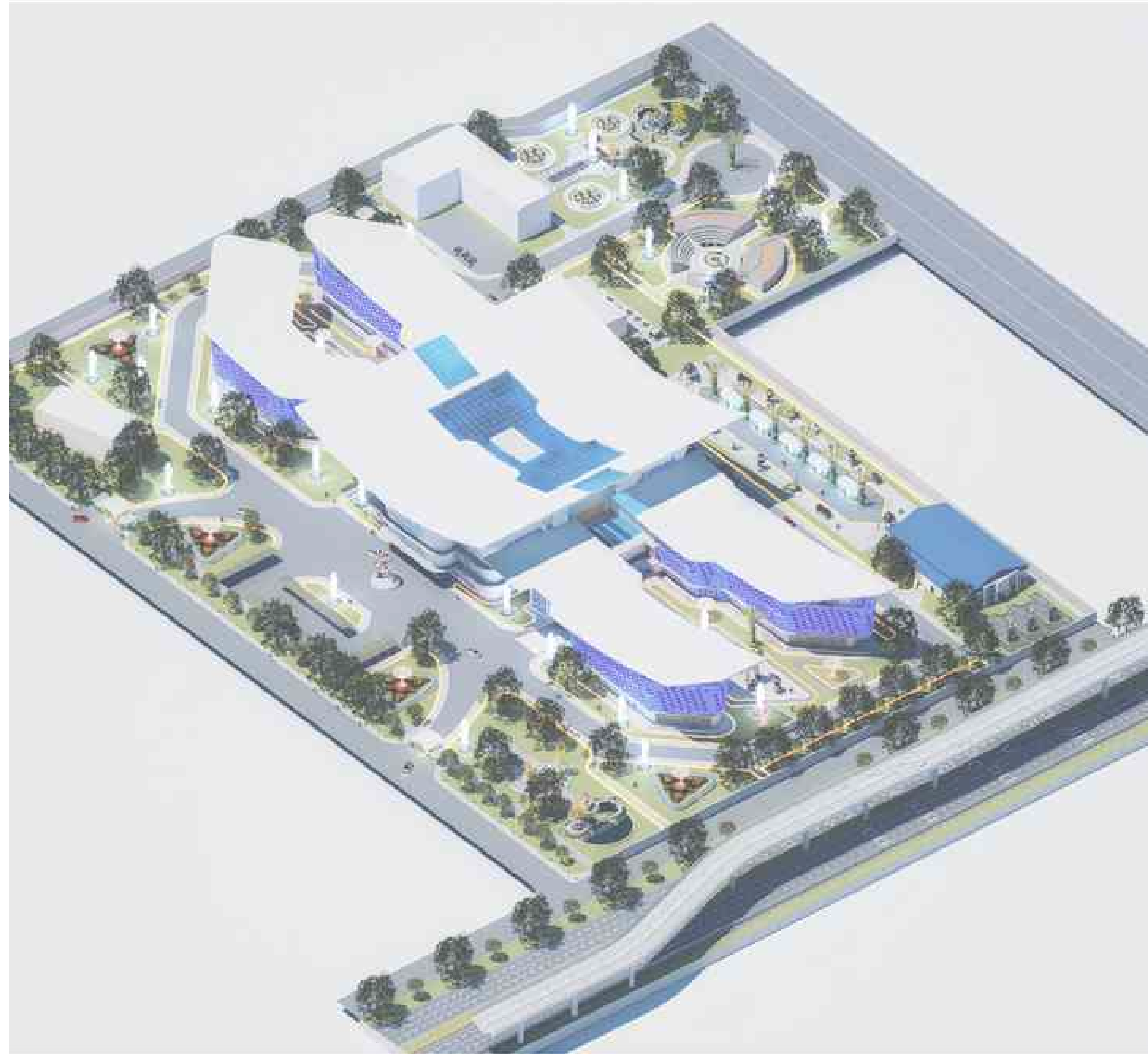
AREA : 115 KANAL

PROJECT: 2024

PROJECT TYPE : FOSTERING TECHNOOGICAL GROWTH AND INNOVATIONS

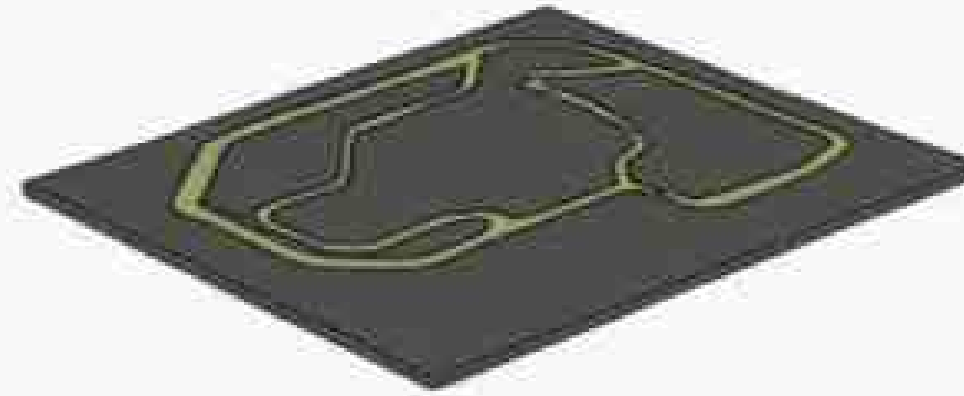
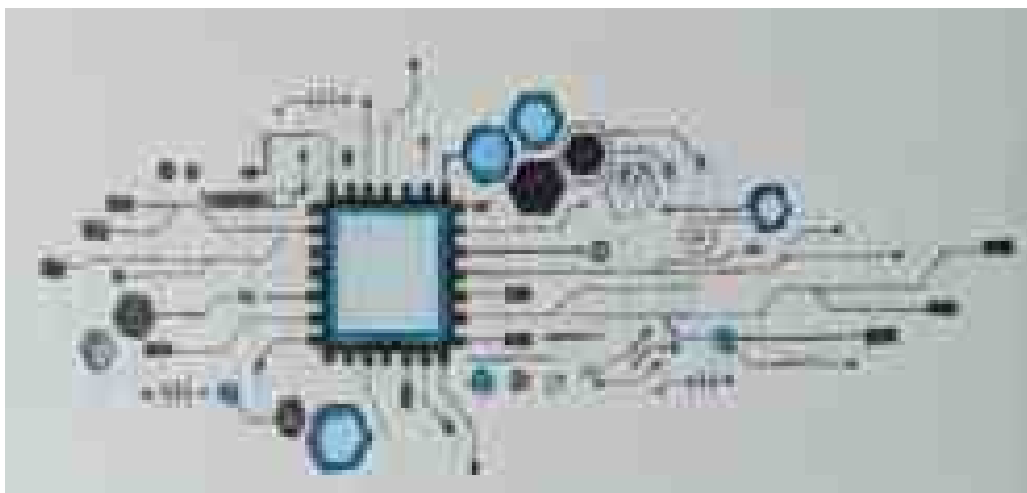
“Rising the Tech” — a Techno Innovation Hub — envisions architecture as a bridge between technology, experience, and emotion, creating interactive spaces that spark innovation and human connection. The project focuses on fostering technological growth, entrepreneurship, and creative collaboration in Pakistan by providing an inspiring and resource-rich environment where young innovators can transform ideas into impactful tech solutions. Through phenomenological and sensory design, the hub promotes innovation not just as a process, but as an immersive architectural experience that embodies the spirit of progress and interaction.

Rising The Tech : Crafting Future Hubs for AI, Creativity, and Technological Gro



DESIGN CONCEPT

The form of my Techno Innovation Hub is inspired by the structured flow of circuit lines, reflecting the essence of technology and smart connectivity. I used vertical, horizontal, and diagonal lines to shape a dynamic layout that encourages movement, interaction, and flexibility. Just like a circuit connects components to perform intelligently, the building's form links spaces to foster collaboration, creativity, and innovation—making the architecture itself a symbol of technological synergy.



2. Programmatic Zoning

- Allocate central block for circulation based on their importance
- Vary heights to define hierarchy (e.g. auditorium, gymnasium, arms, etc.)
- Integrate internal courtyard/ or plazas for social interaction

📍 **Result:** 'H' or 'cross shaped' layout emerges with a strong central core and functional arms.



5. Integration with Landscape

- Soften building edges with green buffers and landscape mounds
- Add courtyards, plazas, seating areas and water elements
- Urban furniture and lighting

📍 **Result:** Architecture and nature merged in a balanced environment.

1. Site Context Analysis

- Analyze natural site boundaries and access roads
- Identify central node for circulation
- Define green buffer zones around site

📍 **Result:** Site divided into zones—central core, surrounding public space, parking, and landscape zones.



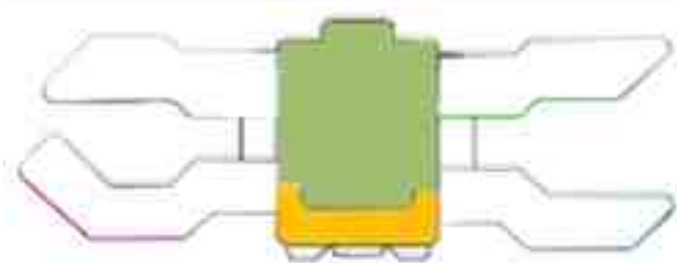
4. Circulation and Looping

- Add vehicular loop around site perimeter
- Include drop-off points near main entrance and services zones
- Pedestrian paths and green loops enhance user movement experience.

📍 **Result:** Clear, efficient flow for both vehicles and pedestrians.



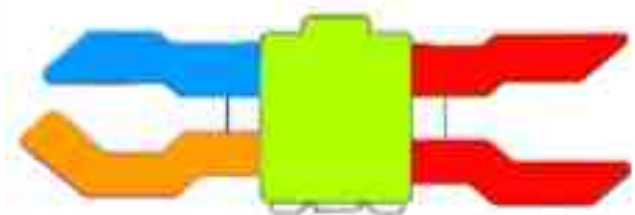
ZONING



ENTERTAINMENT ZONE
ADMIN ZONE



BUSINESS ZONE
INNOVATIVE ZONE
COMMUNITY AND SOCIAL ZONE
R & D ZONE



TRAINING AND EDUCATION ZONE
AI ZONE
R & D ZONE
COLLABORATION ZONE

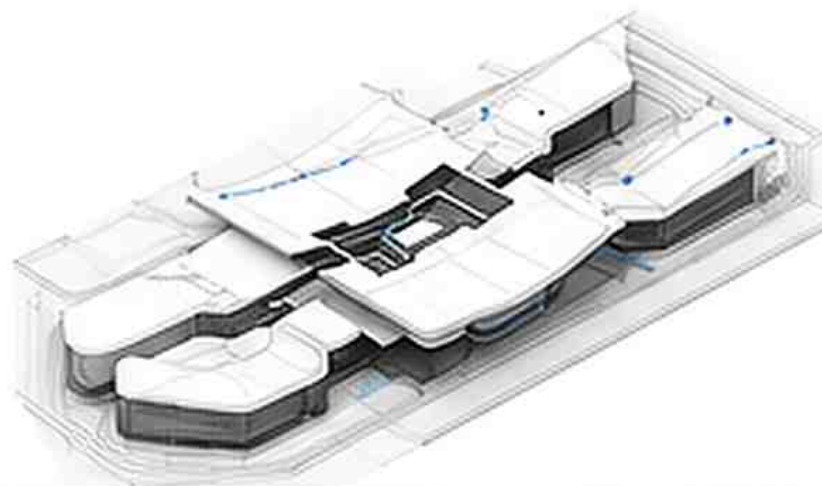


MOBILITY, SPORTS AI & ROBOTICS ZONE
AI COMMUNICATION CENTER ZONE
R & D ZONE
COLLABORATION ZONE

CONCEPT DIAGRAMS

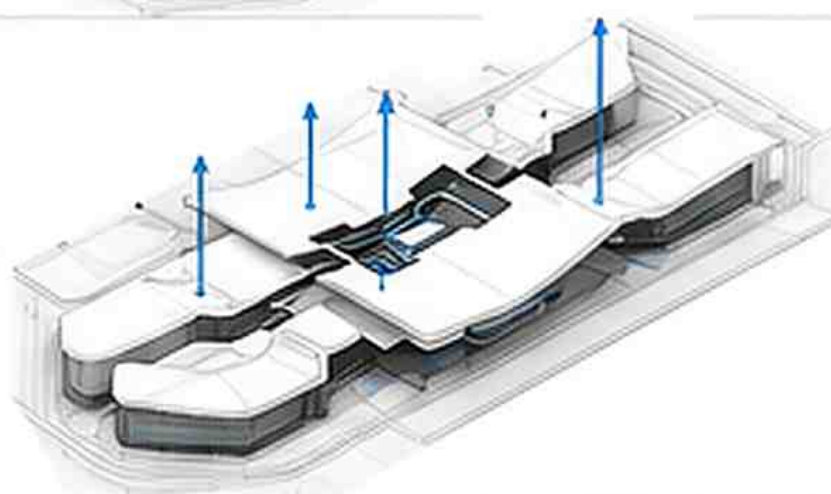
1. SITE & CONTEXT

The building is positioned to maximize visibility and accessibility while responding to the surrounding urban fabric and circulation.



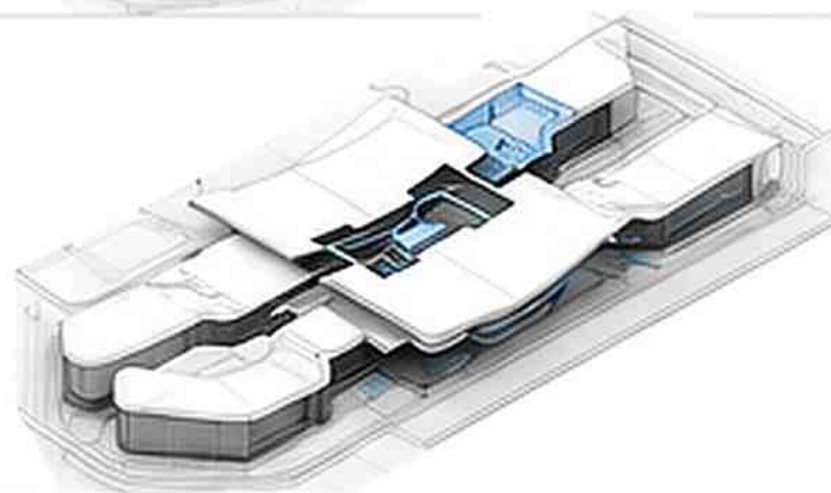
2. MASSING

The form is derived from the site's geometry and functional requirements, creating a fluid and dynamic mass that defines public and private zones.



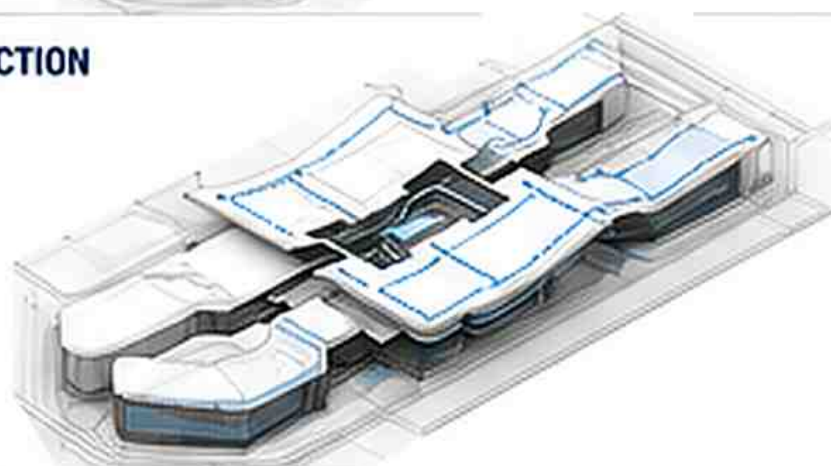
3. FORM DEVELOPMENT

The mass is sculpted to create courtyards and atriums for natural light, ventilation and spatial connectivity.



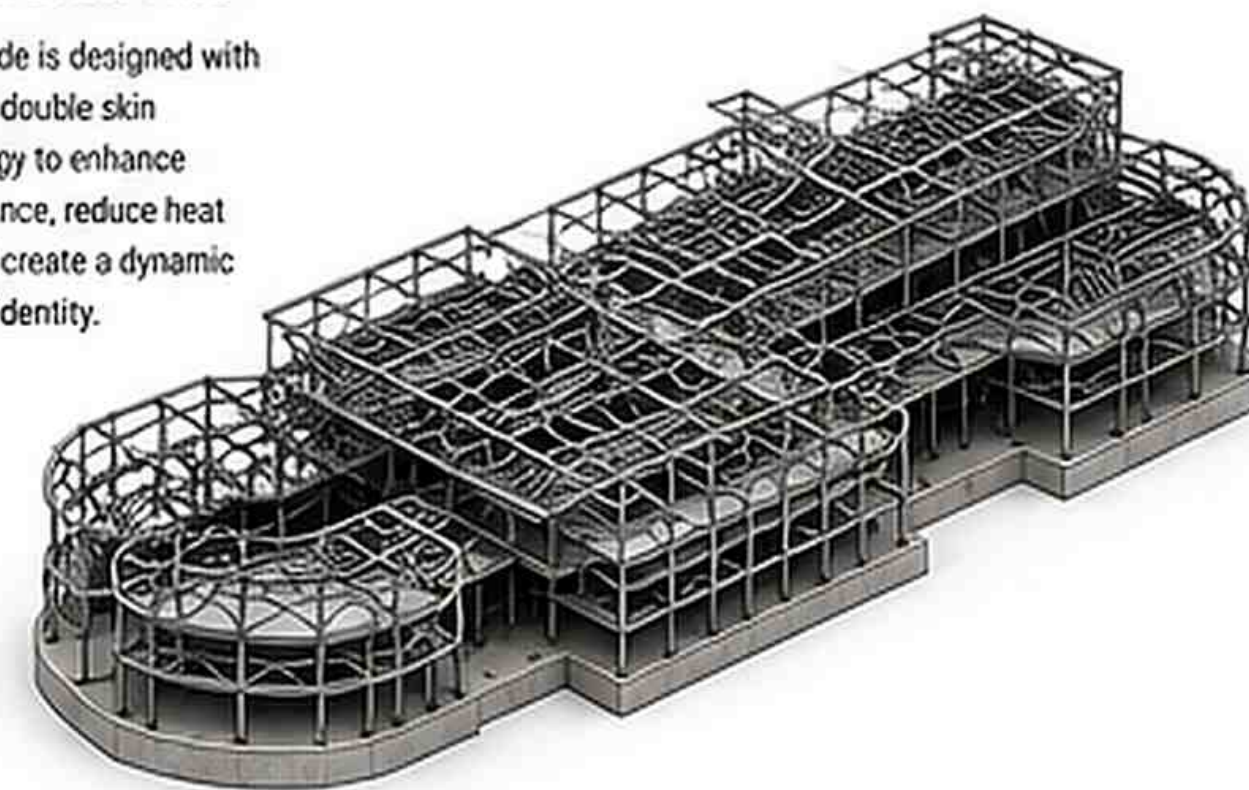
4. CIRCULATION & CONNECTION

Internal circulation flows through atriums and bridging elements, connecting different functions seamlessly.



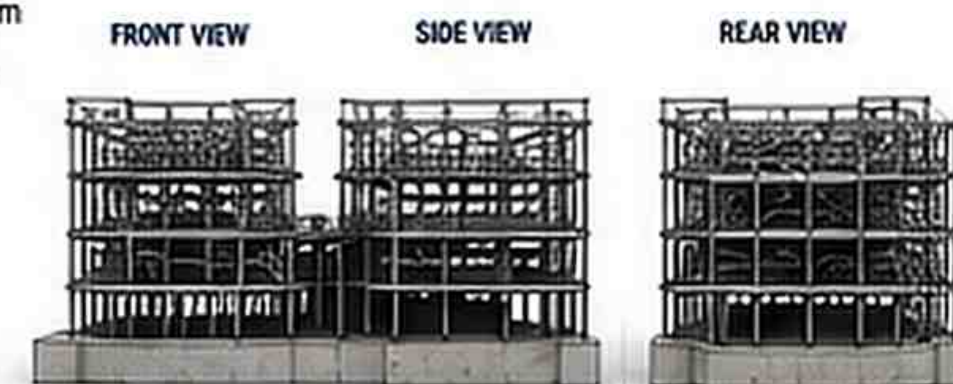
5. FACADE RESPONSE

The facade is designed with 3D glass double skin technology to enhance performance, reduce heat gain and create a dynamic building identity.

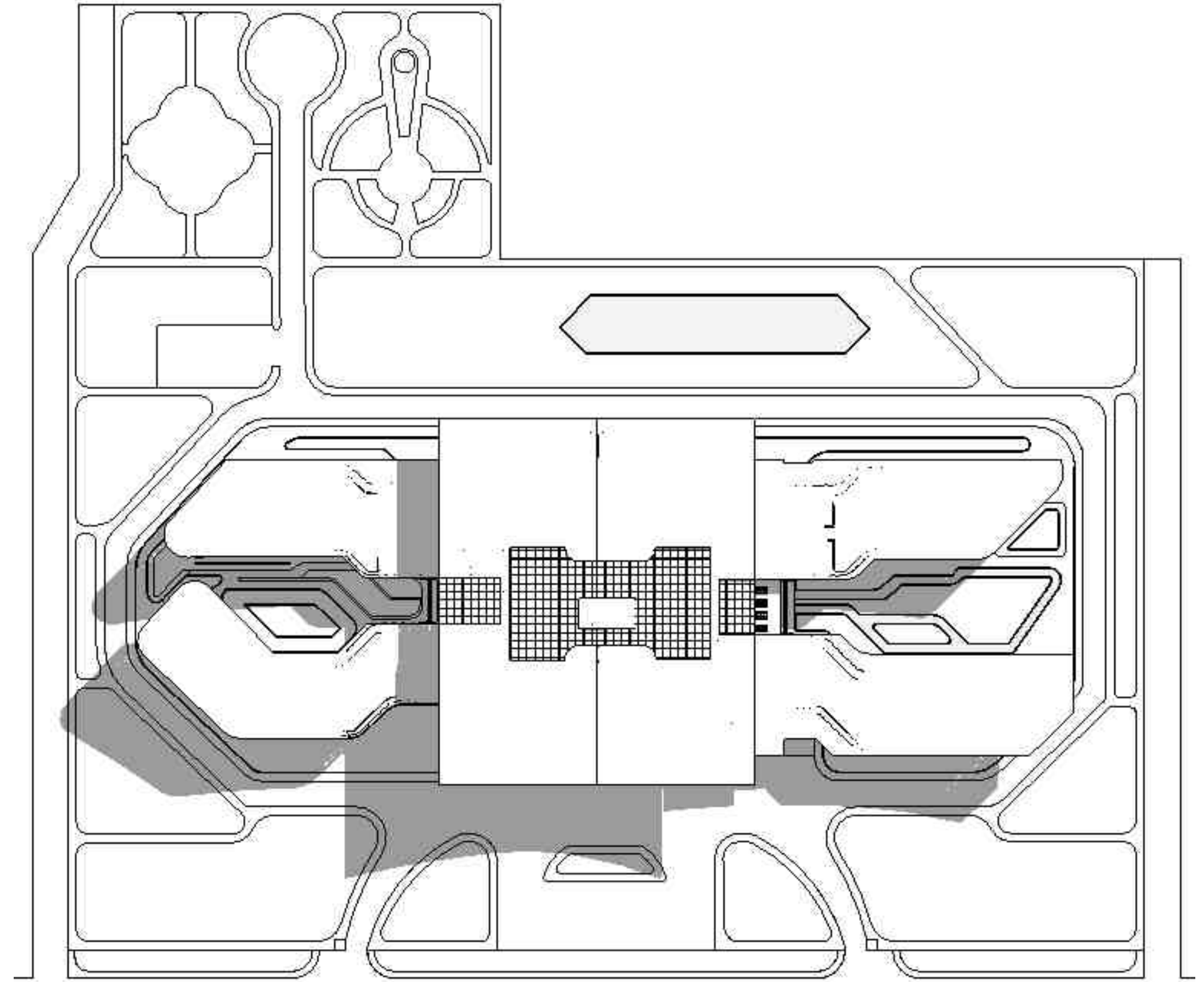
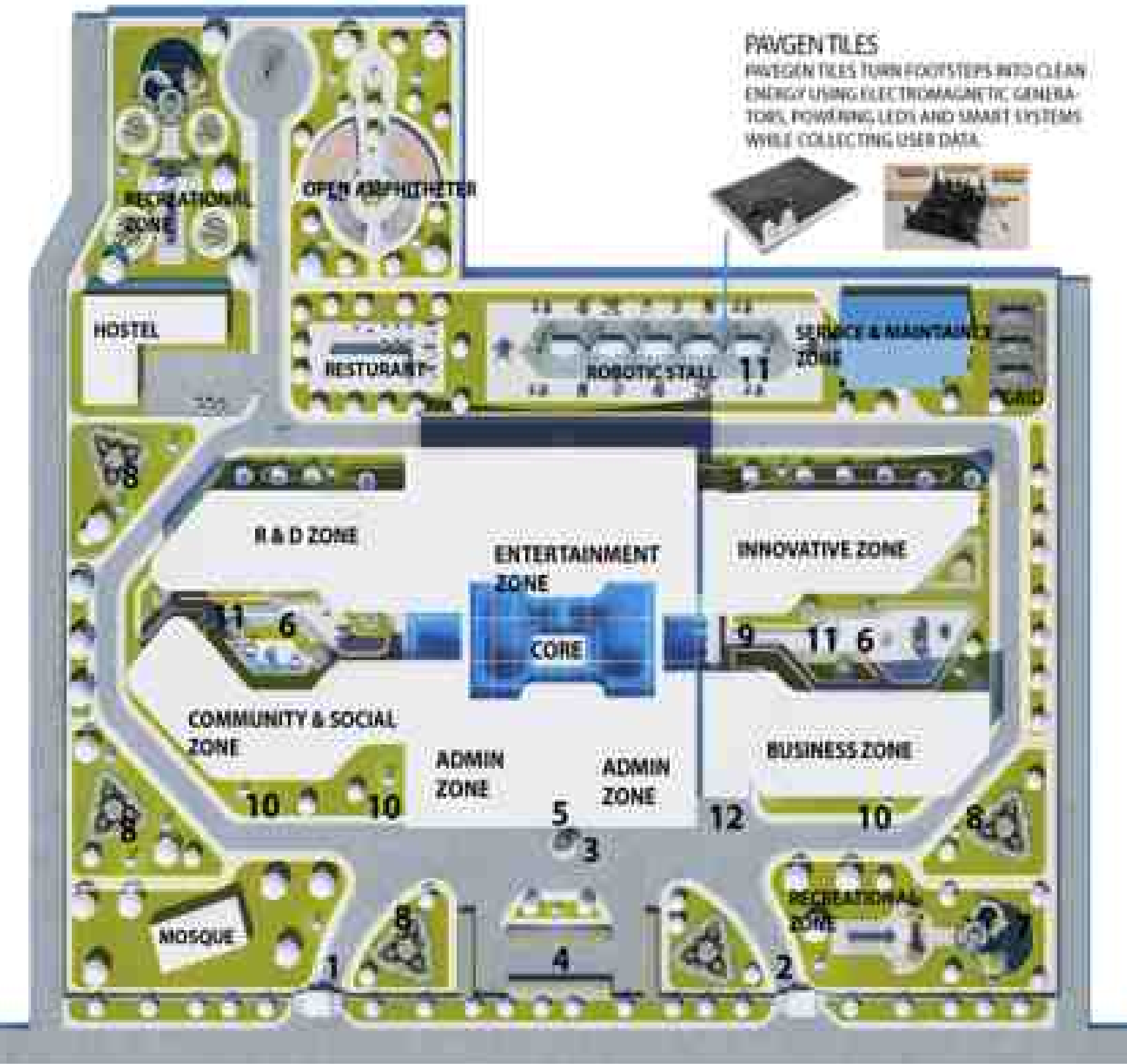


6. FINAL CONCEPT

A dynamic architectural form that integrates technology, sustainability and user experience in a cohesive design.



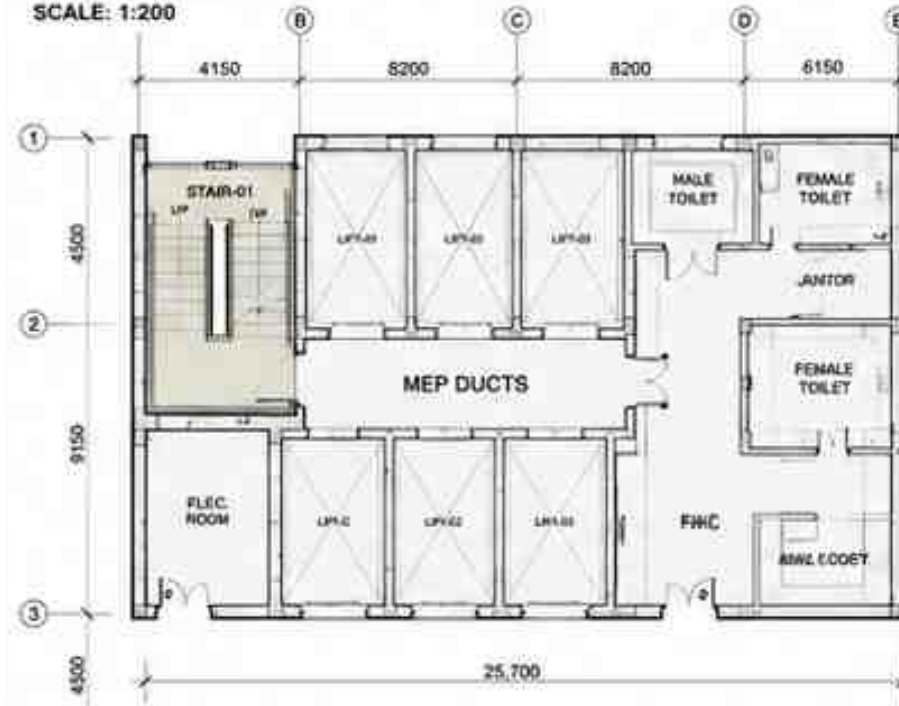
MASTER PLAN



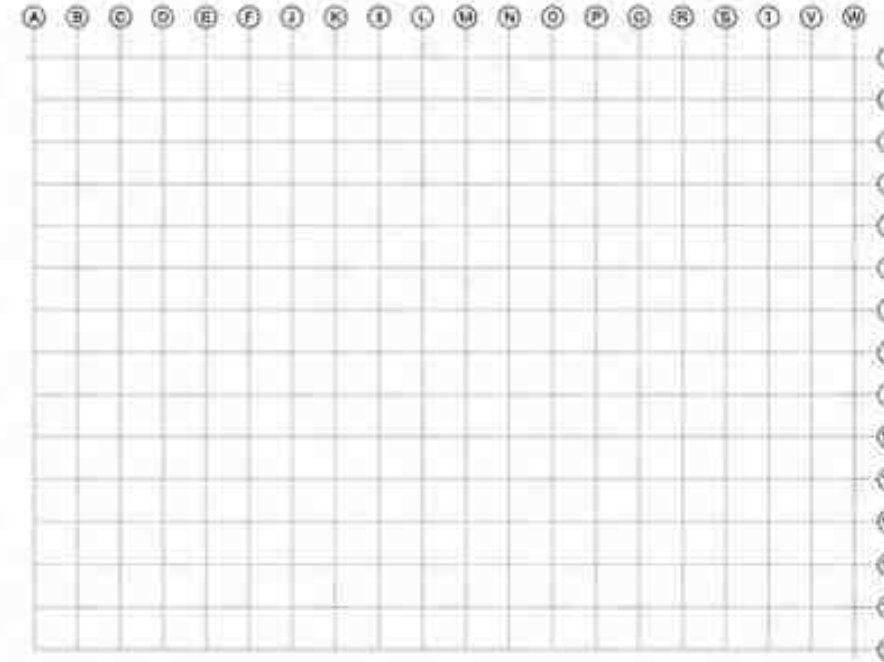
PLANS

DRAWINGS

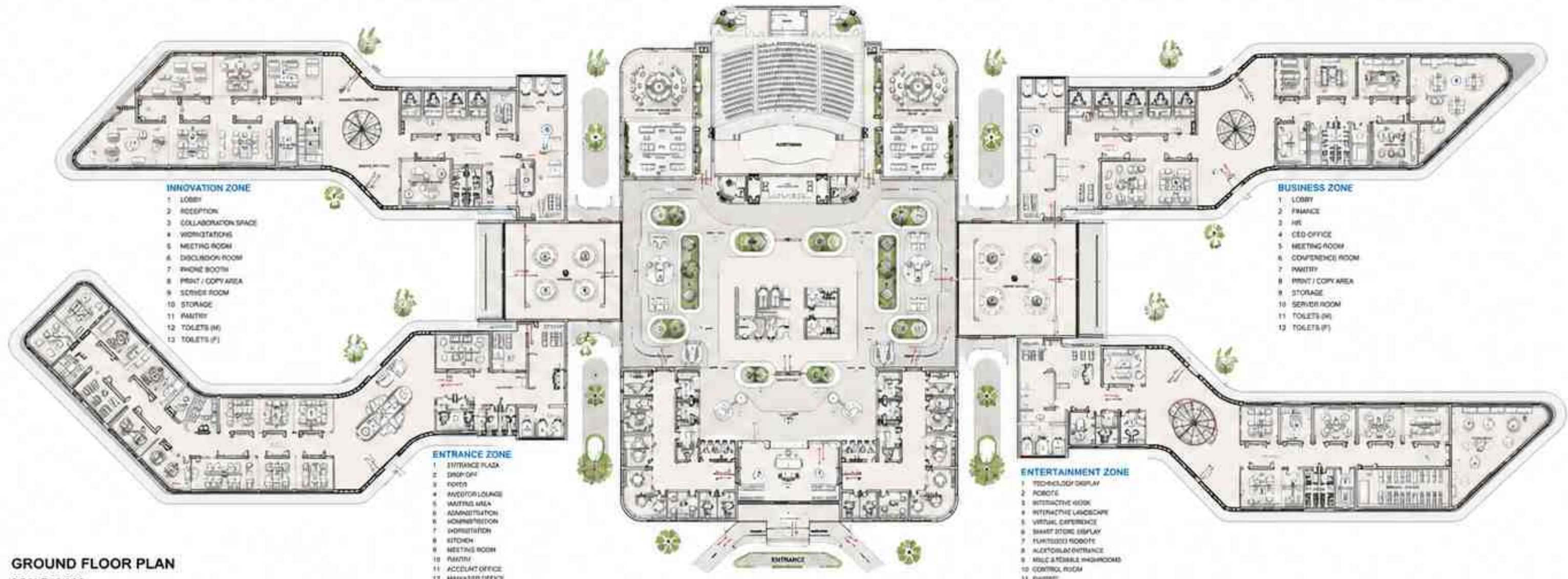
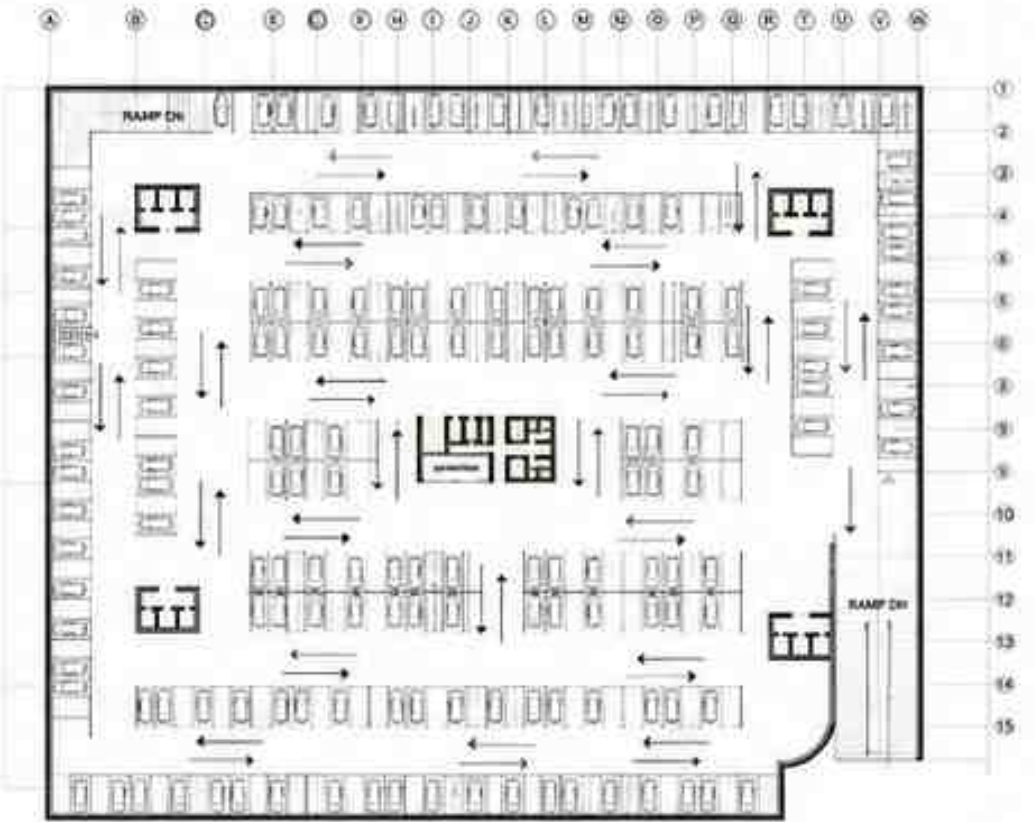
CORE DESIGN PLAN
SCALE: 1:200

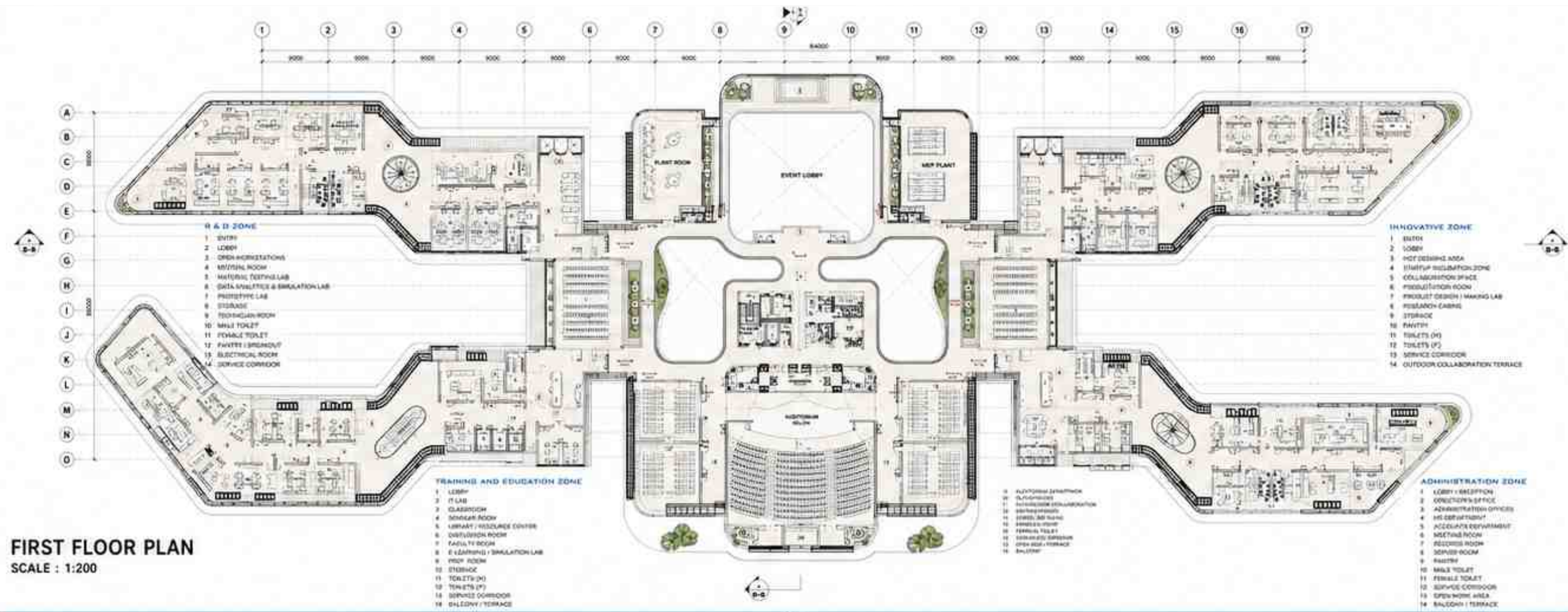


GRID PLAN
SCALE: 1:200

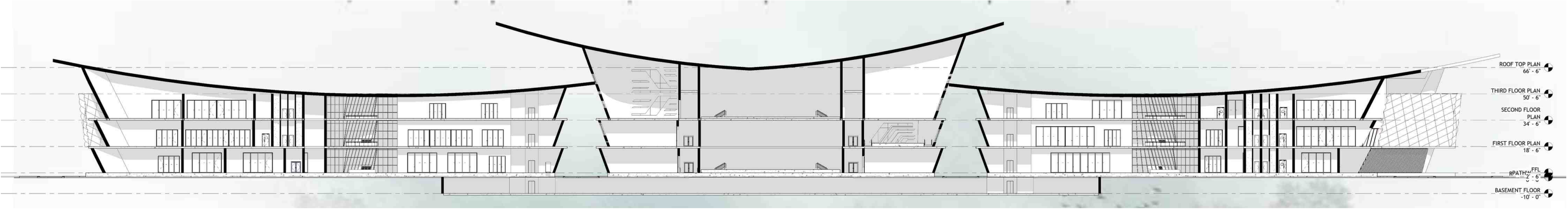


BASEMENT FLOOR PLAN
SCALE: 1:200

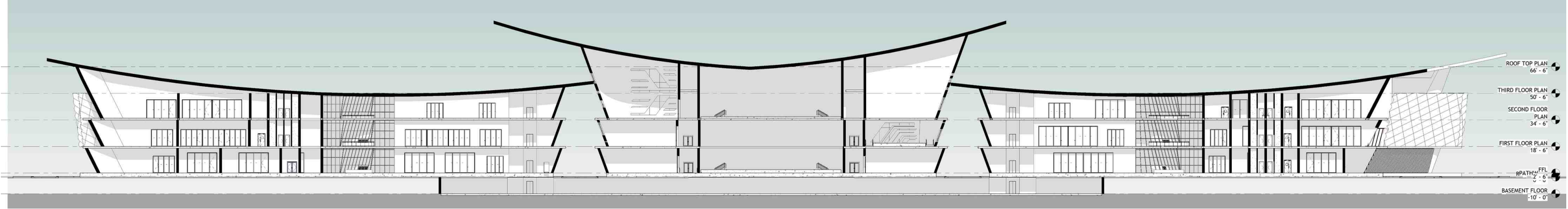




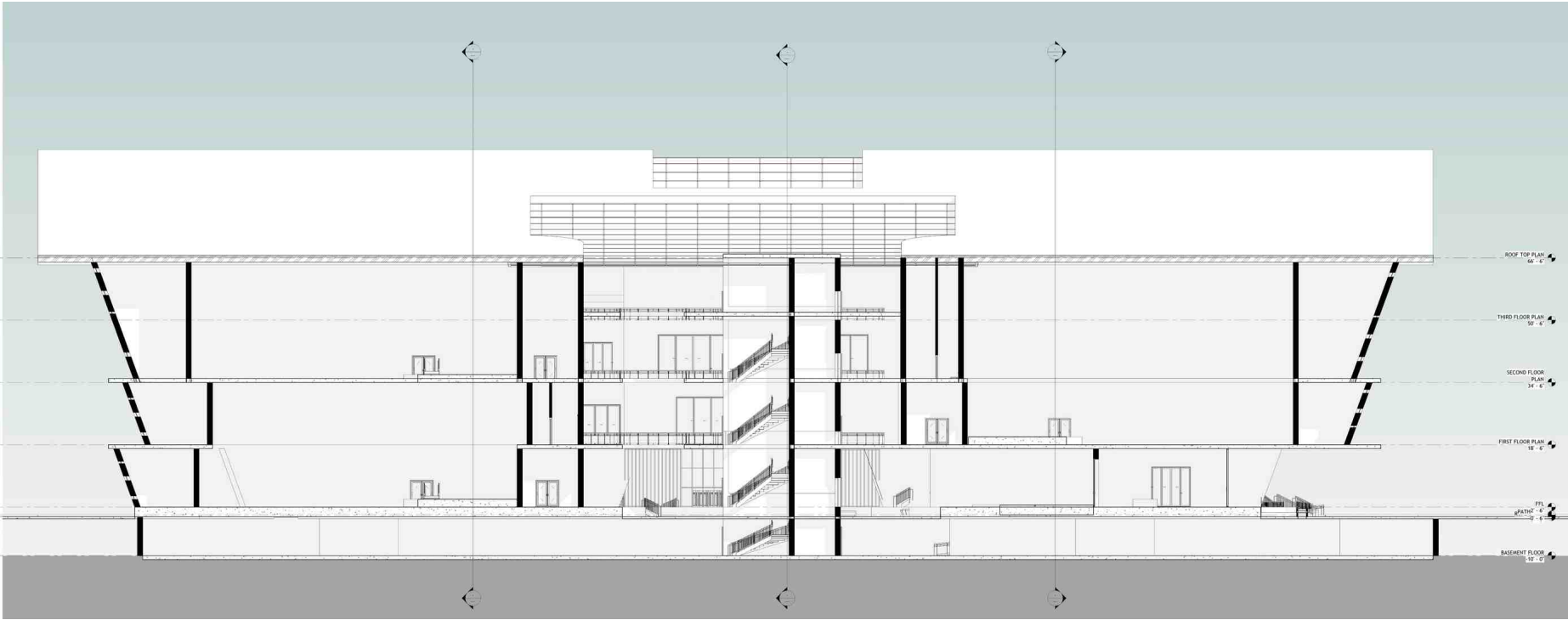
SECTIONS



SECTION AA

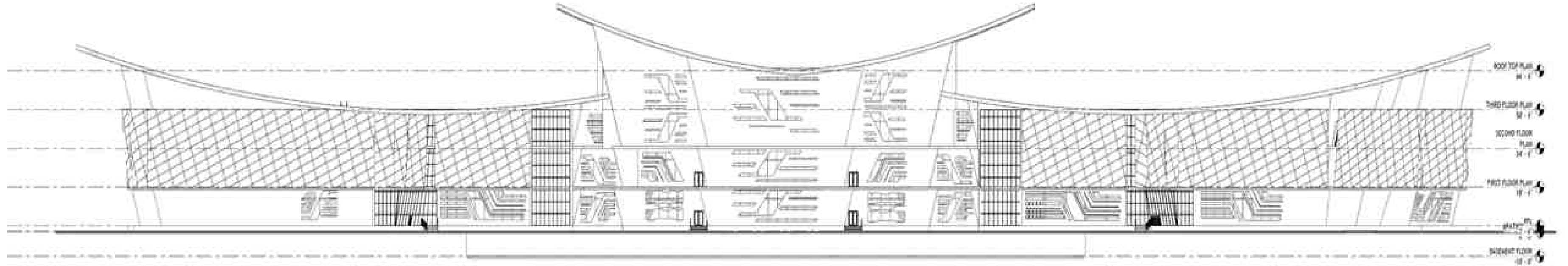


SECTION BB

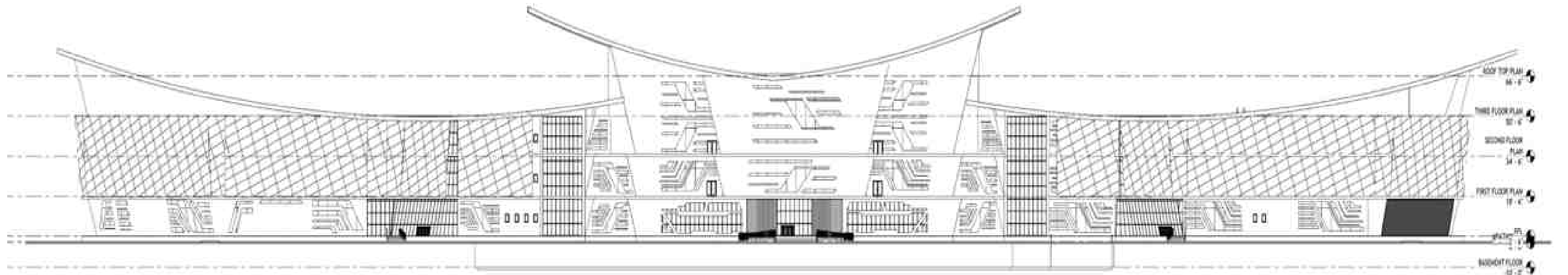


SECTION CC

ELEVATIONS

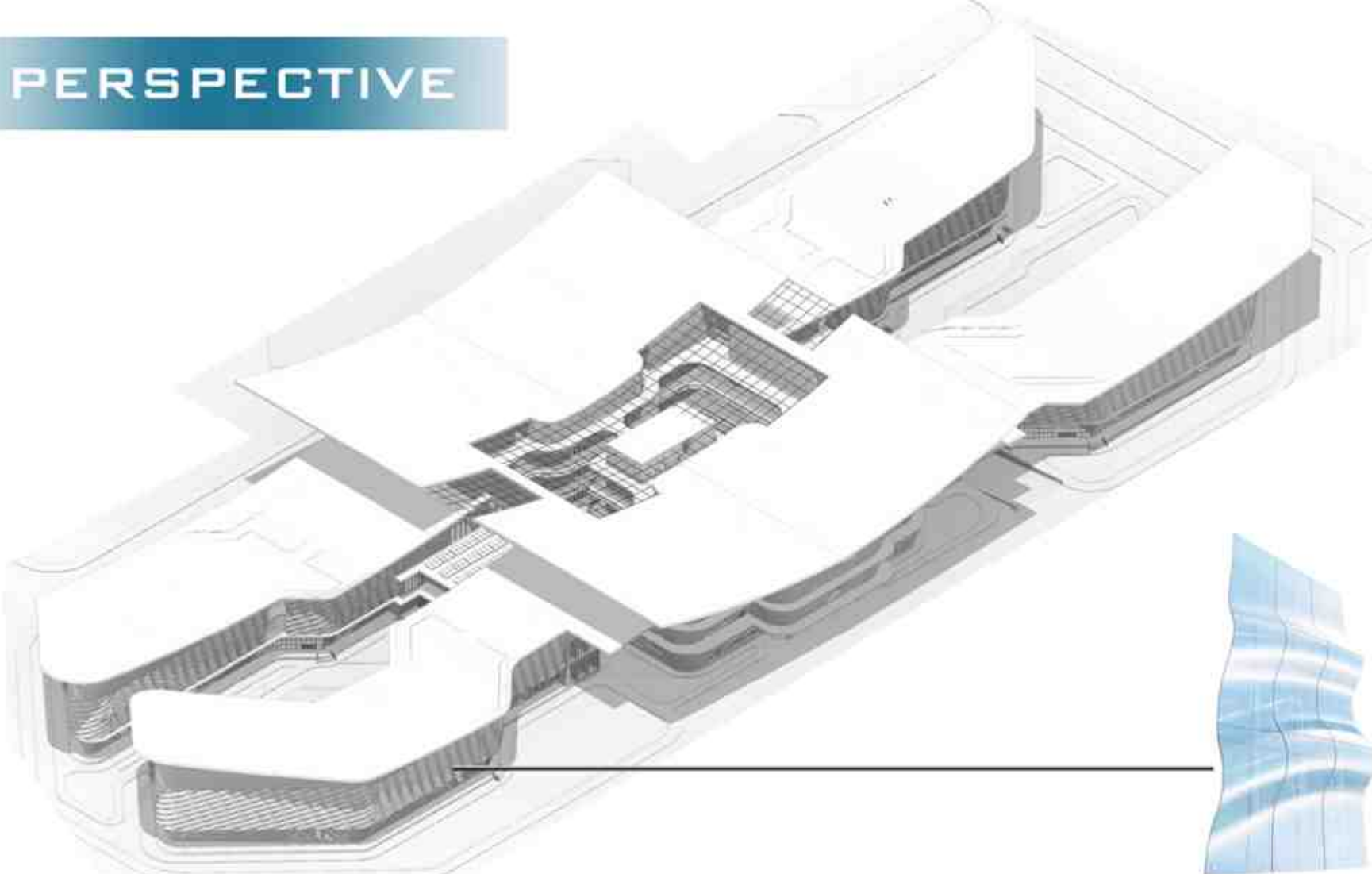


NORTH ELEVATION
SCALE : 1/16"=1'=0"



SOUTH ELEVATION
SCALE : 1/16"=1'=0"

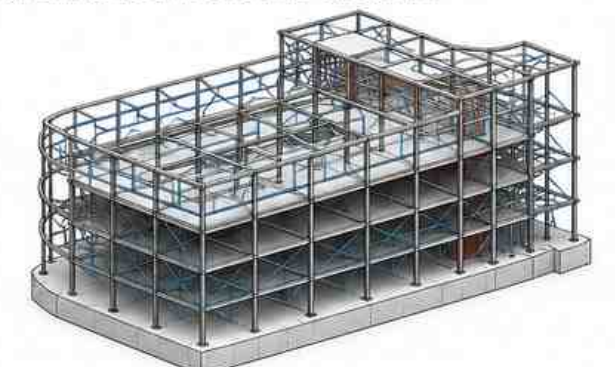
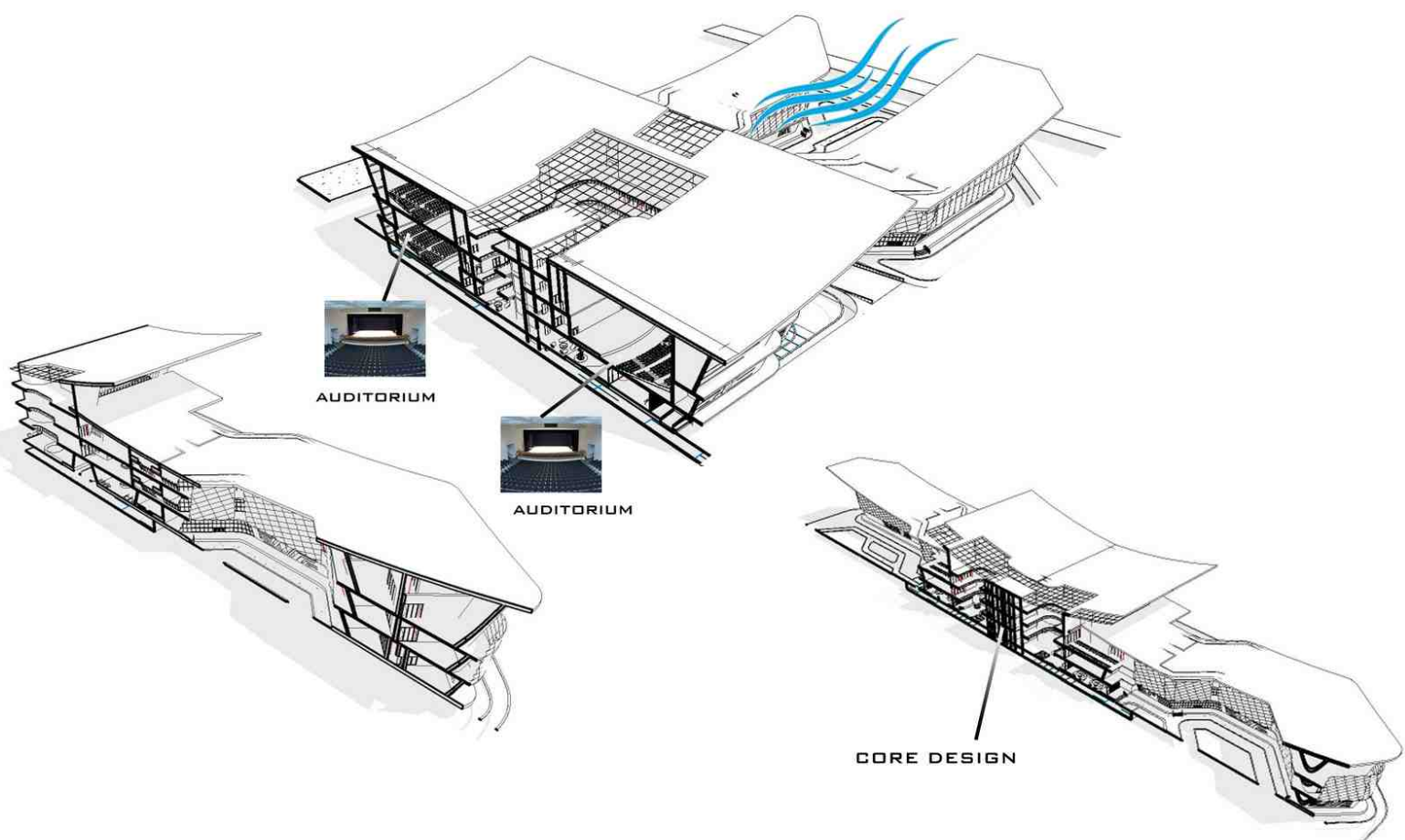
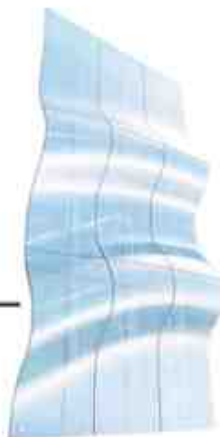
PERSPECTIVE



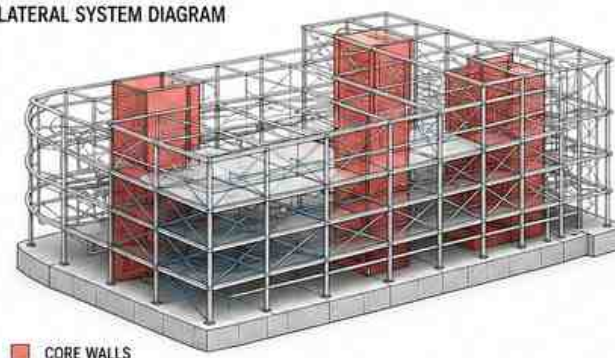
OUR
REVOLUTIONARY
TECHNOLOGY.

Introducing 3D GLASS - a revolutionary, patented technology that lets you shape three-dimensional, smoothly changing glass surfaces and facades.

Thermally formed 3D GLASS fills the market gap of complex architectural glass forms built with geometrically varied elements.

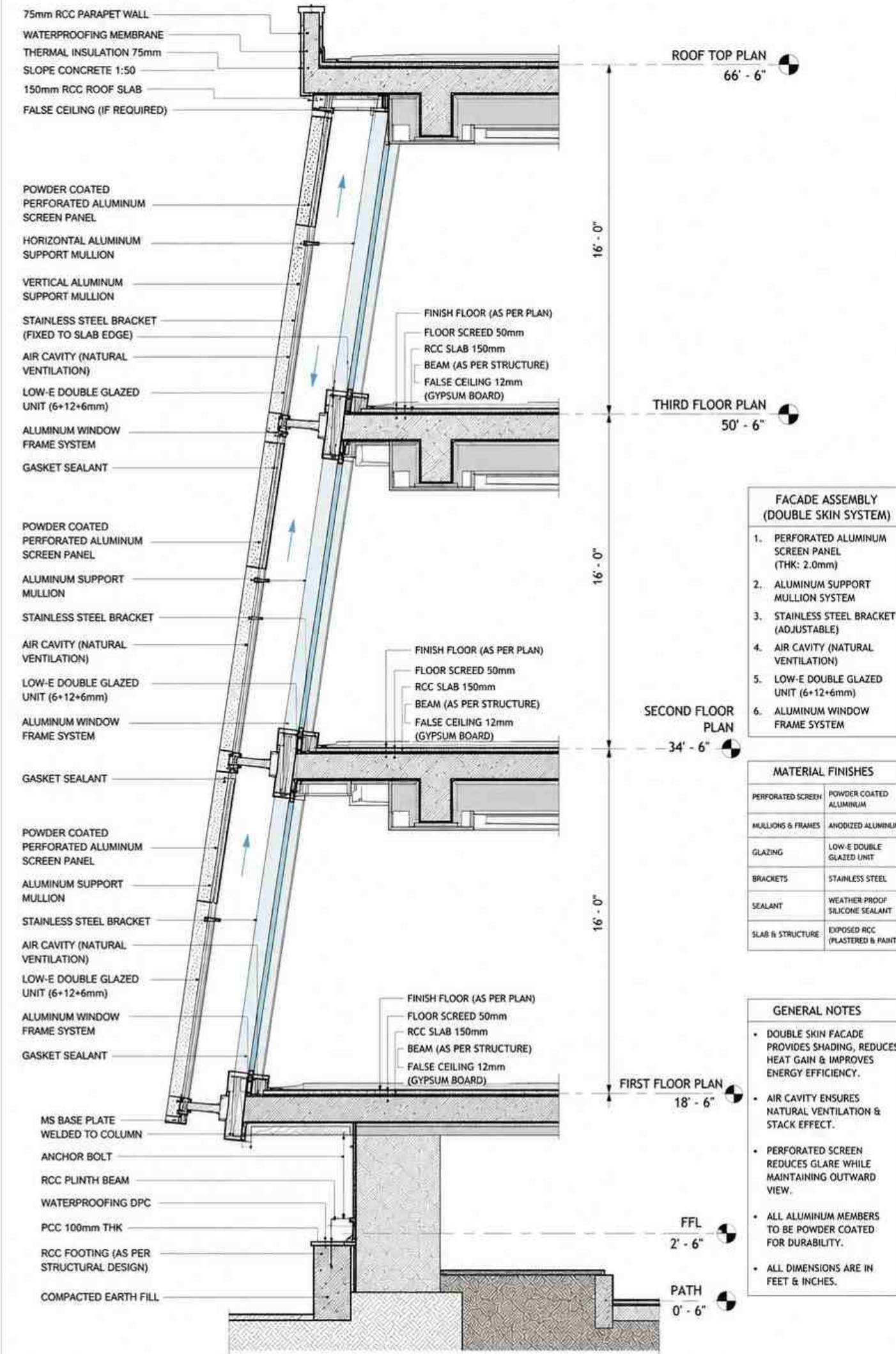


- PRIMARY STRUCTURE
(Columns & Beams)
- SECONDARY STRUCTURE
(Framing)
- LATERAL SYSTEM
(Core, Bracing & Shear Walls)
- FOUNDATION SYSTEM
(Raft & Piles)



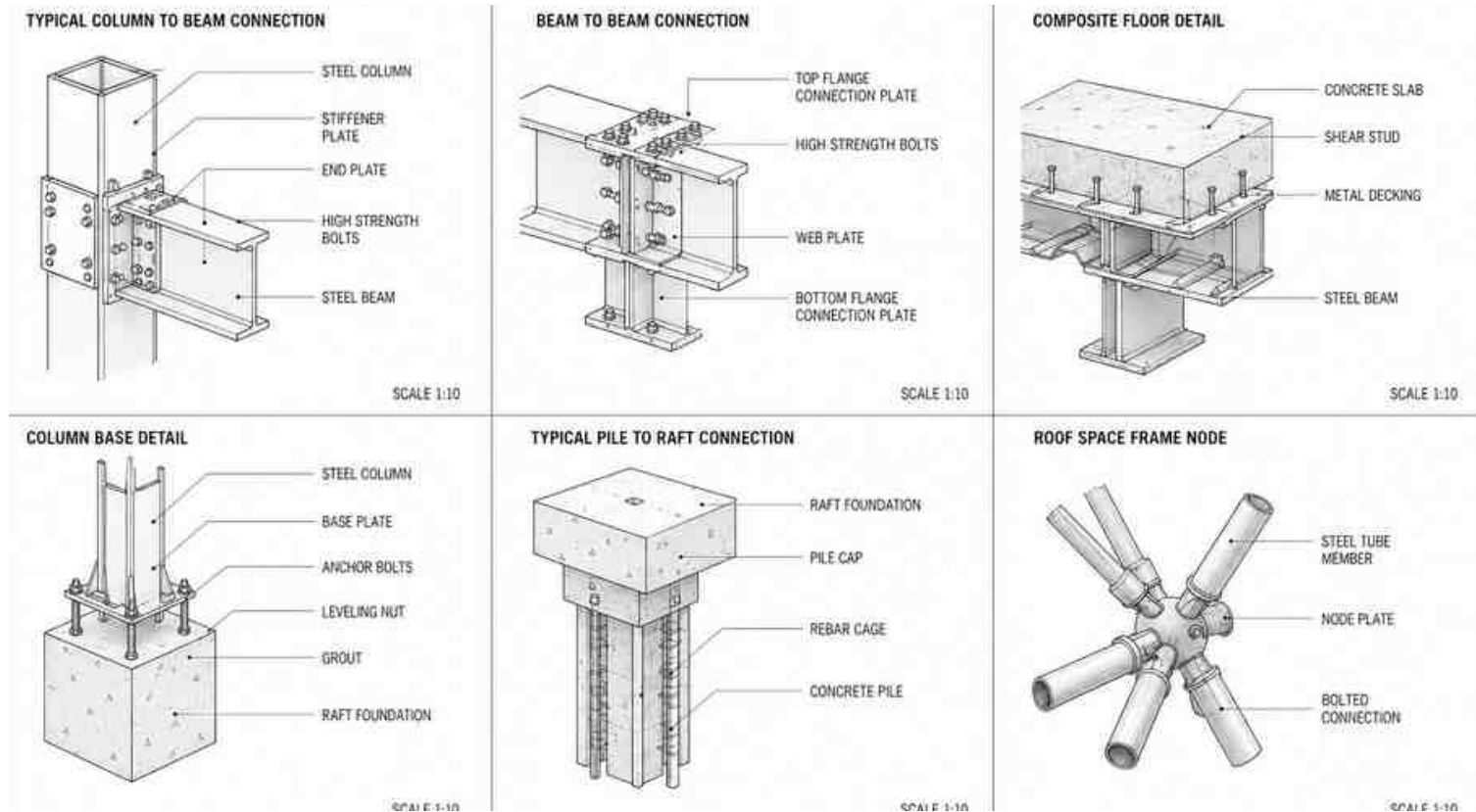
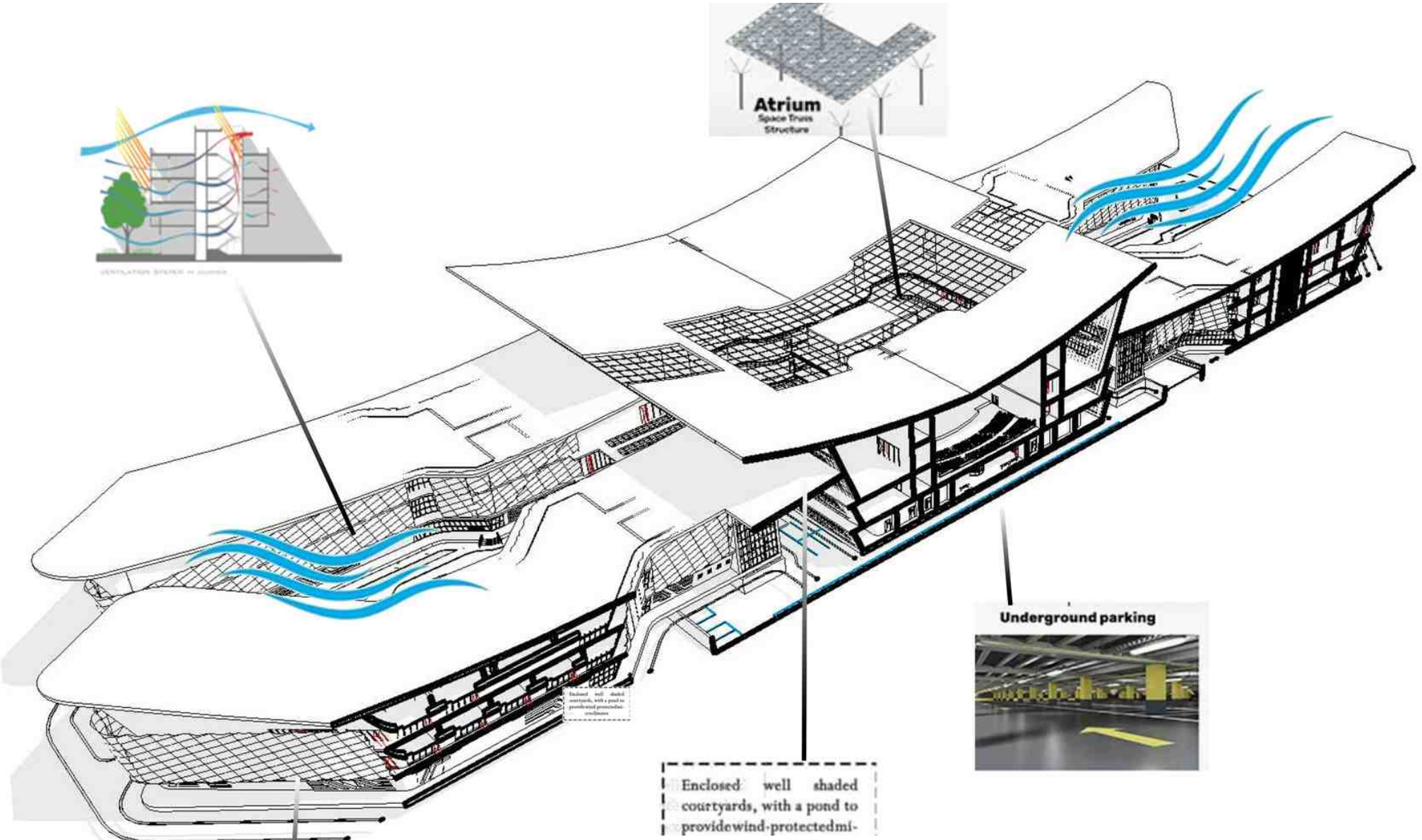
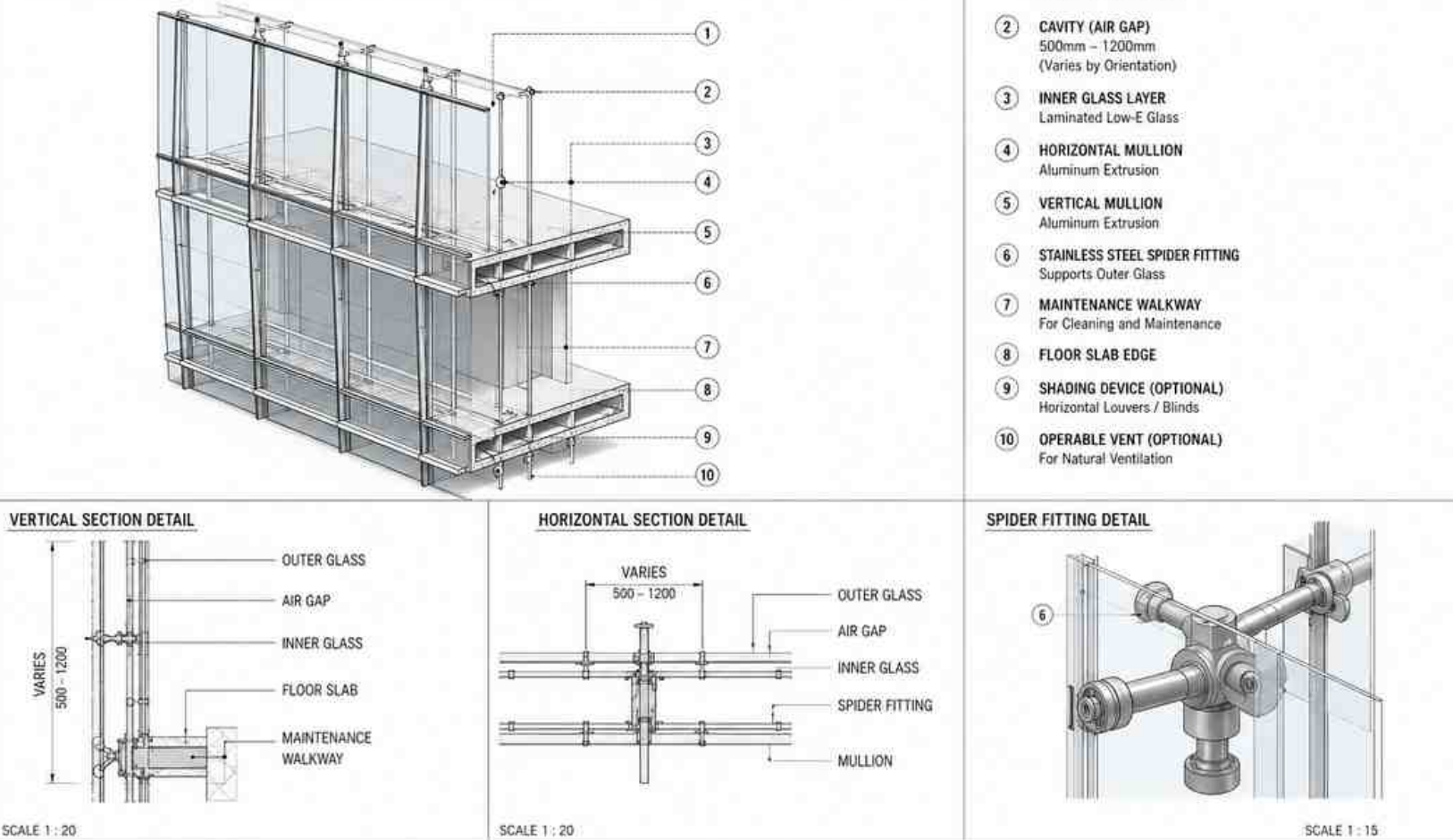
- CORE WALLS
- ×

 STEEL BRACING

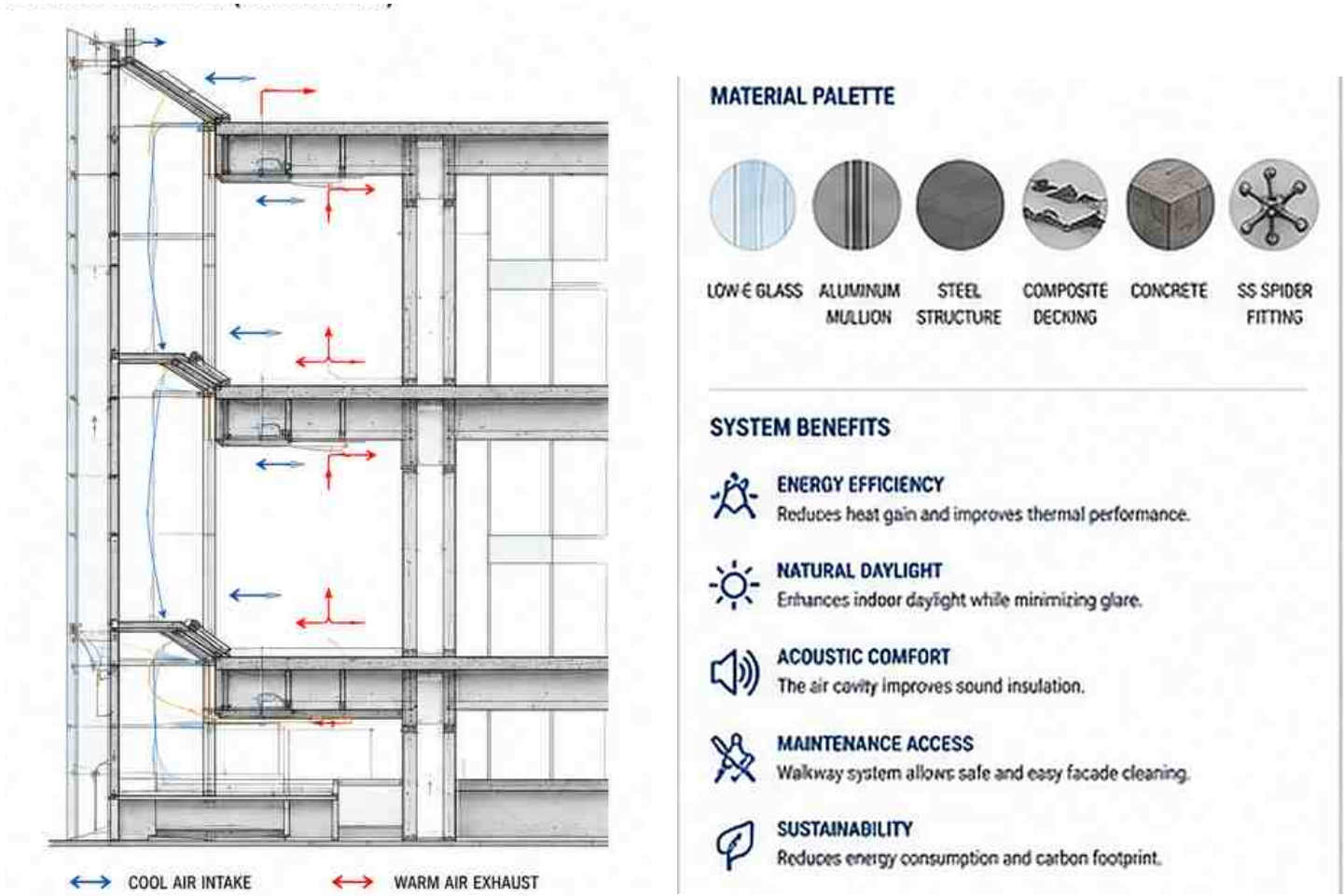


3D SECTIONS DETIALS

4. FACADE - DOUBLE SKIN FACADE SYSTEM



3D SECTIONS DETIALS



GROUND FLOOR SHEDULE

A. DOOR SCHEDULE

MARK	TYPE	MATERIAL / FINISH	SIZE	FRAME	REMARKS
D-01	Main Entrance	Double glazed/aluminium	TBC	TBC	Public entrance
D-02	Internal Double	Timber/laminate	TBC	TBC	Major rooms / halls
D-03	Internal Single	Timber/laminate	TBC	TBC	Offices / labs
D-04	Service Door	Metal	TBC	TBC	MEP / service areas
D-05	Toilet Door	Timber/laminate	TBC	TBC	Toilets
D-06	Emergency Exit	Steel / fire-rated	TBC	TBC	Exit routes

B. WINDOW / GLAZING SCHEDULE

MARK	TYPE	FRAME / GLAZING	SIZE	SILL / LEVEL	REMARKS
W-01	Curtain Wall	Aluminium + Low-E glazing	TBC	TBC	Main public façades
W-02	Fixed Glazing	Aluminium + Low-E glazing	TBC	TBC	Feature / atrium areas
W-03	Operable Window	Aluminium + Low-E glazing	TBC	TBC	Naturally ventilated rooms
W-04	Service/Louver	Aluminium louver	TBC	TBC	MEP / utility rooms

C. WALL TYPE SCHEDULE

MARK	TYPE	CONSTRUCTION / BUILD-UP	THK.	REMARKS	APPLICATION
WT-01	External Opaque Wall	Insulated wall + external finish	TBC	Exterior envelope	
WT-02	Internal Partition	Gypsum/drywall partition	TBC	General rooms	
WT-03	Wet Area Wall	Moisture-resistant partition / finish	TBC	Toilets	
WT-04	Service Wall	Fire/acoustic rated partition	TBC	MEP / plant areas	
WT-05	Core Wall	Reinforced concrete	TBC	Lift / stair core	
WT-06	Double-Skin Façade Interface	Inner glazing + cavity + outer glazing	TBC	Façade perimeter	

FIRST FLOOR SHEDULE

A. DOOR SCHEDULE

MARK	TYPE	MATERIAL / FINISH	SIZE	FRAME	REMARKS
D-01	Main / Public Double	Glazed/aluminium	TBC	TBC	Public circulation
D-02	Internal Double	Timber/laminate	TBC	TBC	Labs / learning spaces
D-03	Internal Single	Timber/laminate	TBC	TBC	Offices / support
D-04	Service Door	Metal	TBC	TBC	MEP / service
D-05	Toilet Door	Timber/laminate	TBC	TBC	Toilets
D-06	Emergency Exit	Steel / fire-rated	TBC	TBC	Exit routes

B. WINDOW / GLAZING SCHEDULE

MARK	TYPE	FRAME / GLAZING	SIZE	SILL / LEVEL	REMARKS
W-01	Curtain Wall	Aluminium + Low-E glazing	TBC	TBC	Main façades
W-02	Fixed Glazing	Aluminium + Low-E glazing	TBC	TBC	Atrium / feature areas
W-03	Operable Window	Aluminium + Low-E glazing	TBC	TBC	Offices / teaching
W-04	Service/Louver	Aluminium louver	TBC	TBC	Service areas

C. WALL TYPE SCHEDULE

MARK	TYPE	CONSTRUCTION / BUILD-UP	THK.	REMARKS	APPLICATION
WT-01	External Opaque Wall	Insulated wall + external finish	TBC	Exterior envelope	
WT-02	Internal Partition	Gypsum/drywall partition	TBC	General rooms	
WT-03	Wet Area Wall	Moisture-resistant partition / finish	TBC	Toilets	
WT-04	Service Wall	Fire/acoustic rated partition	TBC	MEP / support Lift	
WT-05	Core Wall	Reinforced concrete	TBC	/ stair core	
WT-06	Double-Skin Façade Interface	Inner glazing + cavity + outer glazing	TBC	Façade perimeter	

SECOND FLOOR SHEDULE

A. DOOR SCHEDULE

MARK	TYPE	MATERIAL / FINISH	SIZE	FRAME	REMARKS
D-01	Main / Public Double	Glazed/aluminium	TBC	TBC	Public circulation
D-02	Internal Double	Timber/laminate	TBC	TBC	Collaboration / labs
D-03	Internal Single	Timber/laminate	TBC	TBC	Offices / support
D-04	Service Door	Metal	TBC	TBC	MEP / service
D-05	Toilet Door	Timber/laminate	TBC	TBC	Toilets
D-06	Emergency Exit	Steel / fire-rated	TBC	TBC	Exit routes

B. WINDOW / GLAZING SCHEDULE

MARK	TYPE	FRAME / GLAZING	SIZE	SILL / LEVEL	REMARKS
W-01	Curtain Wall	Aluminium + Low-E glazing	TBC	TBC	Main façades
W-02	Fixed Glazing	Aluminium + Low-E glazing	TBC	TBC	Atrium / feature areas
W-03	Operable Window	Aluminium + Low-E glazing	TBC	TBC	Offices / collaboration
W-04	Service/Louver	Aluminium louver	TBC	TBC	Service areas

C. WALL TYPE SCHEDULE

MARK	TYPE	CONSTRUCTION / BUILD-UP	THK.	REMARKS	APPLICATION
WT-01	External Opaque Wall	Insulated wall + external finish	TBC	Exterior envelope	
WT-02	Internal Partition	Gypsum/drywall partition	TBC	General rooms	
WT-03	Wet Area Wall	Moisture-resistant partition / finish	TBC	Toilets	
WT-04	Service Wall	Fire/acoustic rated partition	TBC	MEP / support Lift	
WT-05	Core Wall	Reinforced concrete	TBC	/ stair core	
WT-06	Double-Skin Façade Interface	Inner glazing + cavity + outer glazing	TBC	Façade perimeter	

RENDERS



FUTUREISTIC DESIGNS



YEAR:
2024

LOCATION :
ISLAMABAD

TYPE :
HIGH RISE

04

VETICAL FARMING

“A Vertical Ecosystem for
Future Cities”

Location: Gulberg Green islamabad Year: 2024
Guided by: Ar. FAZALAMMAR
Semester VII Design Studio Project

This project reimagines the typology of high-rise architecture through the lens of ecological balance and sustainable living. Located in Gulberg Greens, Islamabad, this vertical complex integrates the principles of permaculture into a twisting tower form that embodies the fusion of urban density with environmental consciousness. Inspired by the logic of natural systems and the aesthetics of stepped farming landscapes, the design maximizes green terraces, passive strategies, and vertical farming opportunities. The form not only responds to climatic orientation and urban context but symbolically represents the regeneration of built environments through layered sustainability. Designed as a self-sustaining ecosystem in the sky, the tower is a manifesto for future-ready architecture in harmony with nature.



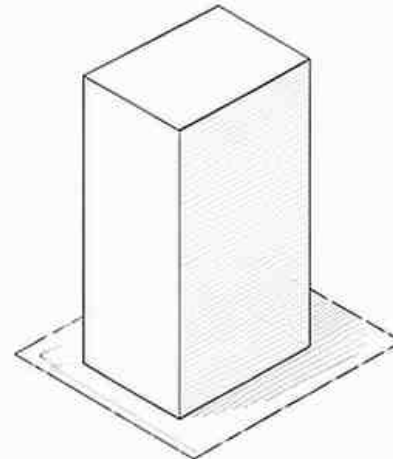
VISION

To design a self sustaining high rise that integrates vertical farming into architecture to promote food production , sustainability and a healthy urban lifestyle



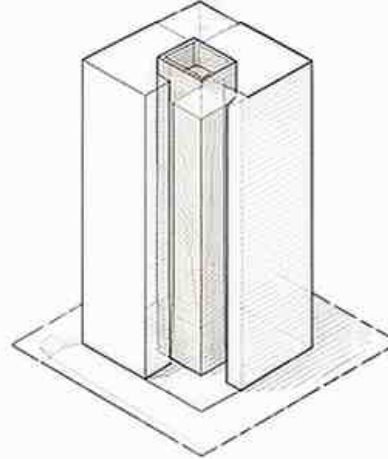
CONCEPT DIAGRAM

1. BASIC MASS



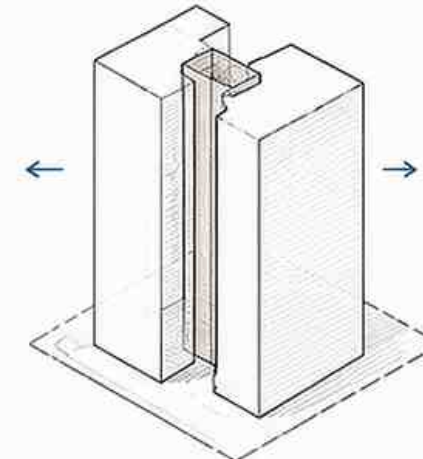
A simple rectangular mass as a starting point.

2. CENTRAL CORE



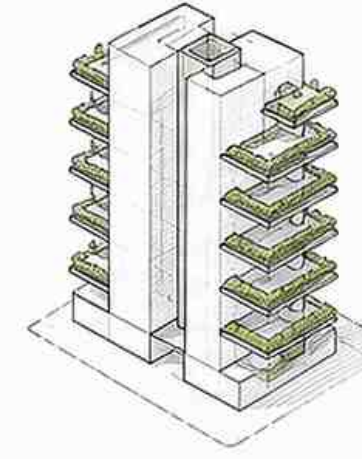
A central core is introduced for structural stability and services.

3. MASS DIVISION



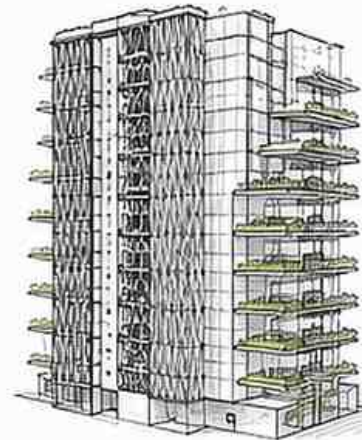
Mass is divided into two parts creating a central void for ventilation and light penetration.

4. TERRACE FORMATION



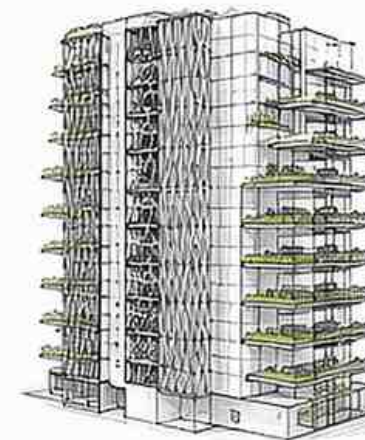
Terraces are carved out on both sides for vertical farming and outdoor green spaces.

5. FACADE DEVELOPMENT



A dynamic facade with diagonal pattern is introduced for shading and aesthetics.

6. VERTICAL FARMING INTEGRATION



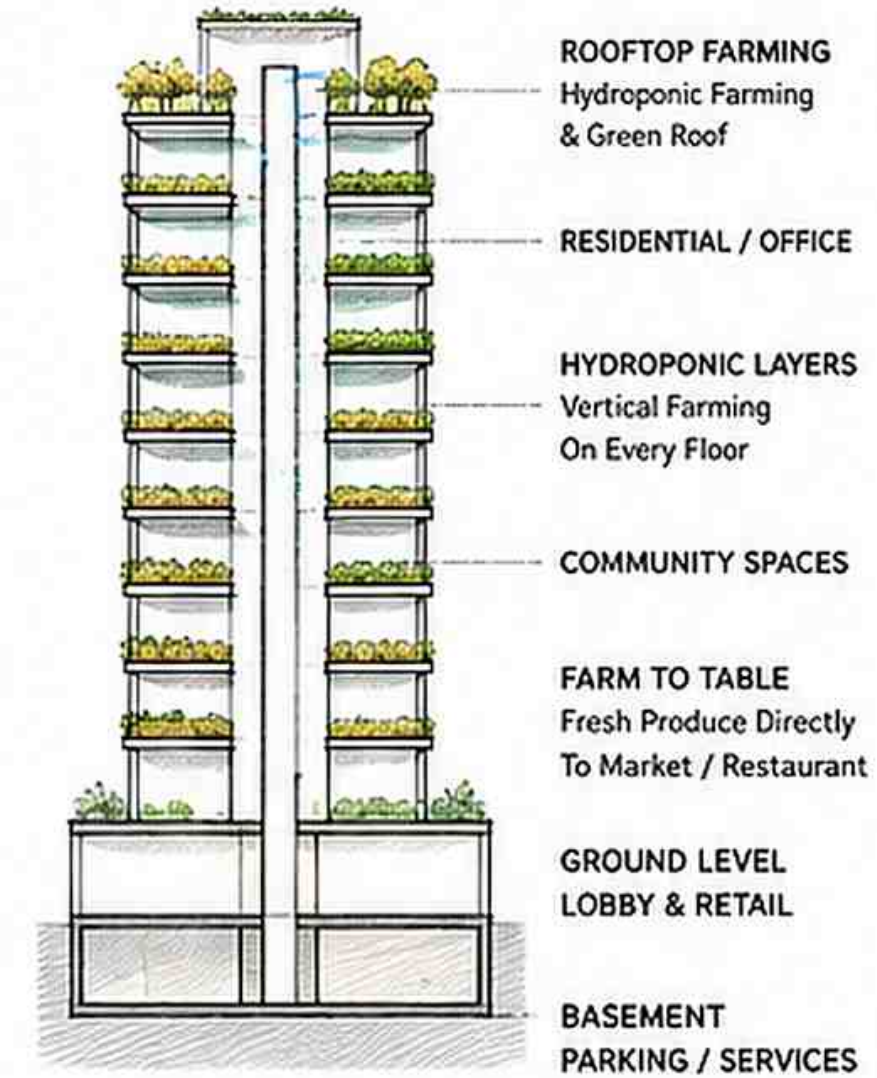
Green layers and planter boxes are integrated on every level for vertical farming.

7. FINAL FORM



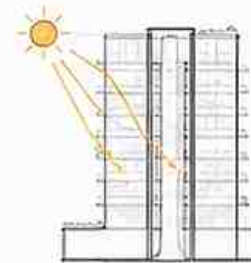
The form is finalized combining architecture with vertical farming for sustainability.

VERTICAL FARMING INTEGRATION



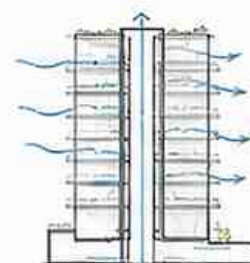
DESIGN STRATEGIES

NATURAL LIGHT



Maximize natural light through central void and open terraces.

NATURAL VENTILATION



Central void and open terraces allow natural airflow and stack ventilation.

VERTICAL FARMING



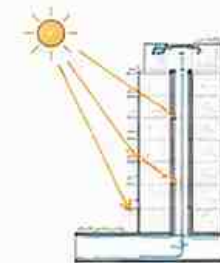
Farming integrated on every level for food production.

WATER HARVESTING



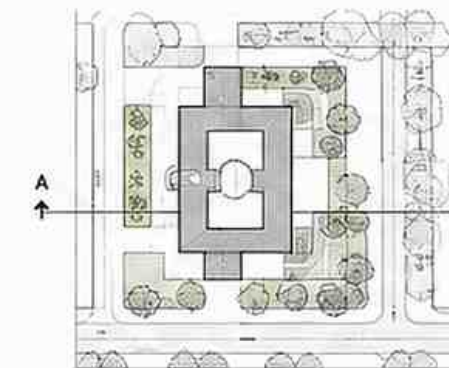
Rain water collected and recycled for irrigation.

SOLAR ENERGY

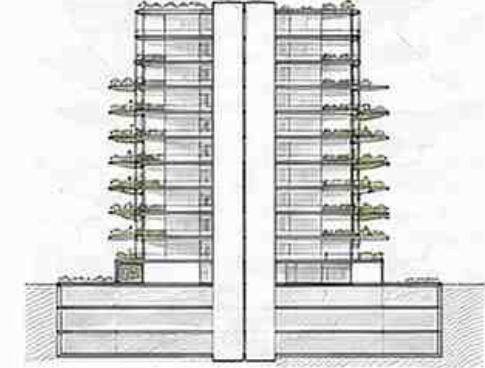


Solar panels on roof for renewable energy.

KEY PLAN



SECTION A-A'

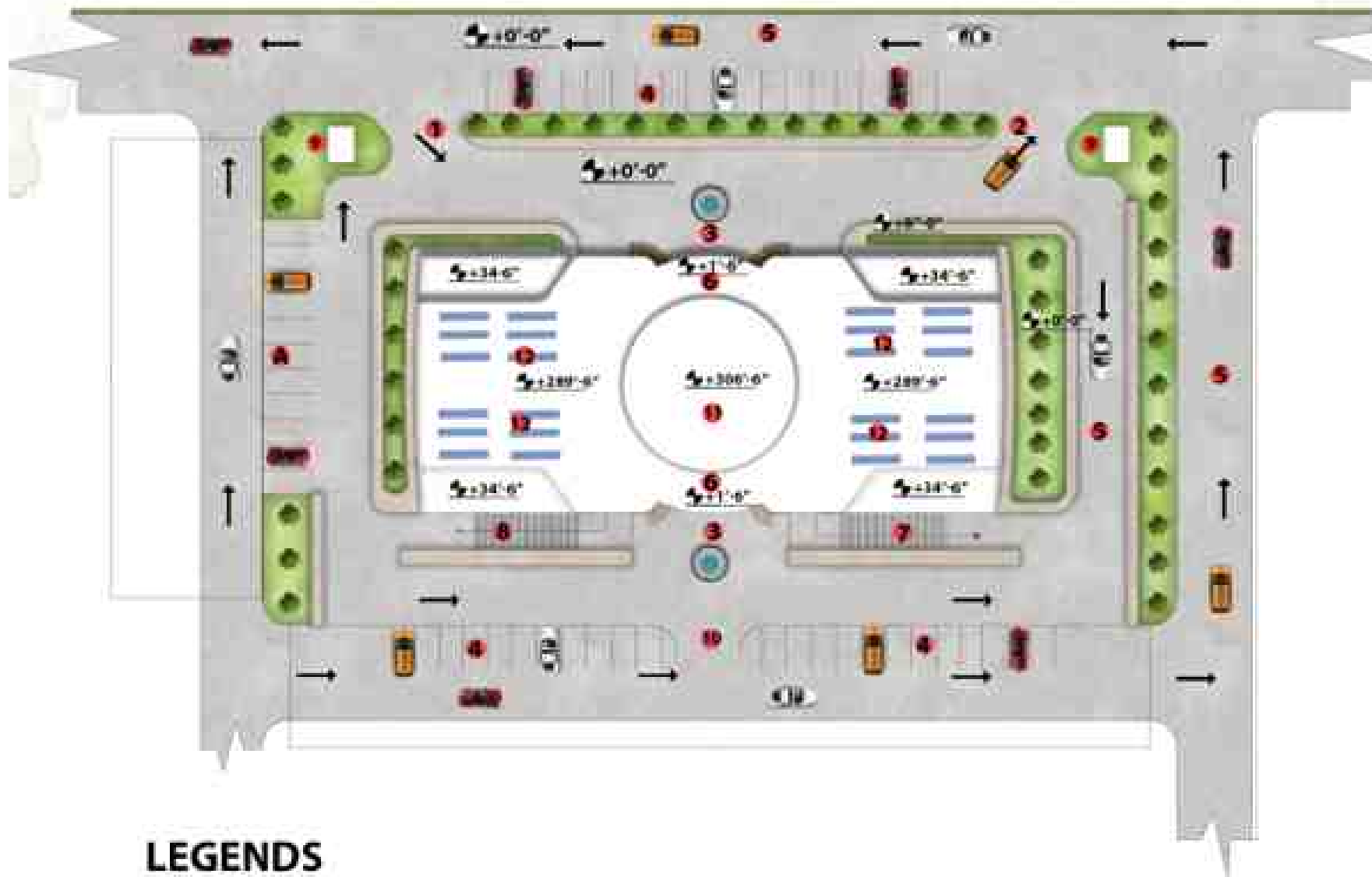


DRAWINGS IN REVIT STYLE

MASTER PLAN

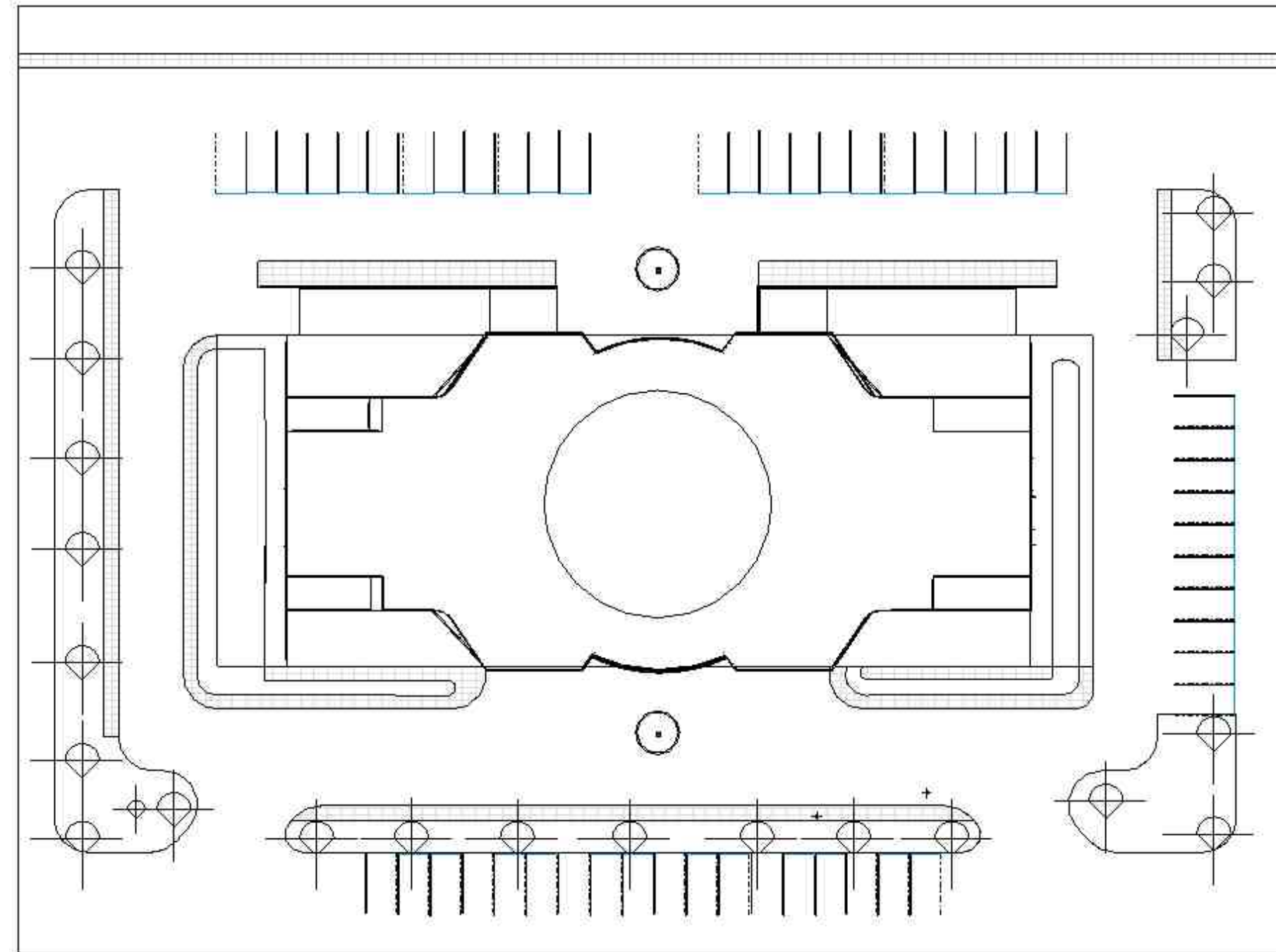
DESIGNED BY: BILAL AHMAD
SEMESTER: 6TH
YEAR: 2023
DURATION: 8 WEEK

FOOT PRINT

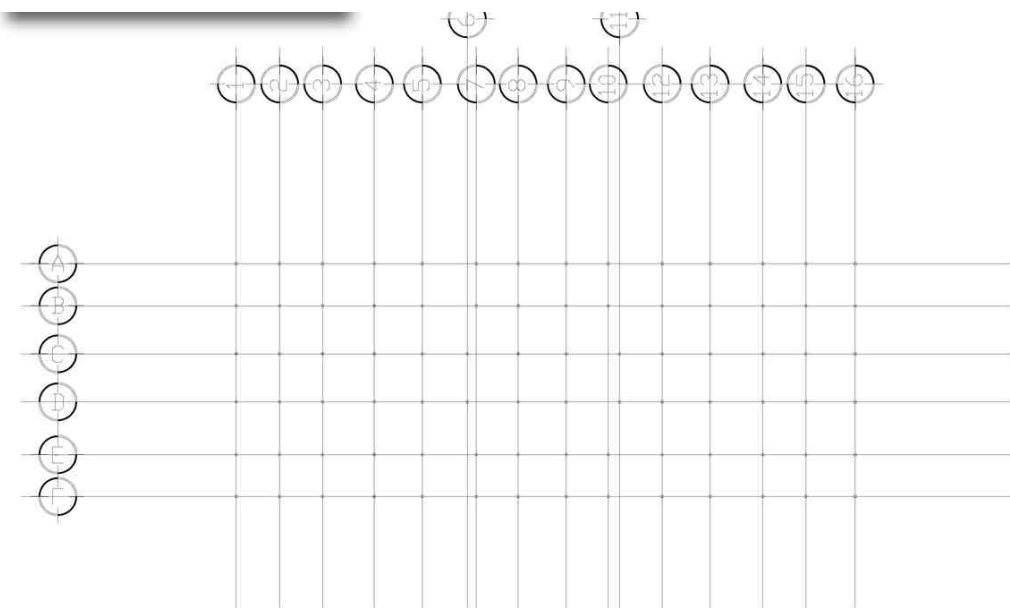


LEGENDS

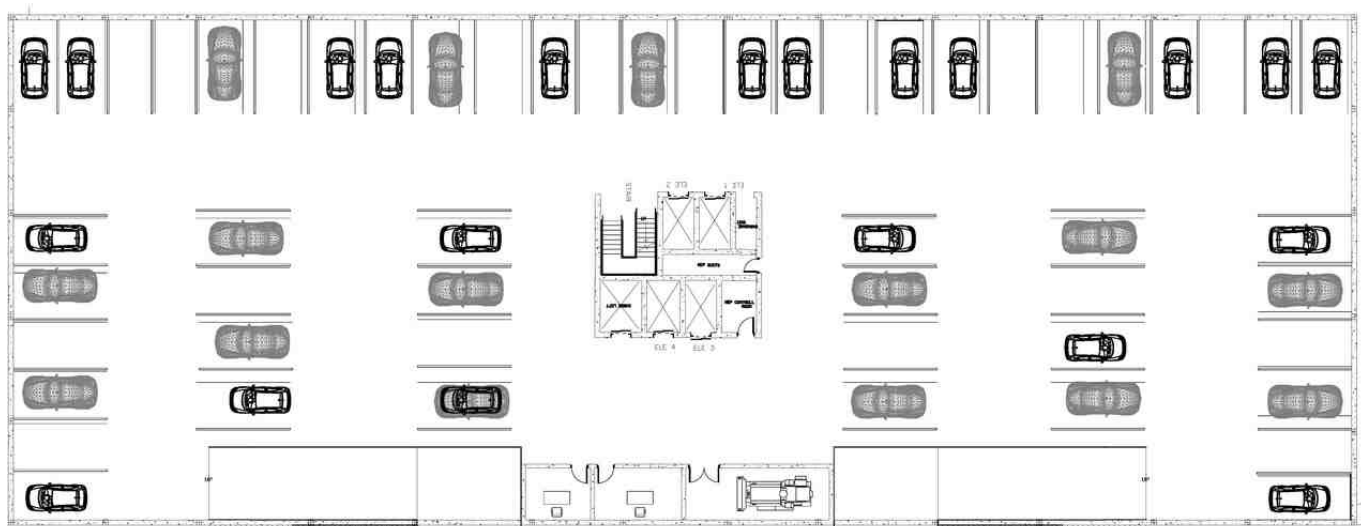
- | | |
|-------------------|-------------------------------|
| 1. ENTRY | 7. BASEMENT IN THROUGH RAMP |
| 2. EXIT | 8. BASEMENT OUT THROUGH RAMP |
| 3. DROPE OF ZONE | 9. SECURITY ROOMS |
| 4. SITE PARKING | 10. LOADER FOR ENTRY AND EXIT |
| A. INSITE PARKING | 11. CORE SERVICE |
| 5. SERVICE ROAD | 12. SOLAR PANELS |
| 6. PODFILIM | |



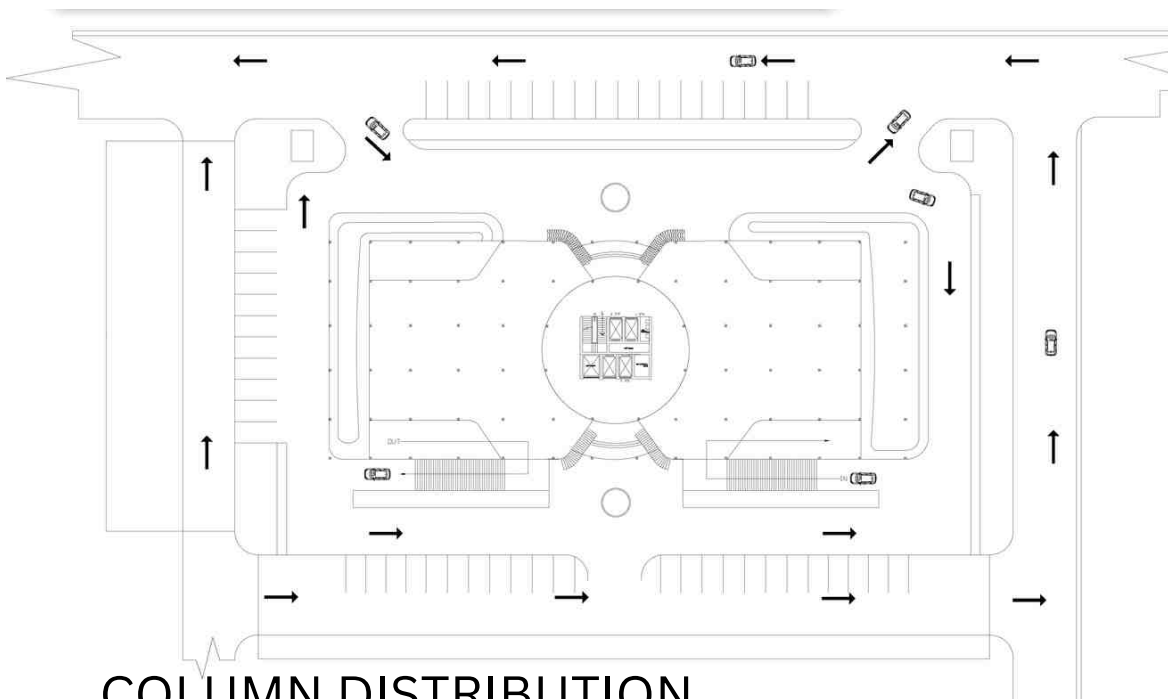
PLANS



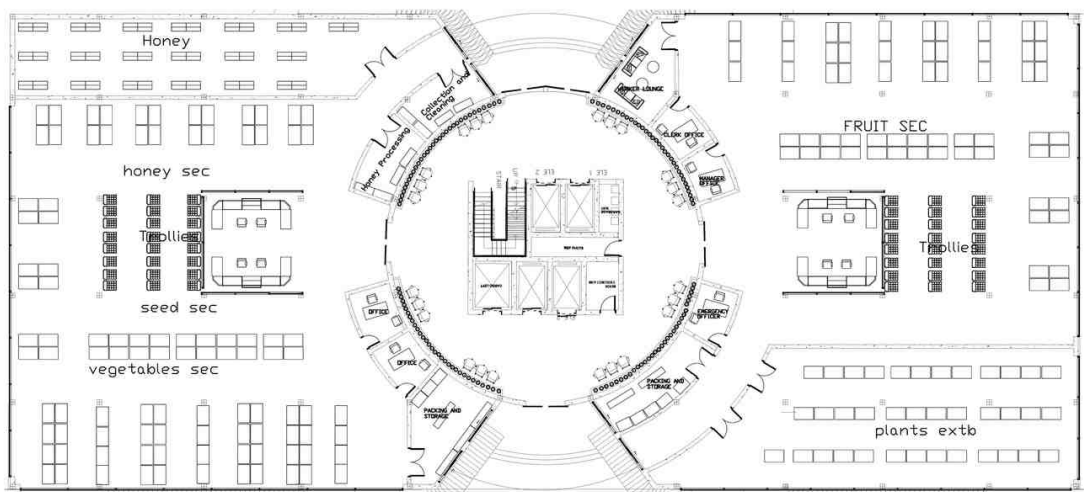
GRID



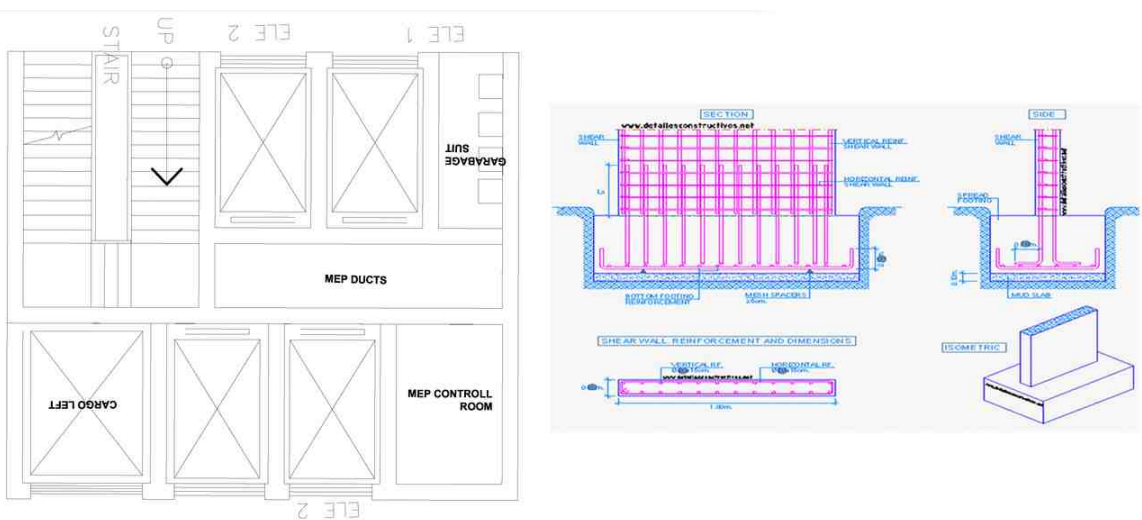
BASEMENT FLOOR



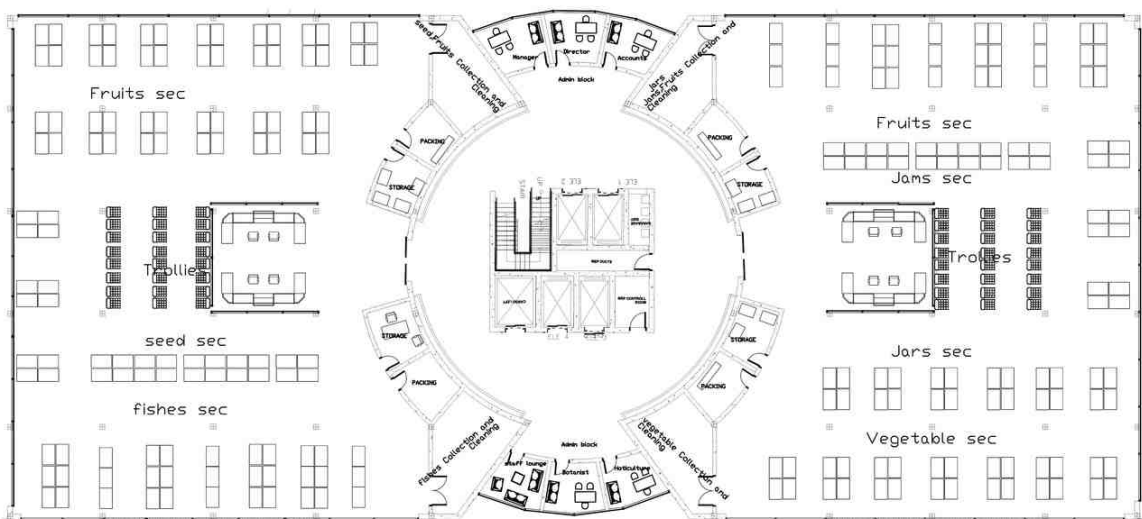
COLUMN DISTRIBUTION



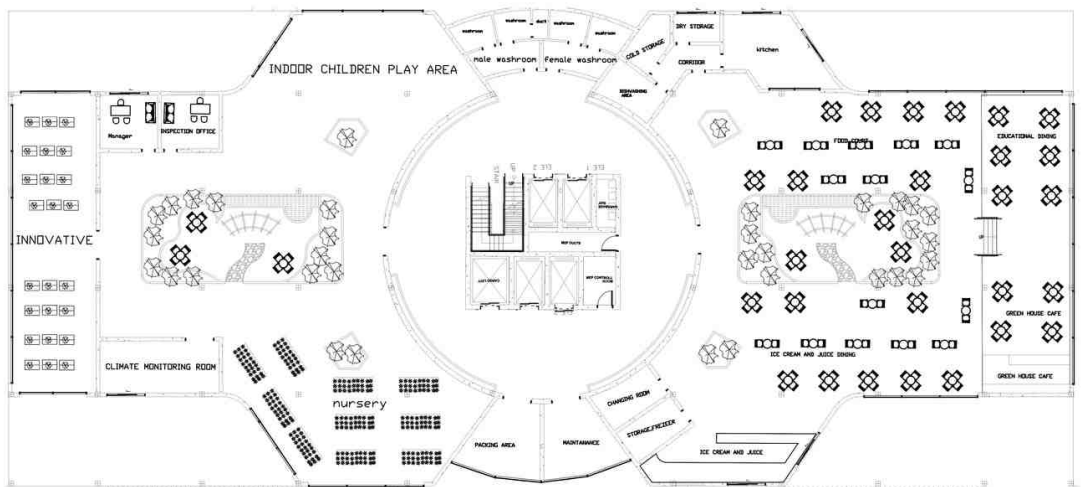
GROUND FLOOR



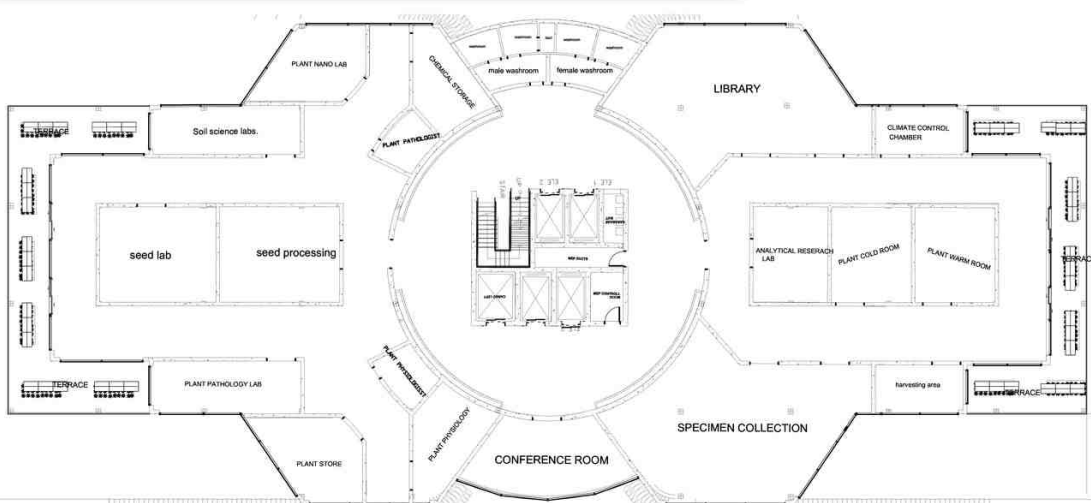
SERVICE CORE



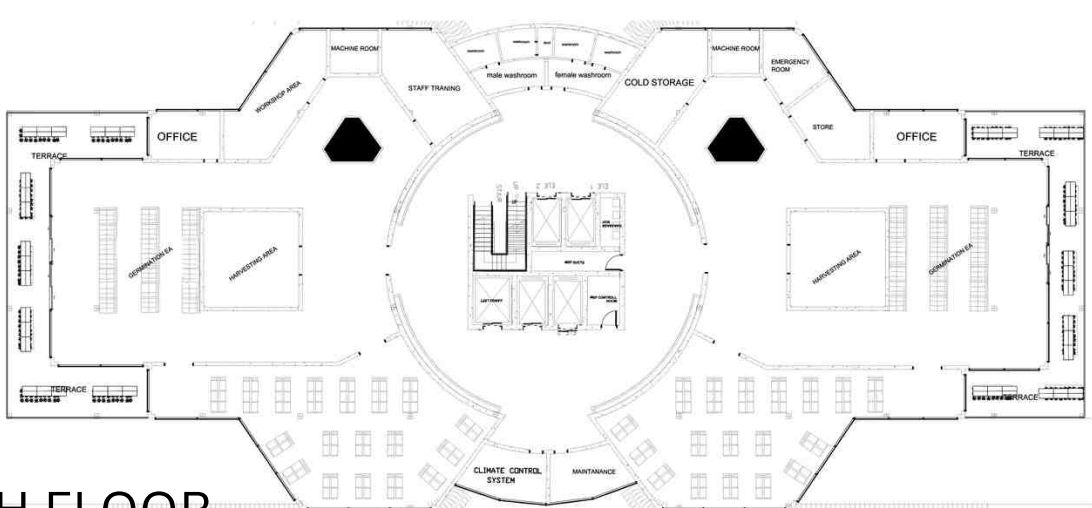
1ST FLOOR



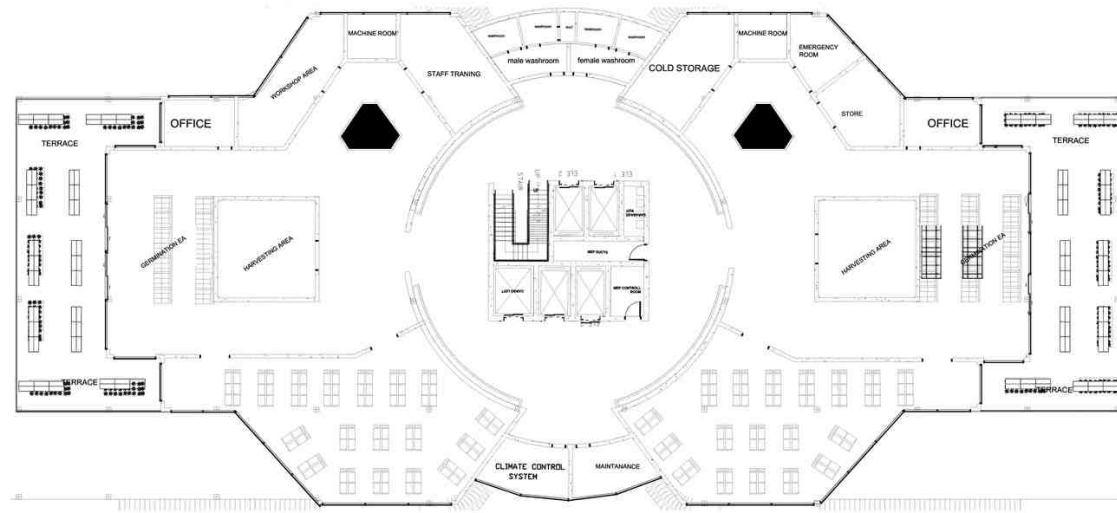
2ND FLOOR



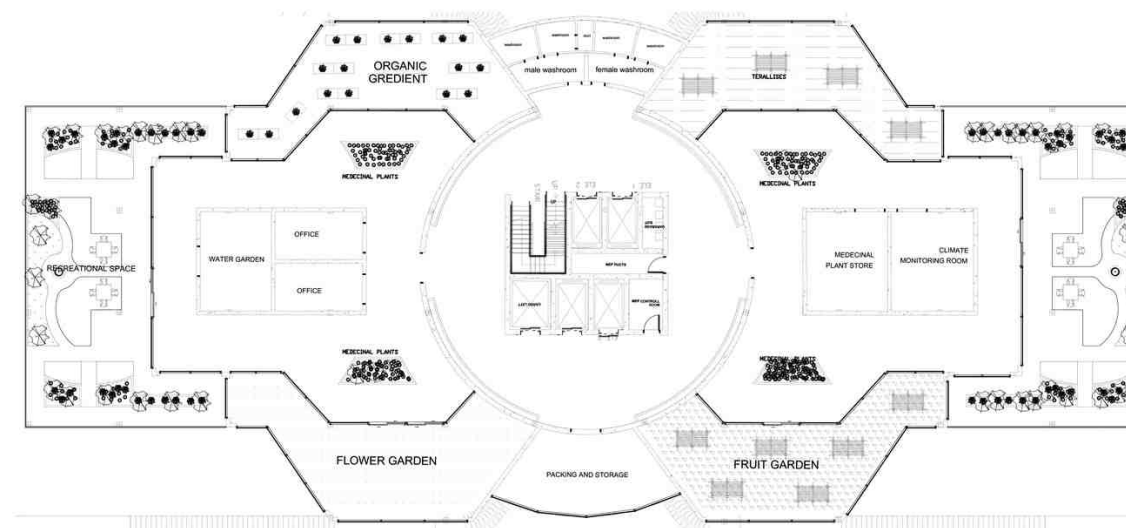
3RD FLOOR



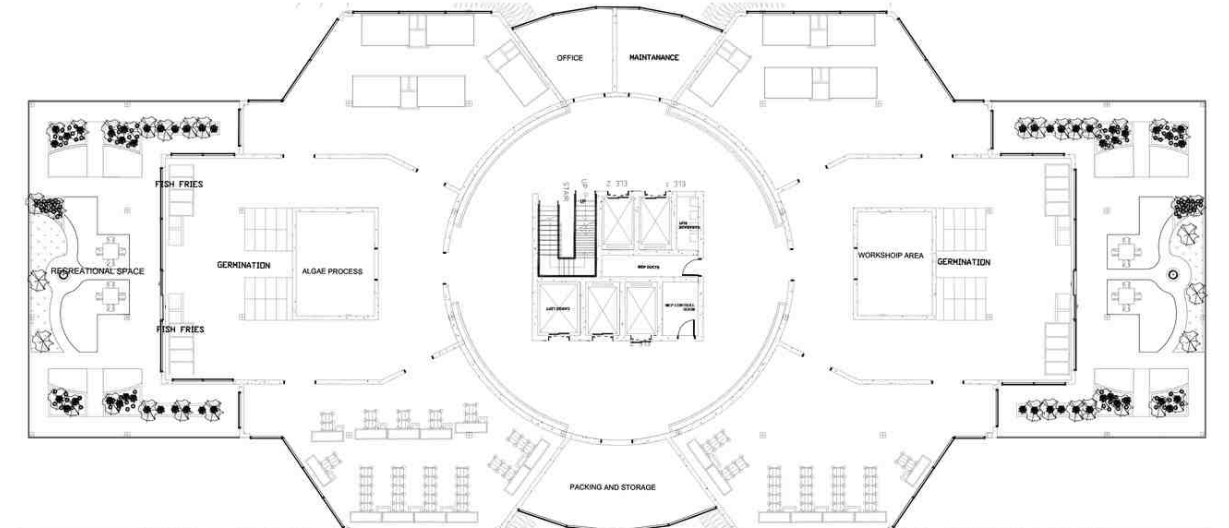
4TH FLOOR



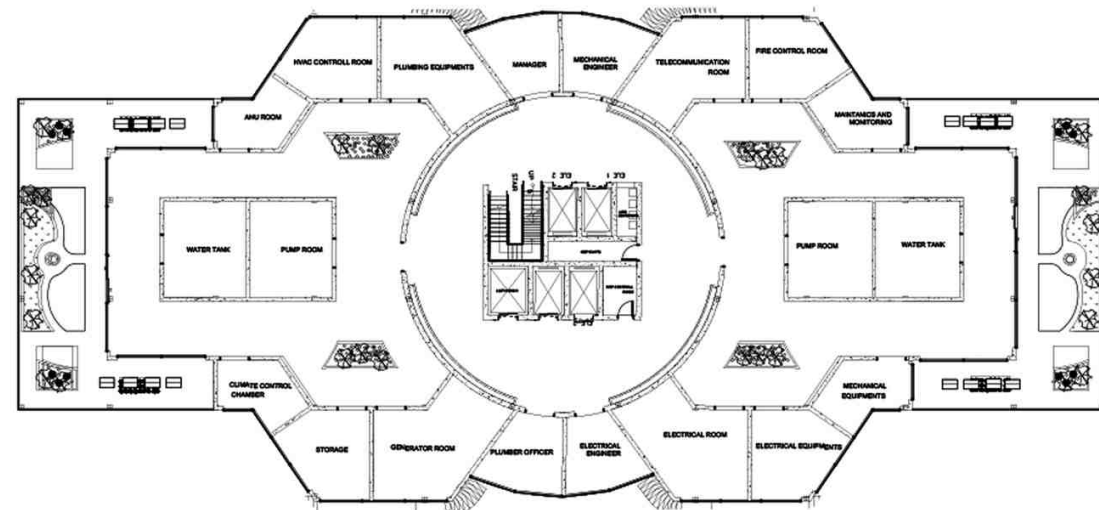
5TH-6TH FLOOR



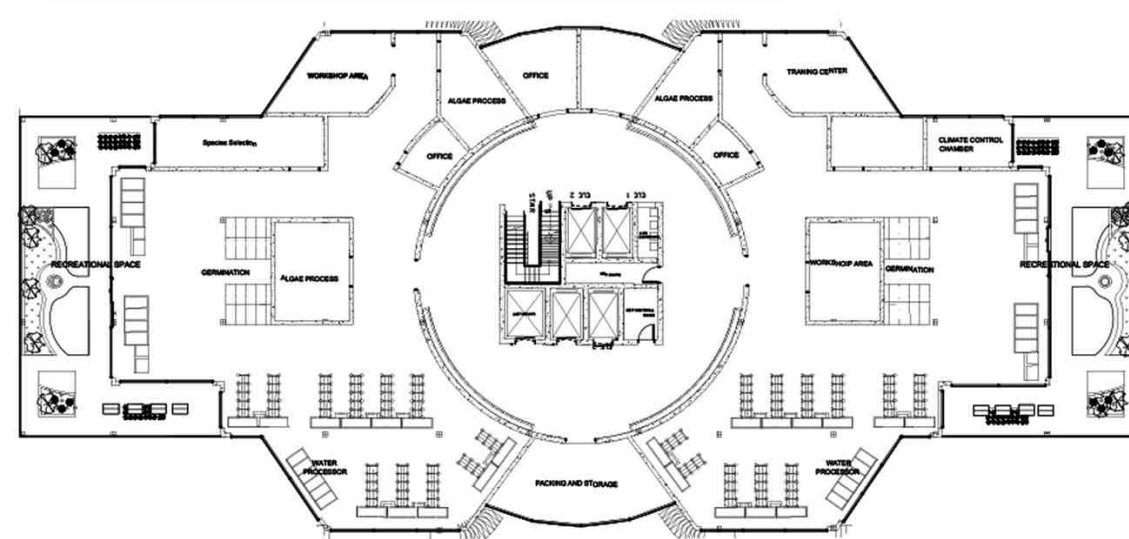
7TH FLOOR



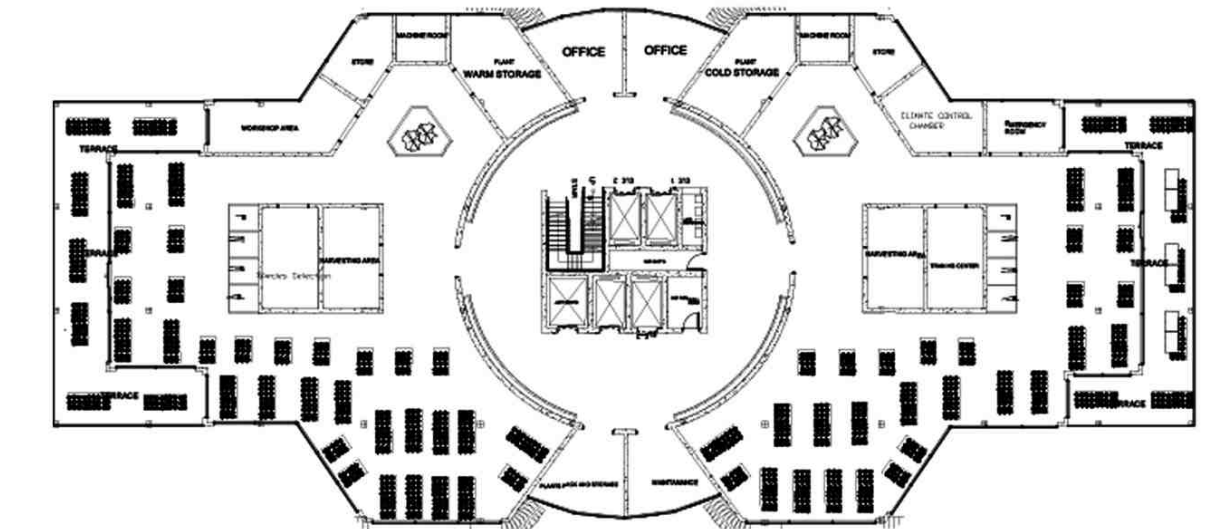
8TH-9TH FLOOR



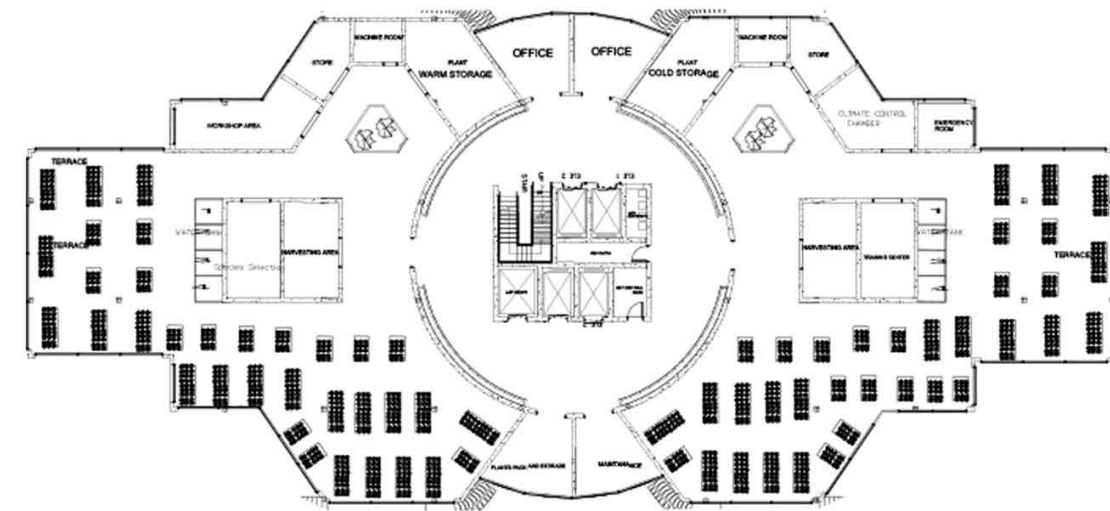
10 TH FLOOR



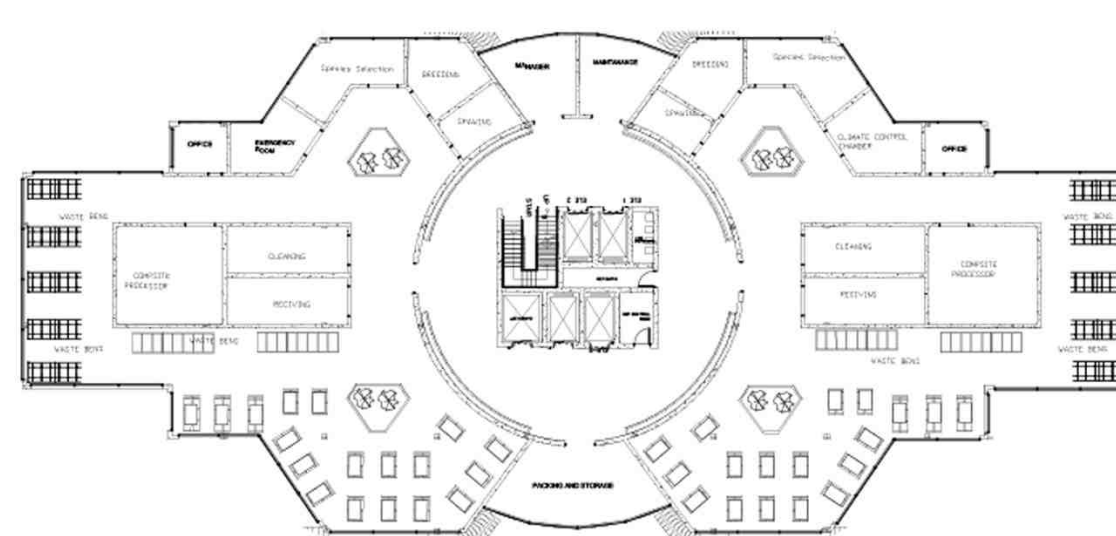
11 TH FLOOR



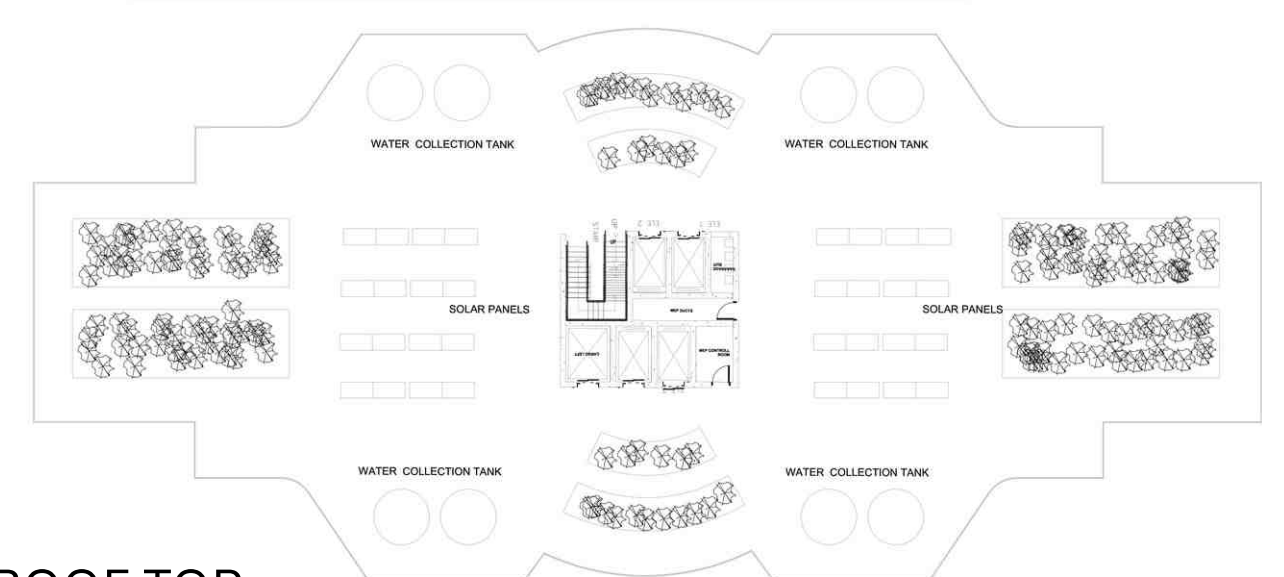
12TH-13TH FLOOR



14 TH FLOOR

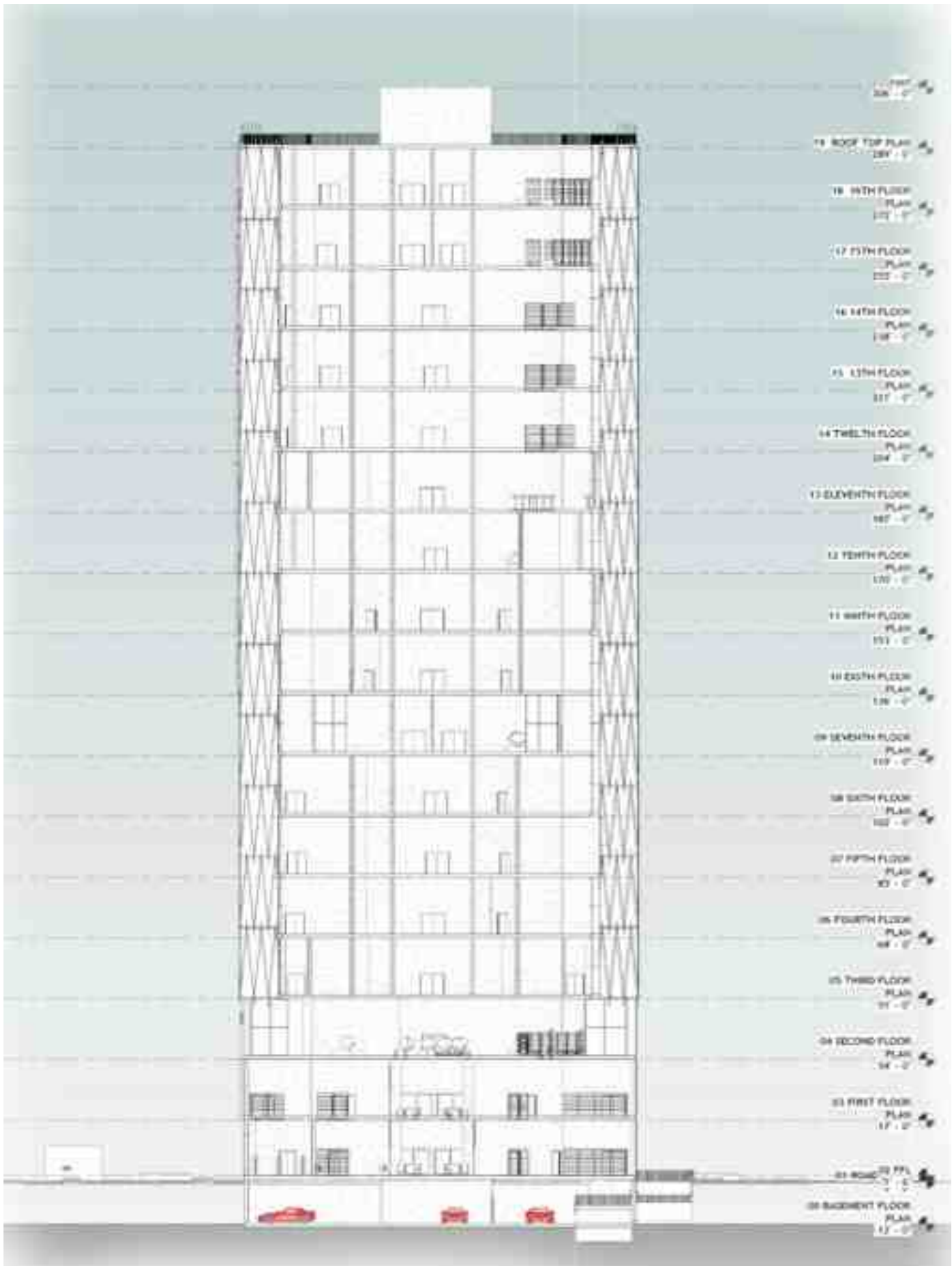


15TH-16TH FLOOR



ROOF TOP

SECTIONS

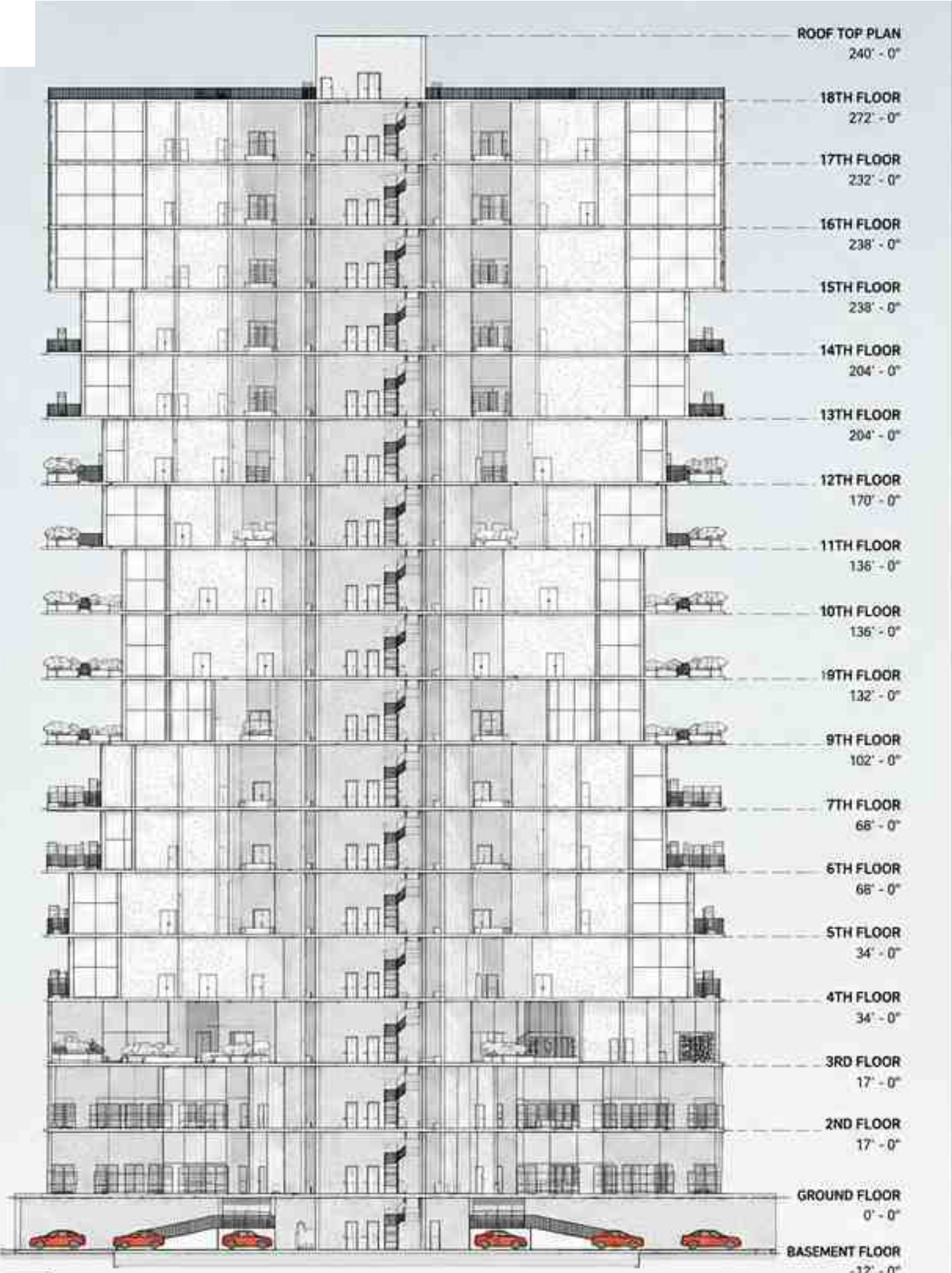


SECTION AA

DETAILS

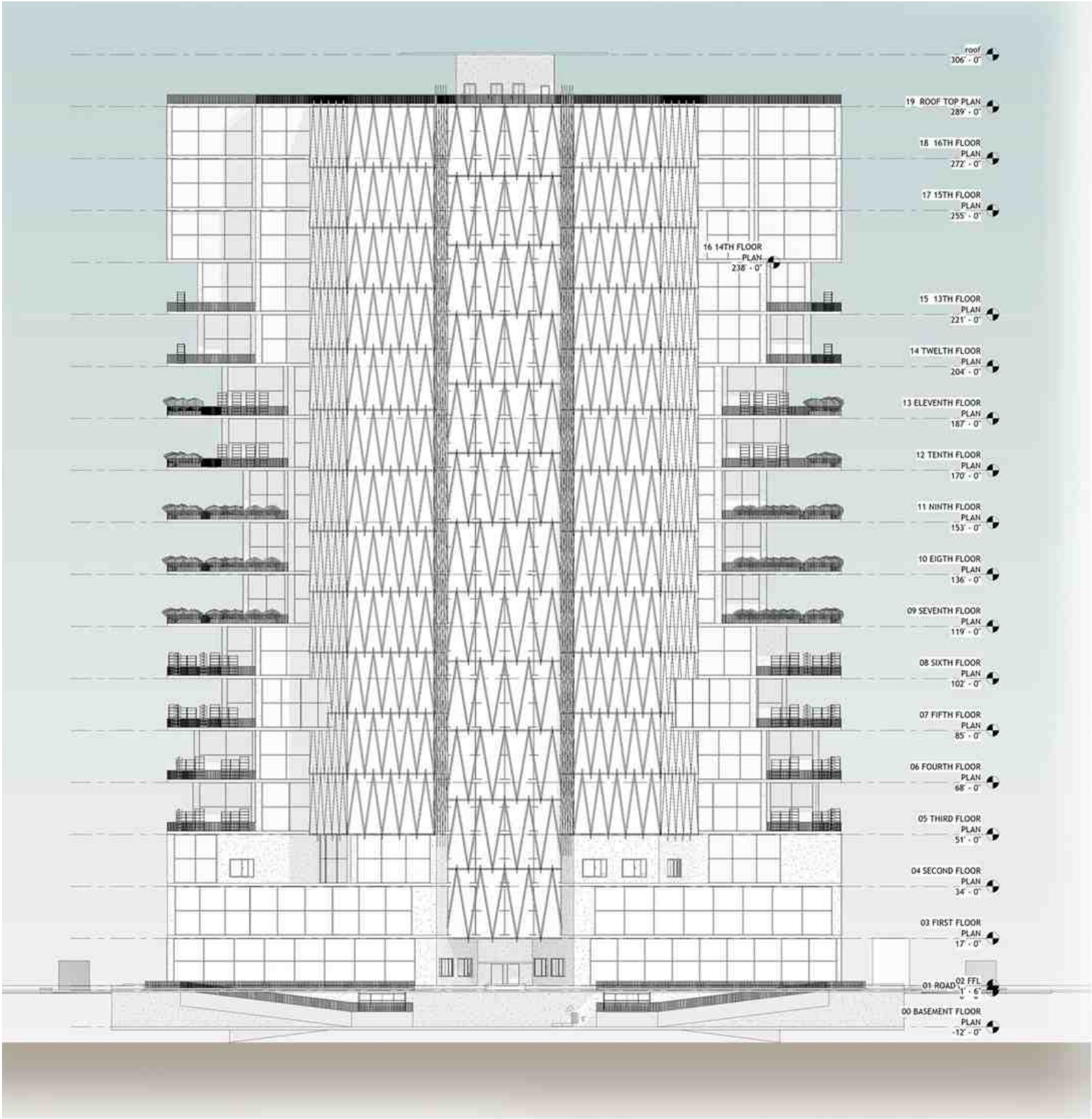
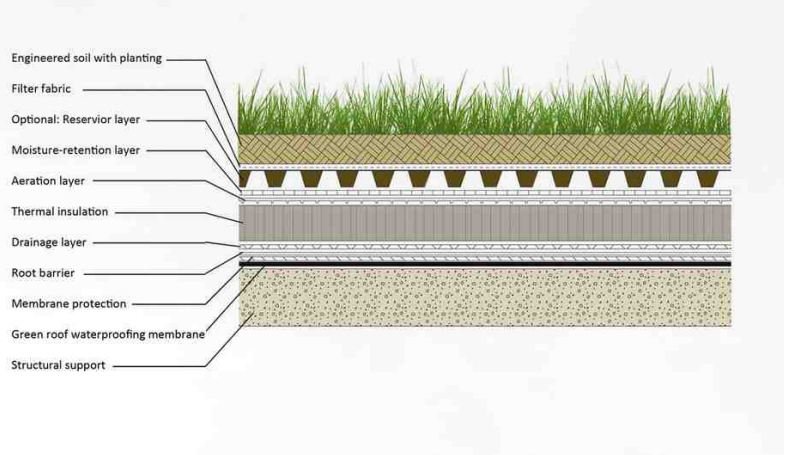
This high rise building is designed with a centralized core system that provides structural stability, efficient vertical circulation, and optimized service distribution. The building consists of basement parking, ground floor amenities, and multiple typical floors with residential units.

- CENTRALIZED CORE**
The core is located at the center of the building and contains all vertical circulation and major services.
- VERTICAL CIRCULATION**
High speed elevators and fire exit stairs are integrated within the core for efficient movement.
- STRUCTURAL SYSTEM**
The core acts as the structural spine, providing lateral stability against wind and seismic loads.
- SERVICE DISTRIBUTION**
Electrical shafts, plumbing stacks, and mechanical ducts are centralized for easy maintenance.
- FLOOR PLAN EFFICIENCY**
Central core allows maximum usable space around the perimeter for apartments with balconies and views.

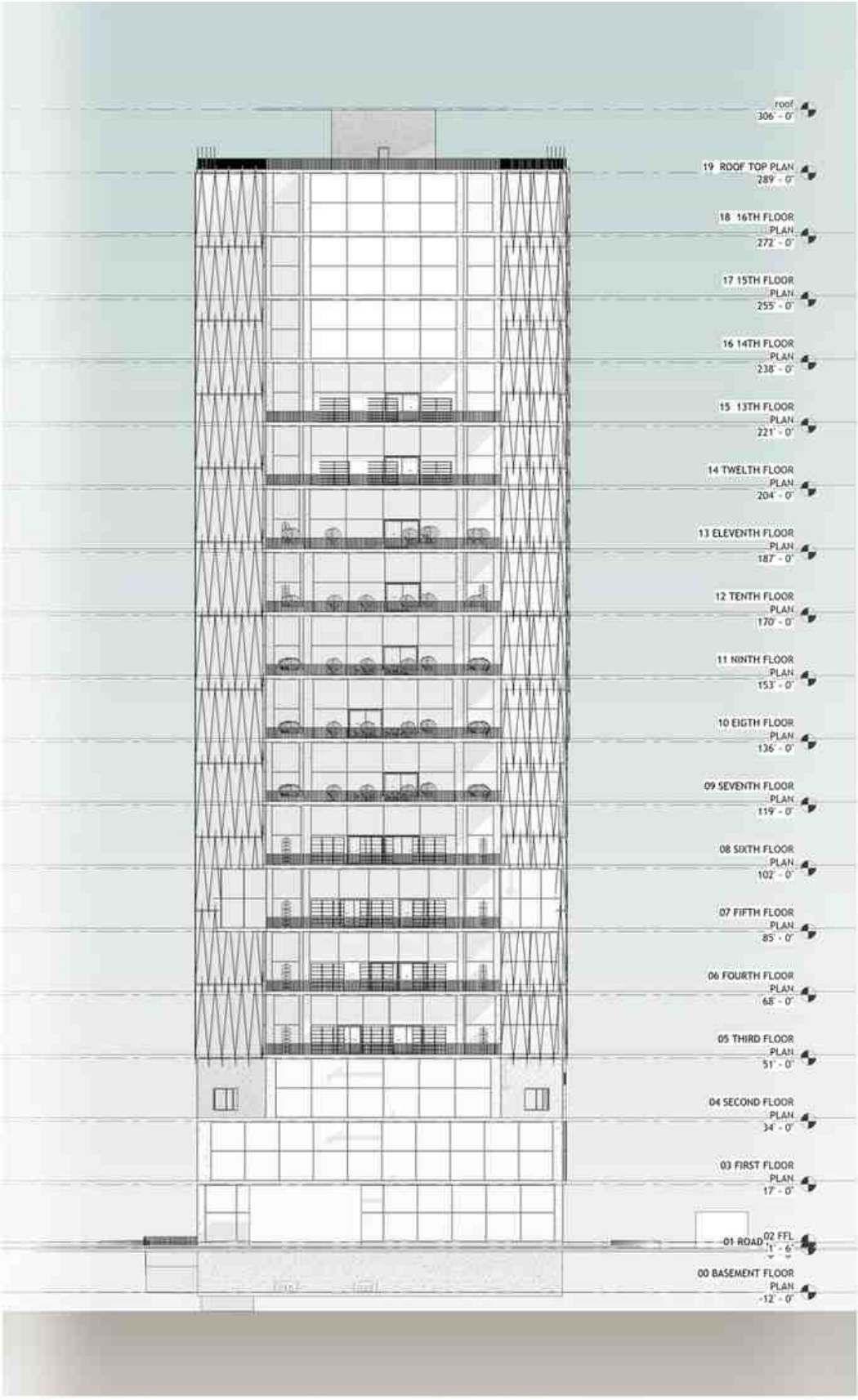


SECTION BB

ELEVATIONS



SOUTH ELEVATION
SCALE : 1/16"=1'-0"

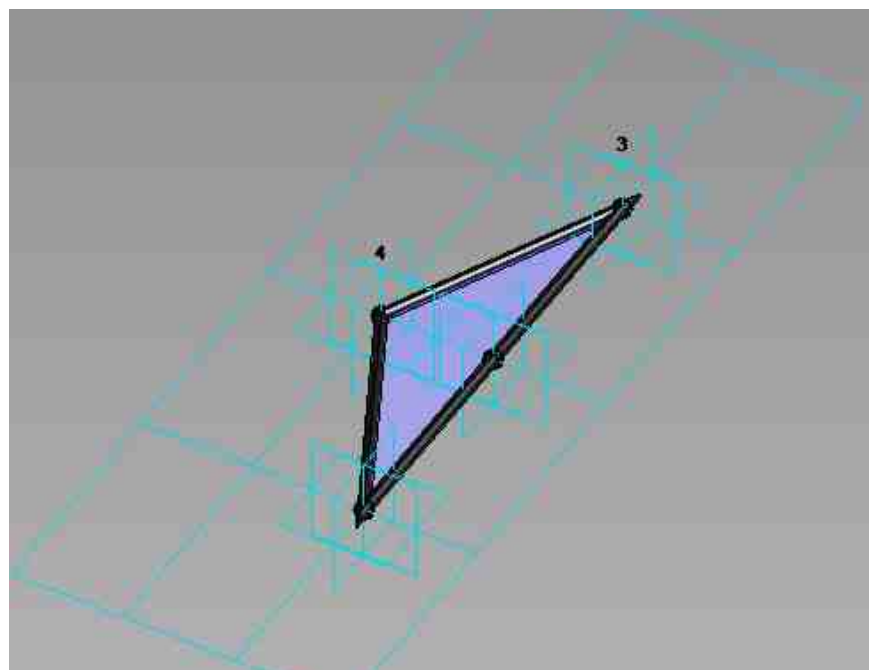
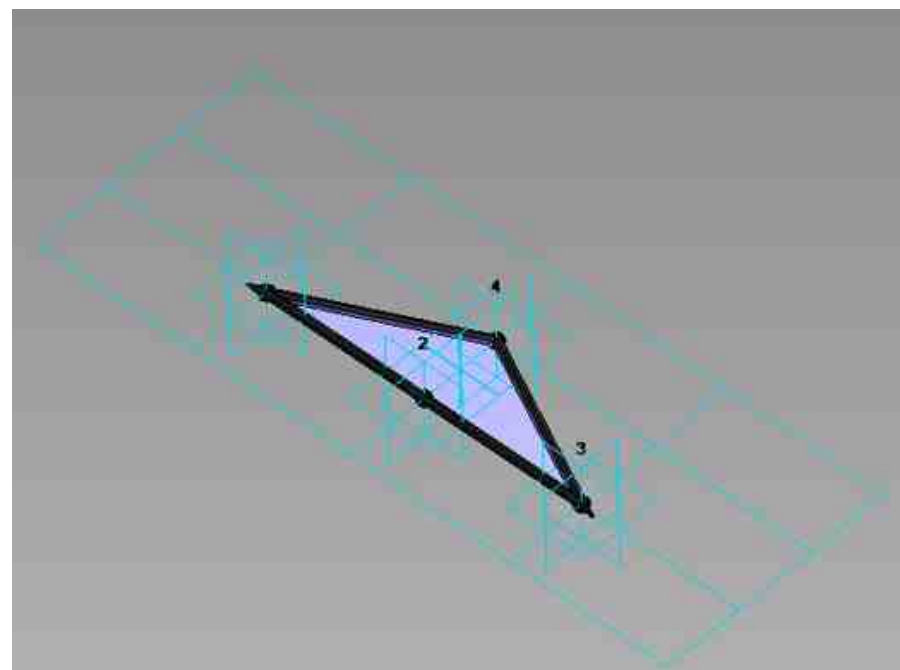
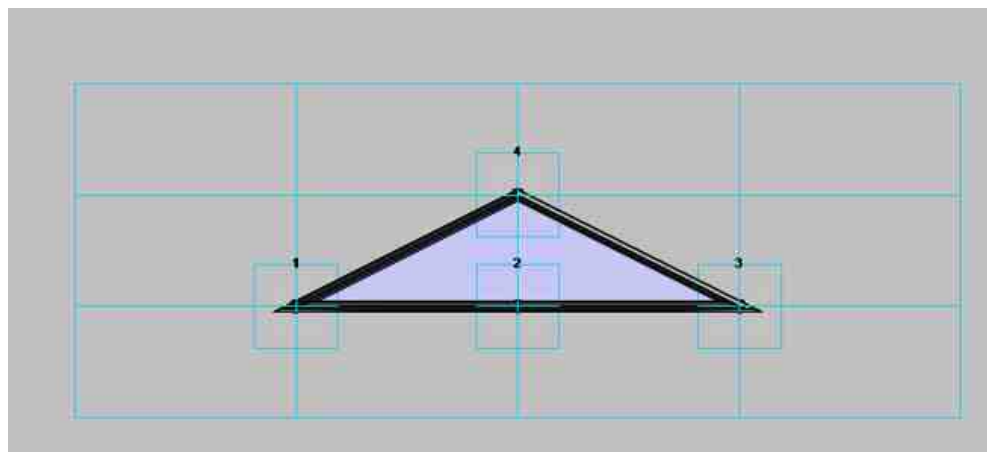
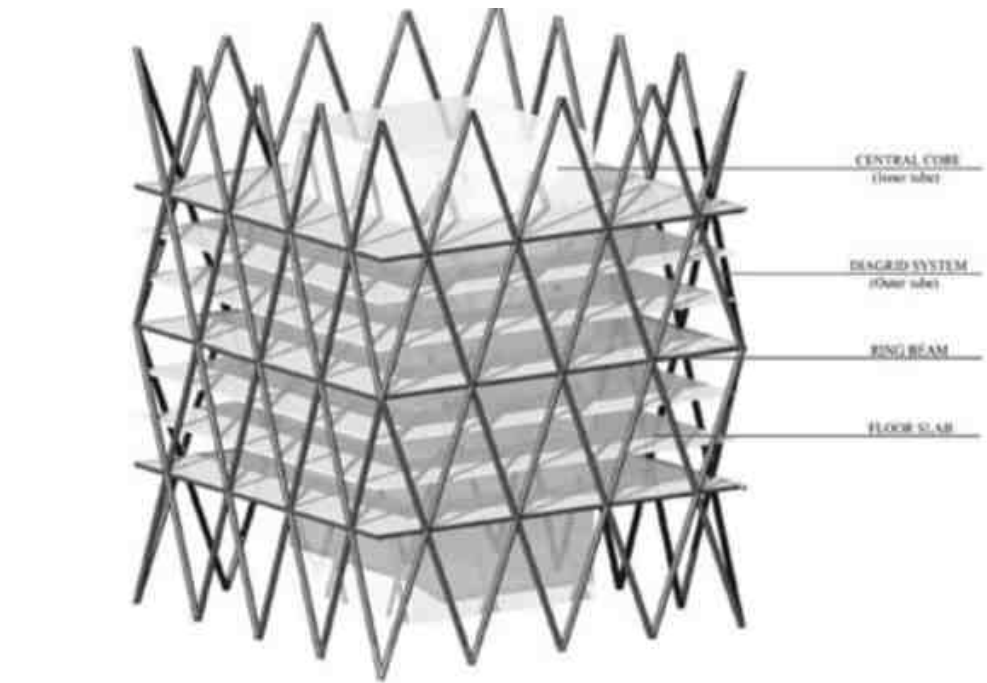
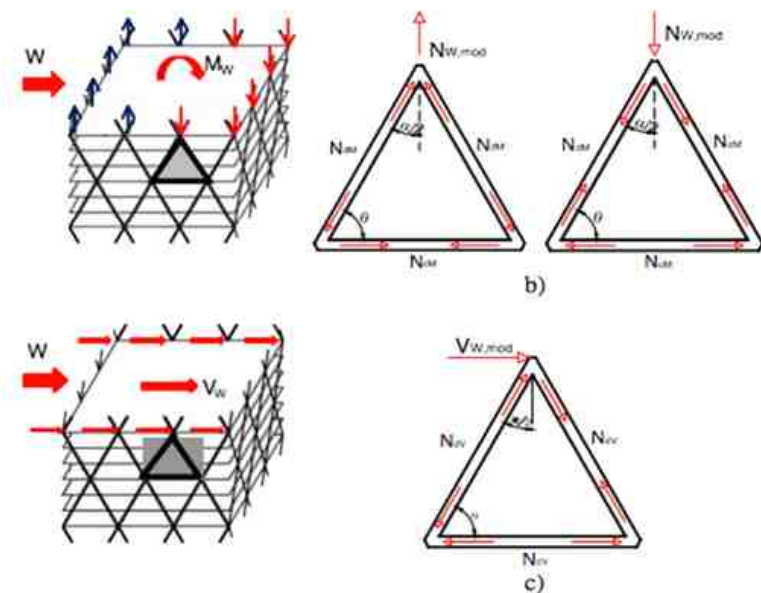
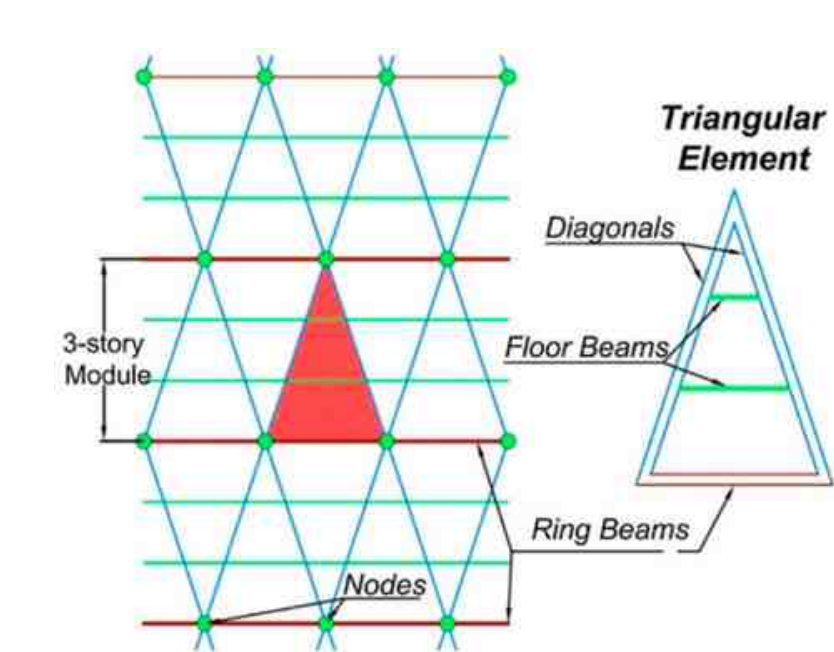


EAST ELEVATION
SCALE : 1/16"=1'-0"

3D SECTIONS VIEWS



DIAGRID FAMILY



HIGH RISE BUILDING

3D ISOMETRIC VIEW

REVIT MODEL



BUILDING INFORMATION

TOTAL FLOORS	G + 18 FLOORS
BASEMENT	1 LEVEL (PARKING)
TOTAL HEIGHT	272'-0" (82.91 m)
FLOOR TO FLOOR HEIGHT	17'-0" (5.18 m)
STRUCTURAL SYSTEM	R.C.C. SHEAR WALL CORE + FLAT SLABS
USE	RESIDENTIAL
PARKING	BASEMENT + GROUND
TOTAL UNITS	90 APARTMENTS (TYP.)

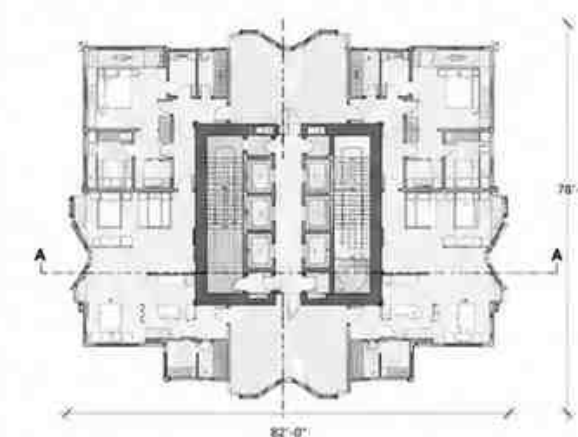
BUILDING FEATURES

- Centralized core for structural stability and efficient vertical circulation.
- Shear wall core resists lateral loads (wind & seismic).
- Column free perimeter for maximum daylight and views.
- Double skin facade for energy efficiency.
- Wide balconies for outdoor living.
- Basement parking for vehicles and services.

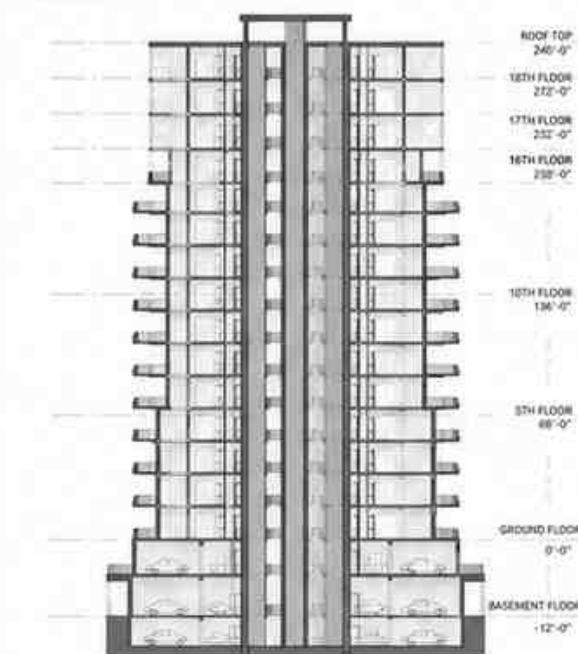
MATERIAL LEGEND

- GLAZING
- DOUBLE SKIN FACADE (DIAGRID)
- R.C.C. SHEAR WALL / CORE
- ALUMINUM LOUVERS
- CONCRETE FINISH
- GLASS RAILING

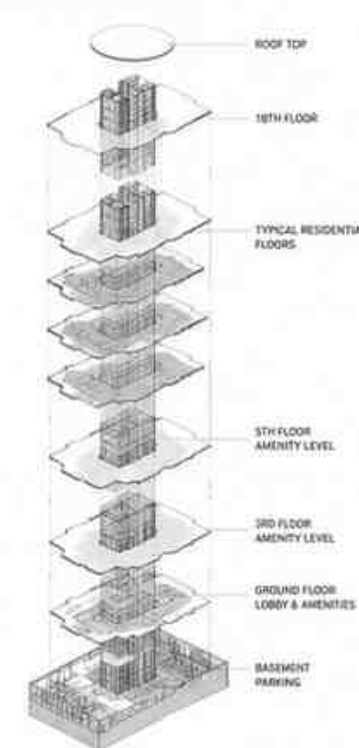
TYPICAL FLOOR PLAN (TYP. RESIDENTIAL FLOOR)



SECTION A-A



EXPLODED 3D DIAGRAM



3D VIEW - FRONT



3D VIEW - BACK



DOOR,WINDOW AND WALL SCHEDULE

VERTICAL FARMING HIGH-RISE — ARCHITECTURAL SCHEDULES

DOOR SCHEDULE

MARK	TYPE / LOCATION	DOOR TYPE	SIZE W×H	LEAF / MATERIAL	FRAME	APPLICATION	FIRE RATING
D-01	Main Entrance	Double glazed door	1800 × 2400	Aluminium + laminated glass	Aluminium	Public / Lobby	TBC
D-02	Secondary Entrance	Double door	1500 × 2400	Aluminium / glass	Aluminium	Public / Retail	TBC
D-03	Internal Public	Single flush door	900 × 2100	Solid core timber / laminate	Timber/steel	Office / public rooms	TBC
D-04	Office / Lab	Single door	1000 × 2100	Solid core / laminate	Steel	Research / office	TBC
D-05	Toilet	Single door	900 × 2100	Moisture-resistant flush Steel	Aluminium/steel	Toilets	TBC
D-06	Service / Utility	Single service door	1000 × 2100	Fire-rated steel	Steel	Plant / service	TBC
D-07	Stair Exit	Fire-rated door	1100 × 2100	Fire-rated steel	Steel	Stair / escape	90 min*
D-08	Fire / Service	Double fire door	1500 × 2100	Aluminium / steel	Steel	Service / escape	90 min*
D-09	Farming Zone	Single glazed/metal door	1000 × 2100	Steel	Aluminium/steel	Hydroponic / aquaponic	TBC
D-10	Basement / Parking	Service door	1000 × 2100		Steel	Basement / back-of-house	TBC

WINDOW & CURTAIN WALL SCHEDULE

MARK	TYPE / LOCATION	WINDOW TYPE	SIZE / MODULE	GLAZING	FRAME	APPLICATION	PERFORMANCE
W-01	Typical External	Fixed/operable window	1200 × 1500	Double-glazed low-E	Aluminium	Office / research	TBC
W-02	Large External	Full-height glazing	1800 × 2400	Double-glazed low-E	Aluminium	Public / amenity	TBC
W-03	Lab / Research	Fixed + operable	1500 × 1500	Low-E insulated glass	Aluminium	Research lab	TBC
W-04	Service	Fixed louver/window	900 × 900	Glass / aluminium	Aluminium	Service rooms	TBC
W-05	Farming Zone	High-transmission glazing	1500 × 1800	Insulated horticultural glazing	Aluminium	Vertical farming	TBC
W-06	Core / Small Rooms	Fixed window	900 × 1200	Double glazing	Aluminium	Support spaces	TBC
CW-01	Curtain Wall	Floor-to-floor curtain wall	1500 module × floor height	Double-glazed low-E IGU	Aluminium mullion	Main glazed façade	TBC
CW-02	Farming Curtain Wall	High-transmission curtain wall	1500 module × floor height	High-transmission IGU	Aluminium mullion	Farming façade	TBC

WALL TYPE SCHEDULE

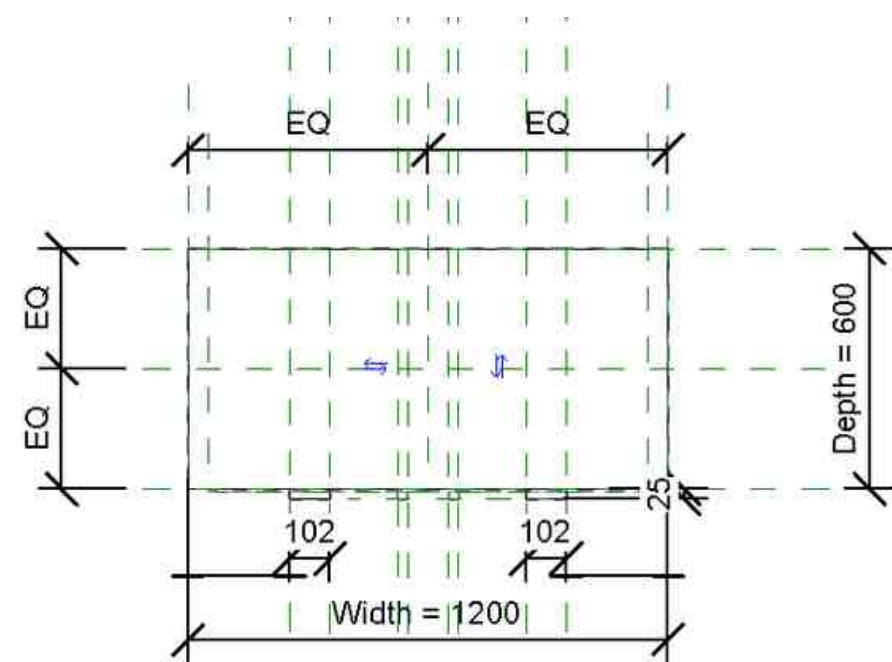
MARK	TYPE	THICKNESS	CONSTRUCTION / BUILD-UP	POSITION	APPLICATION	PERFORMANCE / STATUS
WALL-01	Internal Partition	100	12.5mm gypsum + 75mm metal stud + 12.5mm gypsum	Internal	Office / general rooms	TBC
WALL-02	Wet Area Partition	100	Moisture-resistant gypsum + 75mm stud + MR board	Internal	Toilets / wet rooms	TBC
WALL-03	External Infill Wall	200	AAC/blockwork + insulation + internal/external finish	External	Typical enclosed perimeter	TBC
WALL-04	Core / Shear Wall	300	Reinforced concrete	Structural	Central core / lift / stairs	Engineer verify
WALL-05	Fire-Rated Wall	150	Fire-rated gypsum/steel stud assembly	Internal	Escape/service zones	Code verify
WALL-06	Service Shaft Wall	150	Fire-rated shaft enclosure	Internal	MEP / vertical shafts	Code verify
WALL-07	Basement Retaining Wall	300	Reinforced concrete retaining wall + waterproofing	External	Basement perimeter	Engineer verify
WALL-08	Façade Backing Wall	200	Insulated wall build-up behind façade	External	Diagrid / glazed façade zones	TBC
WALL-09	Green/Farming Backing	200	Water-resistant insulated wall build-up	External	Vertical farming / planter zones	TBC

MARK	TYPE	THICKNESS	CONSTRUCTION / BUILD-UP	POSITION	APPLICATION	PERFORMANCE / STATUS
WALL-10	Plant / Utility Wall	200	Concrete/blockwork with service finish	Internal	Mechanical / utility rooms	TBC

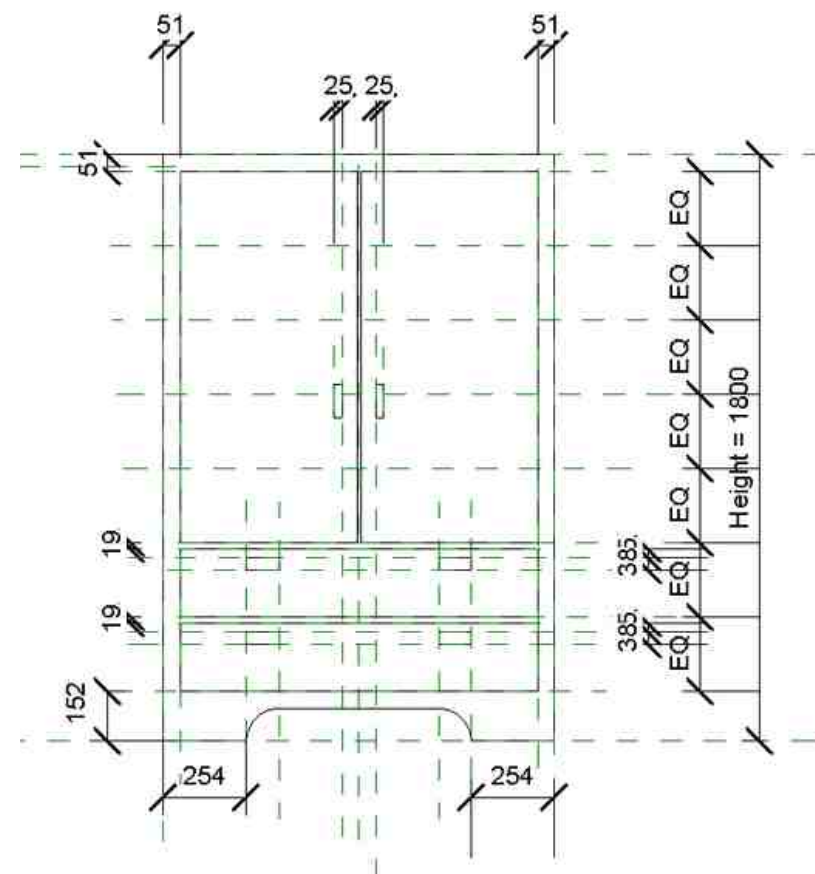
RENDERS



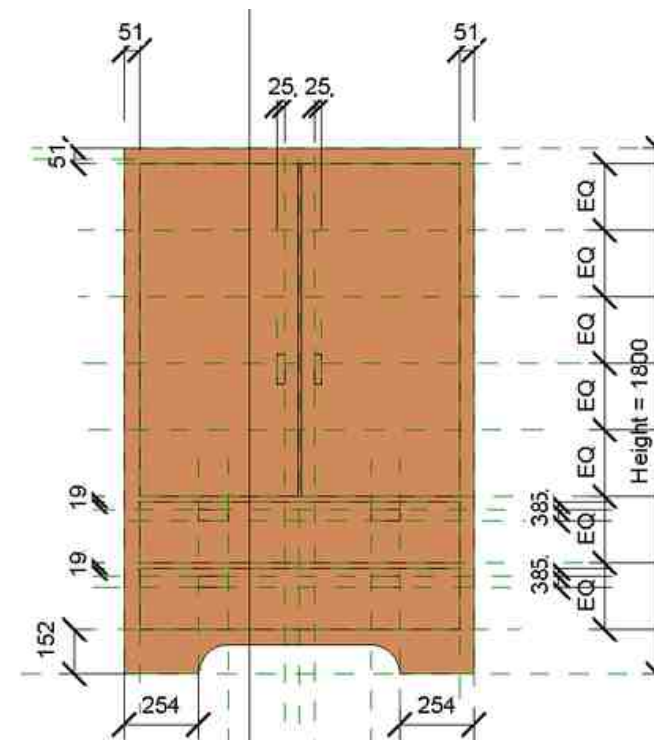
1001



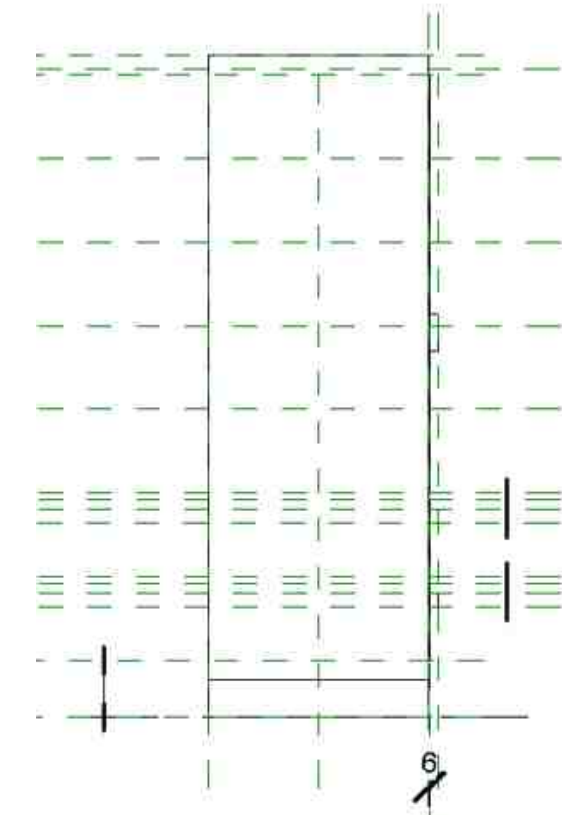
PLAN



FRONT ELEVATION



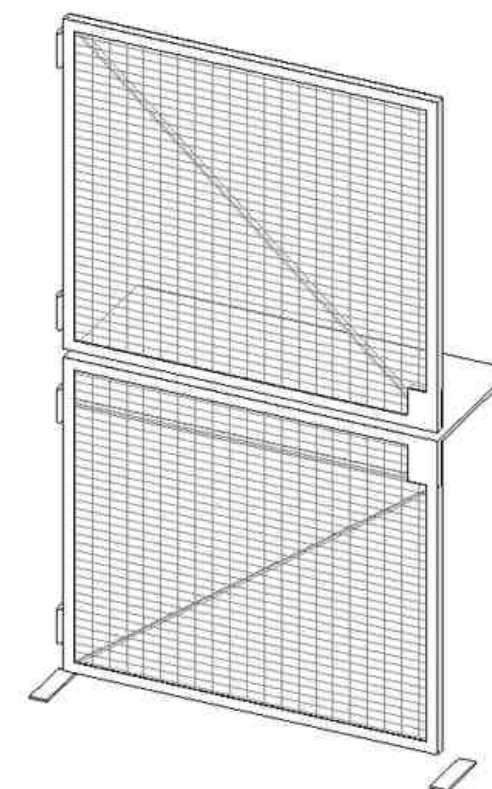
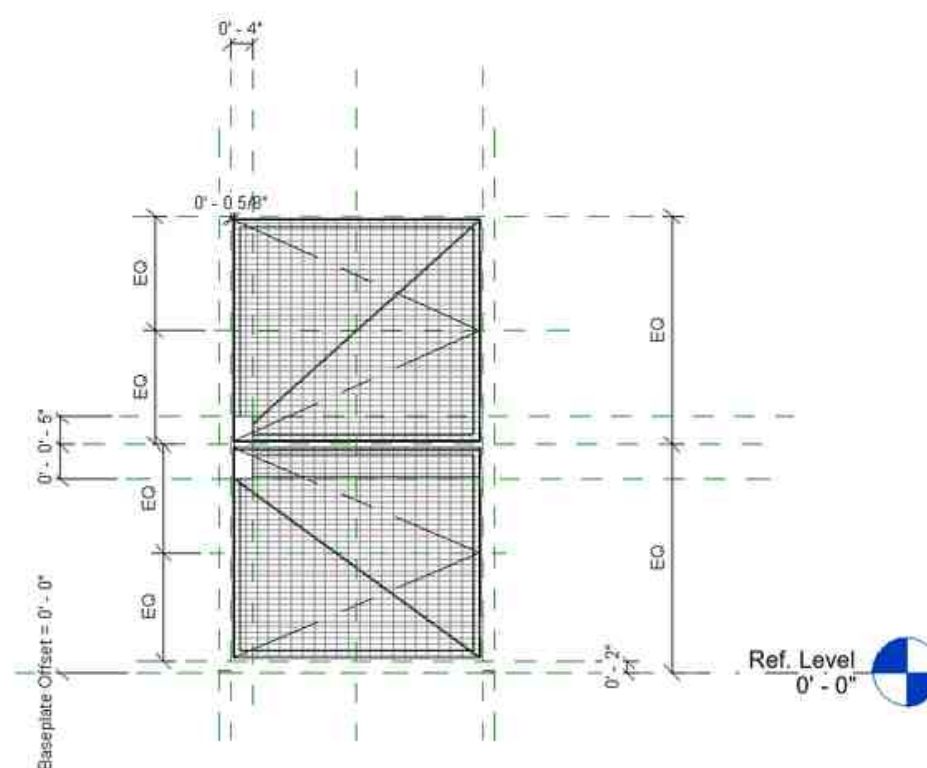
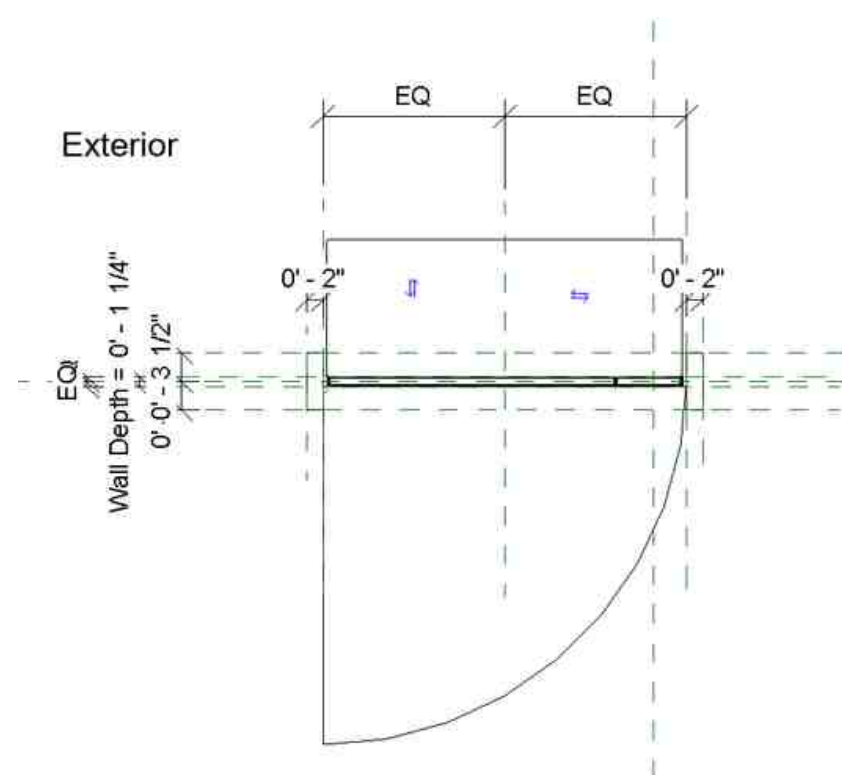
FRONT ELEVATION



LEFT ELEVATION

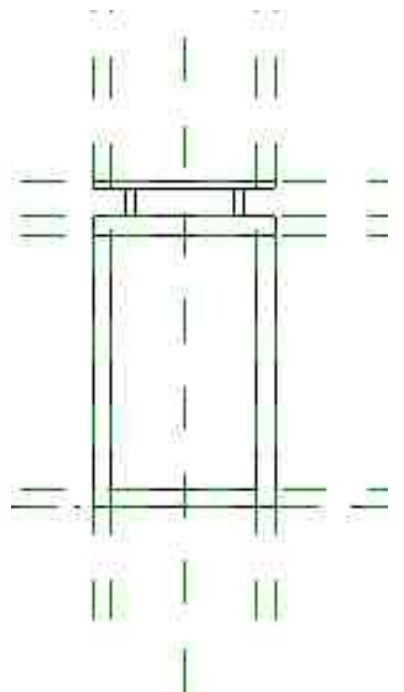
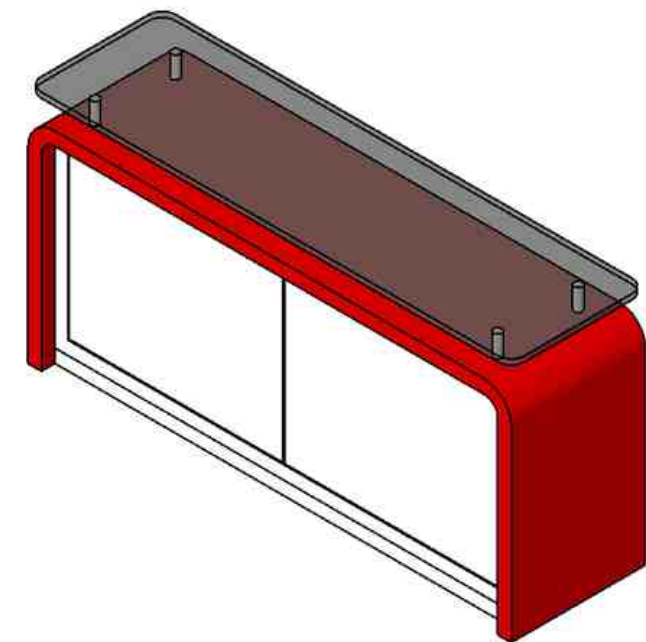
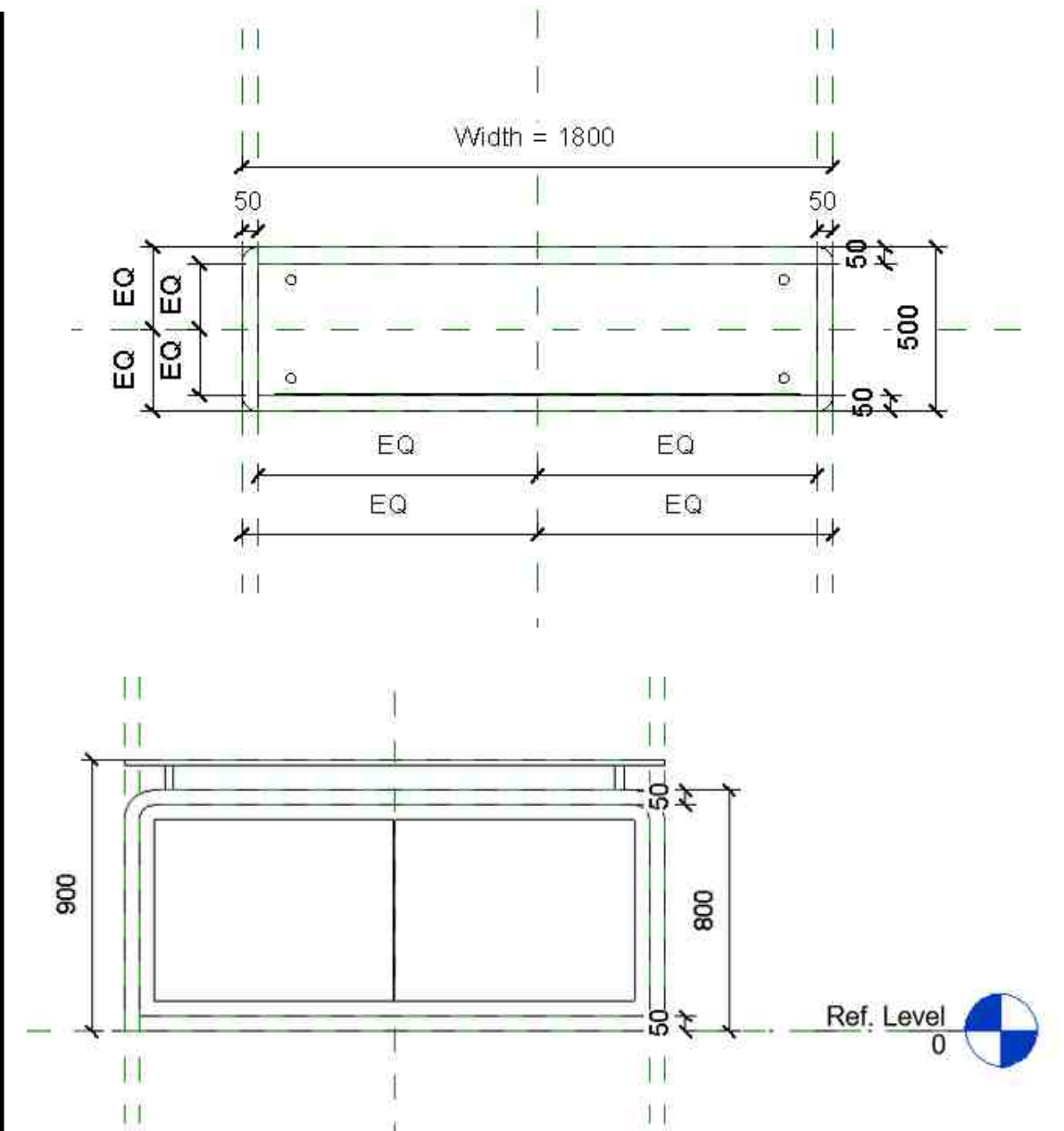
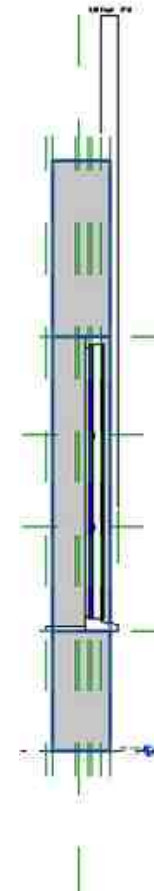
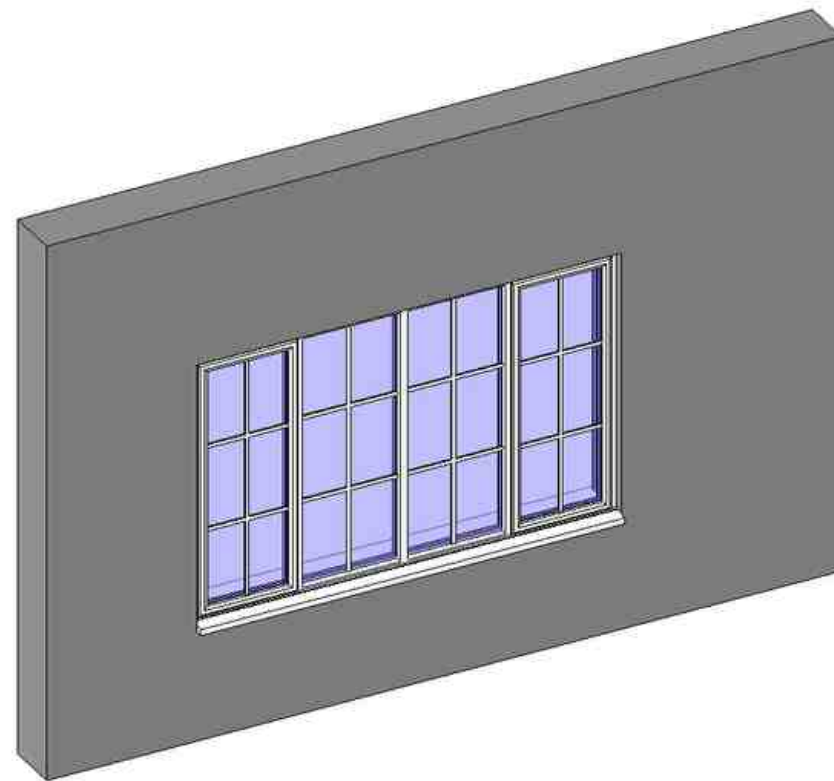
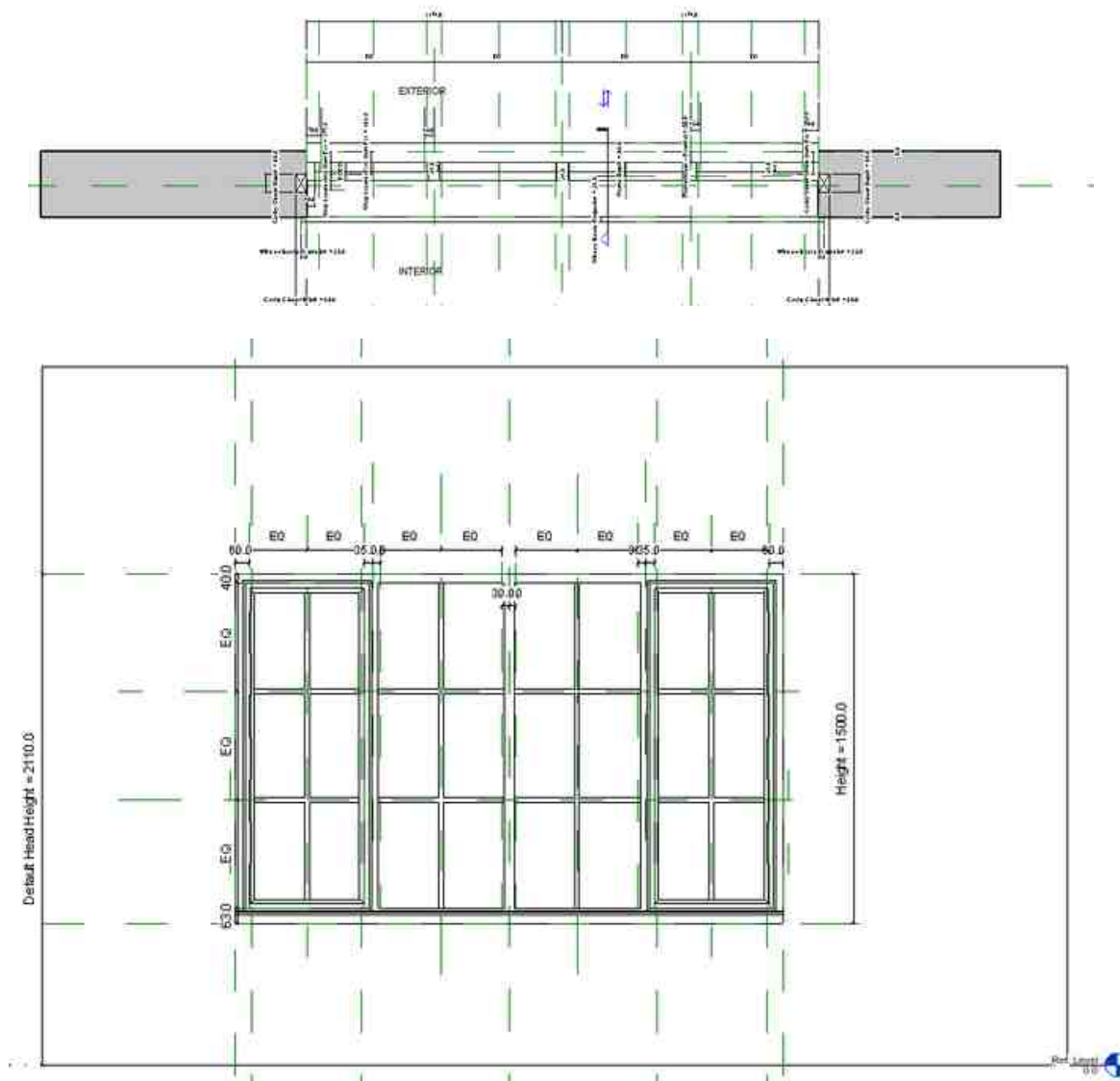
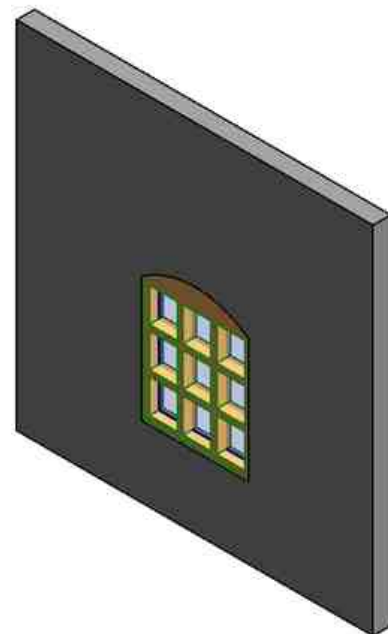
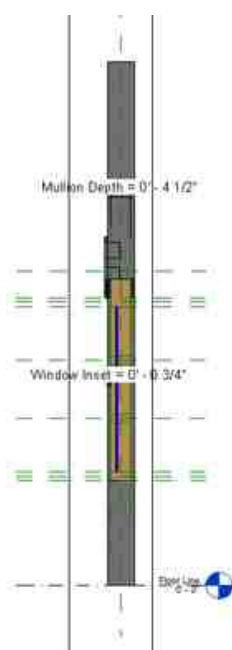
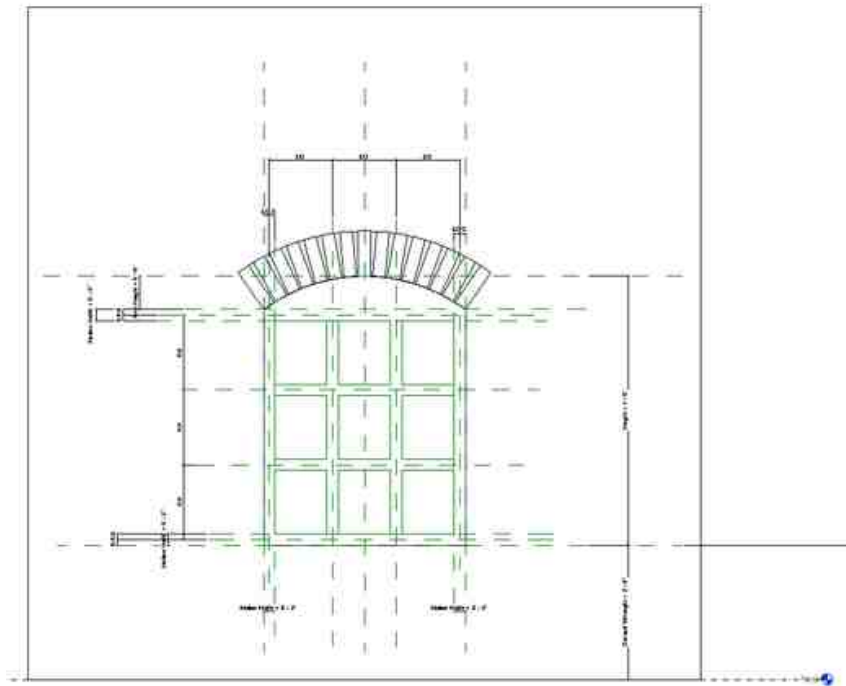
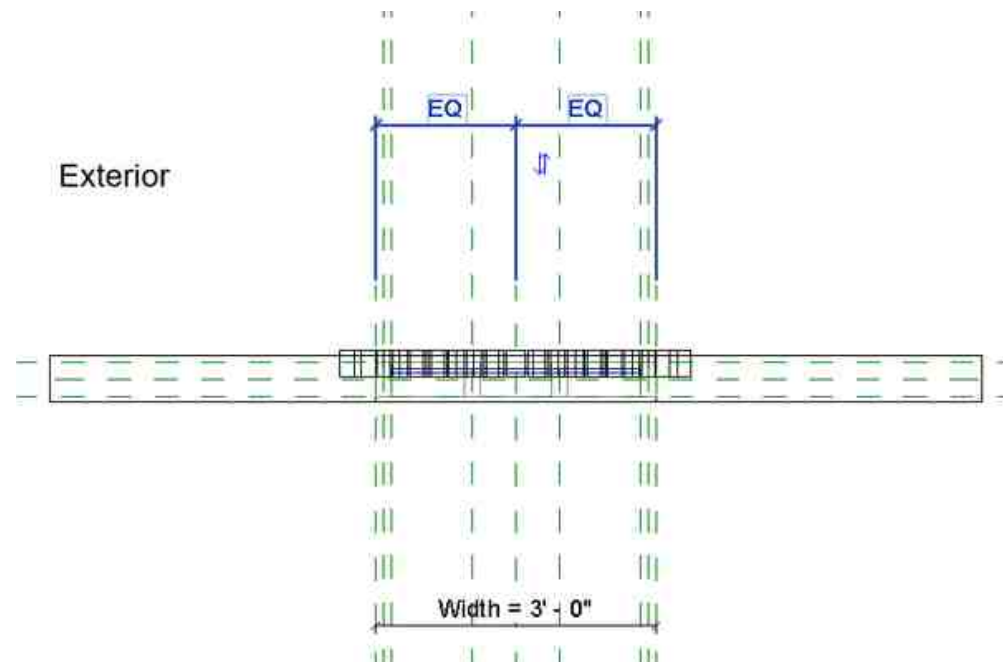


3D

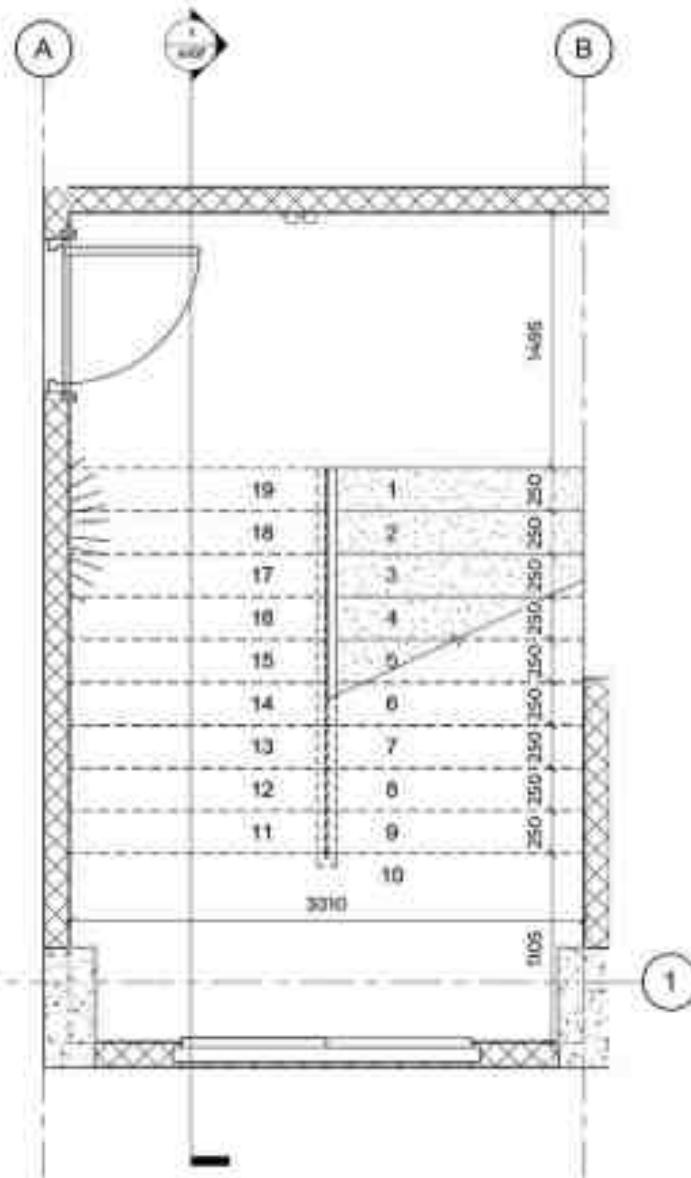


Constraints	
Extrusion End	0' 0 1/4"
Extrusion Start	0' 0 1/8"
Work Plane	Reference Plane
Graphics	
Visible	☒
Visibility/Graphics O...	Edit...
Materials and Finishes	
Material	Metal - WireMesh +...
Identity Data	
Subcategory	Panel
Solid/Void	Solid

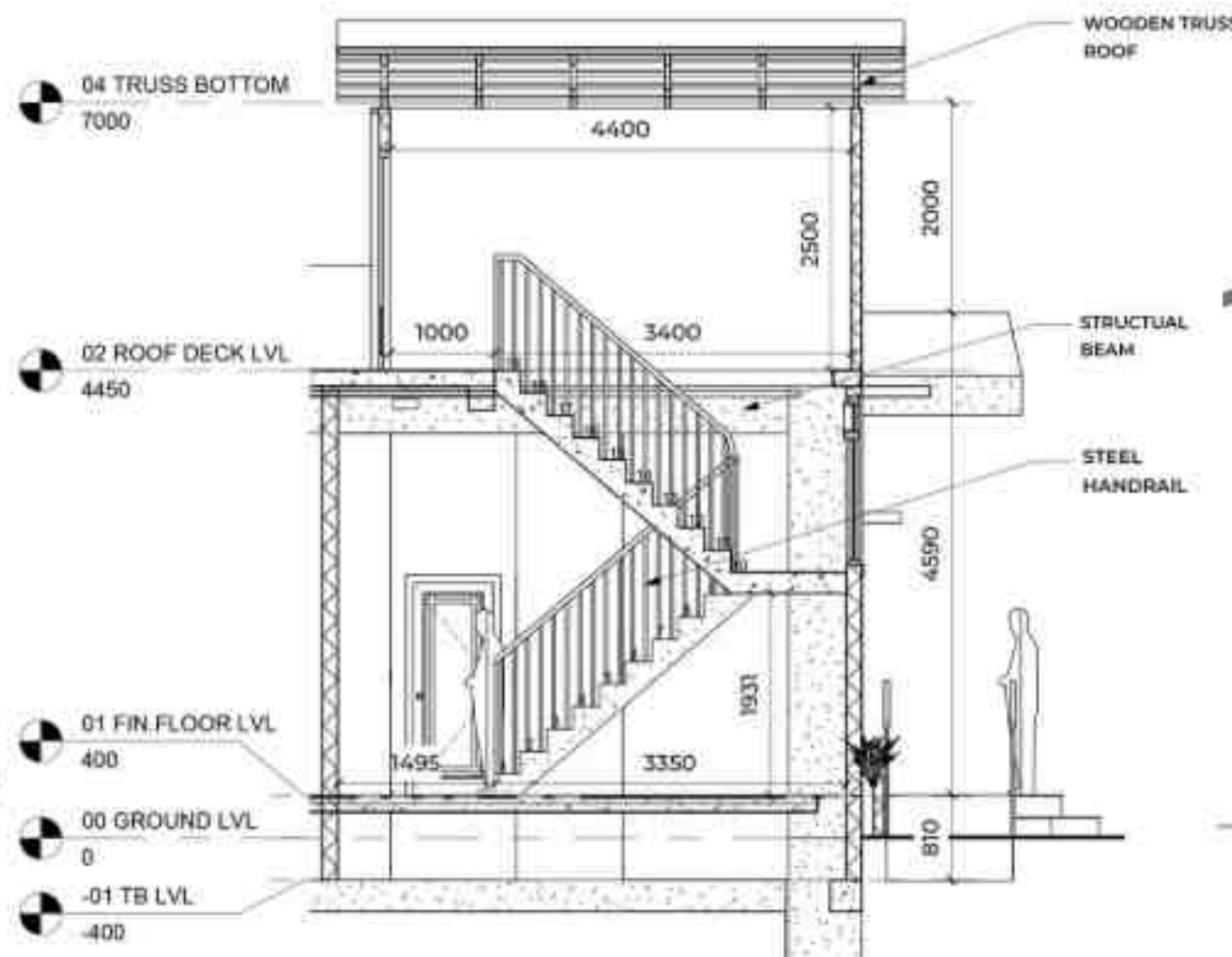
Exterior



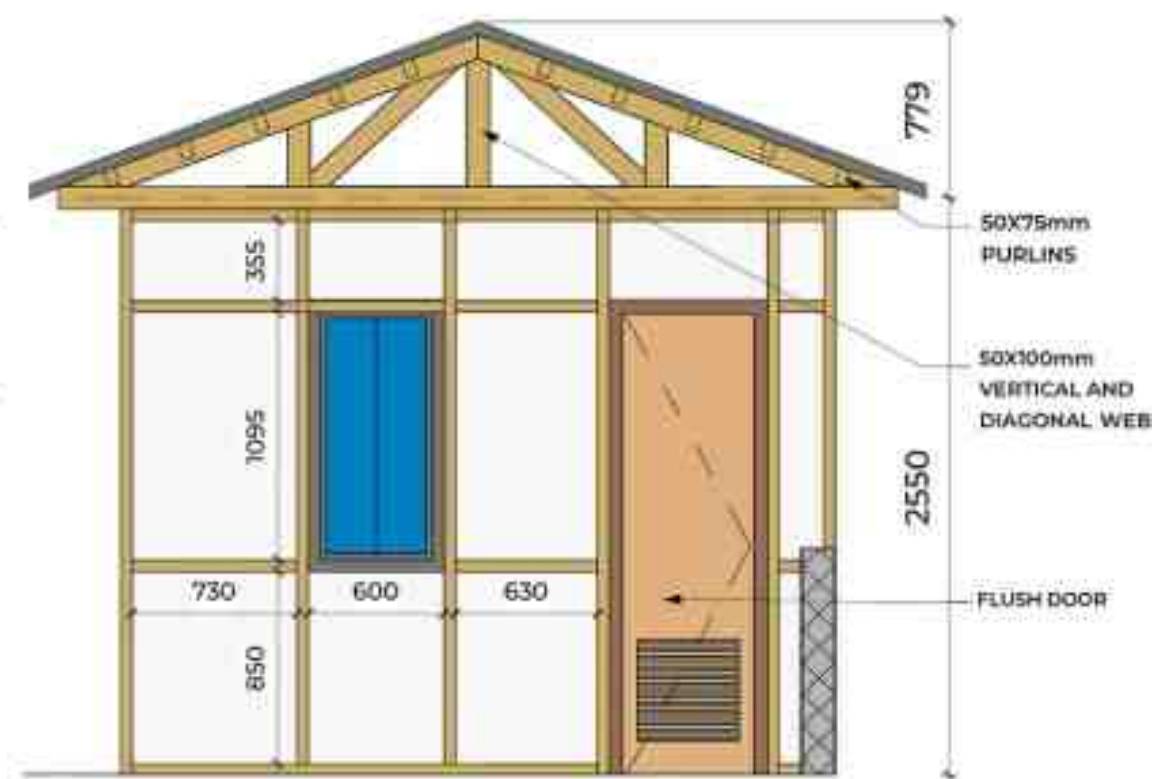
05 DETAILS



STAIRCASE PLAN



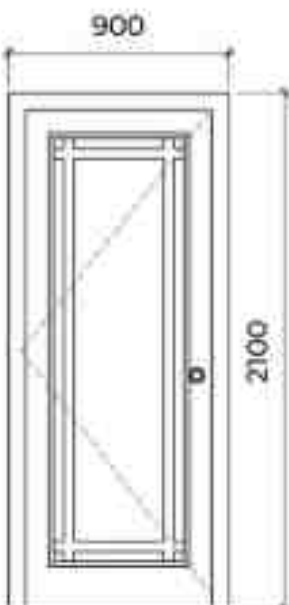
STAIRCASE SECTION



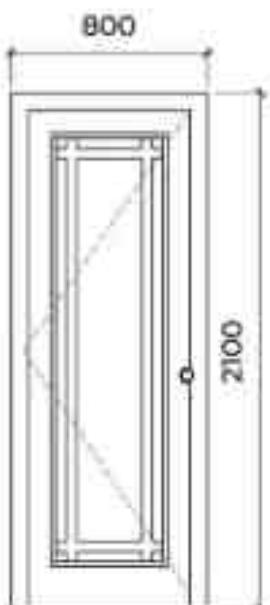
HEADROOM DETAIL



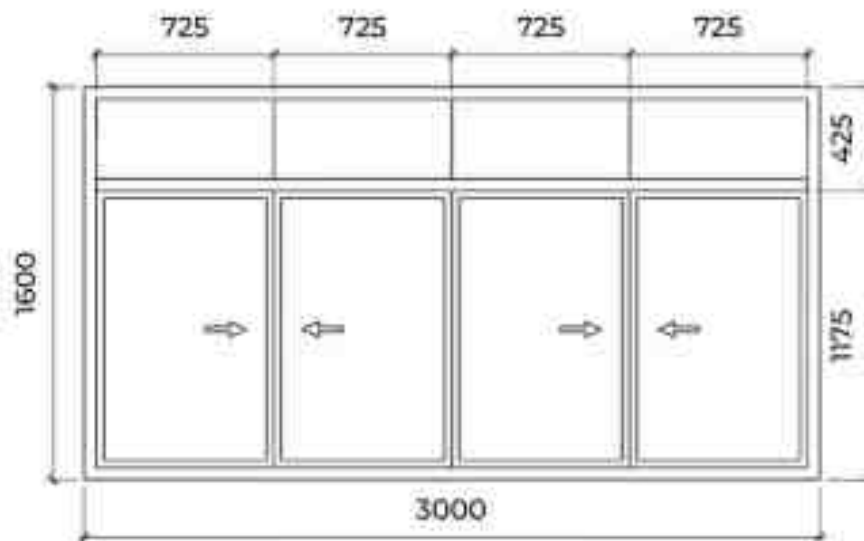
DOOR 1 - DETAIL



DOOR 2 - DETAIL



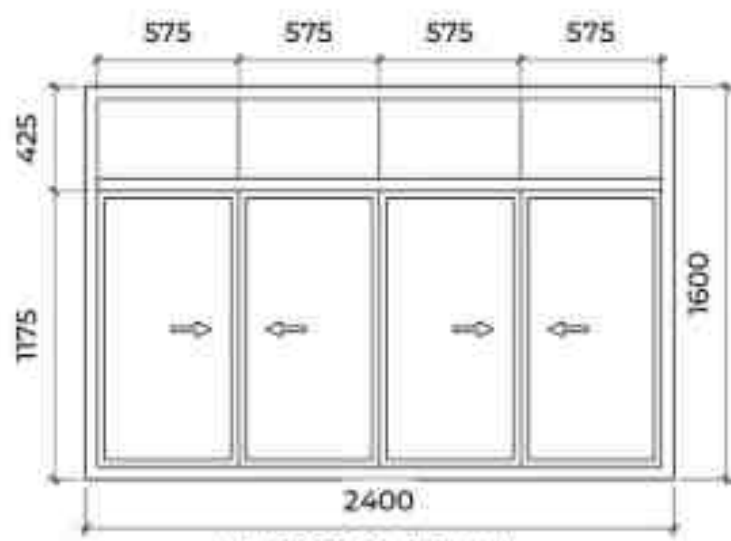
DOOR 3 - DETAIL



WINDOW 1 - DETAIL

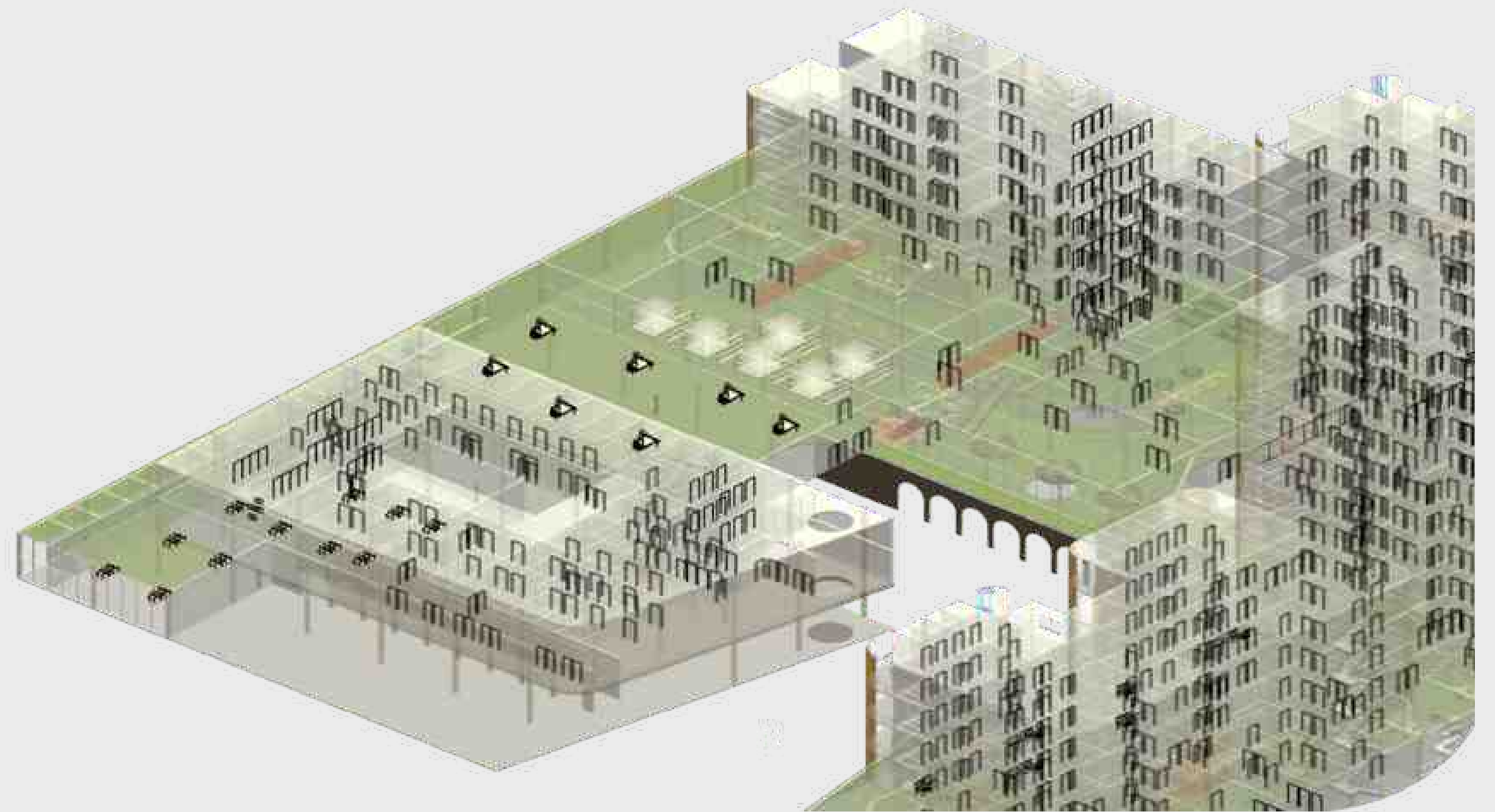
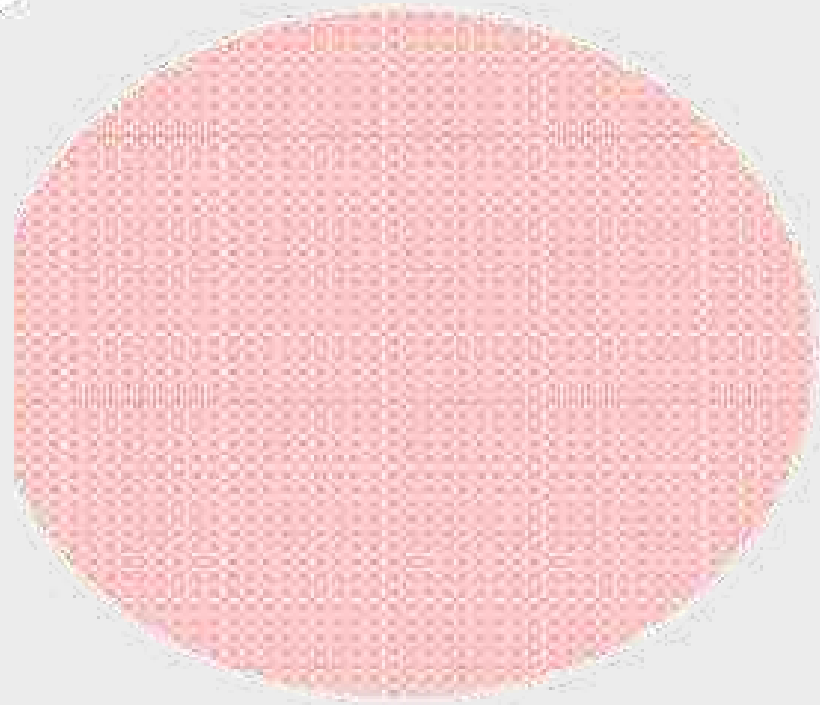


WINDOW 5 - DETAIL



WINDOW 2 - DETAIL

DOOR & WINDOW LEGEND



THANKU

Ar.Bilal Ahmad